ARA: from premises to papers

Version 7 · Executive summary · Evidence retained from v6, through 30 October 2026

ARA separates **L**, a bank of licensed moves, **T**, a frozen source theory, and **H**, proposed targets and disclosed hypotheses. Models propose; the runtime checks provenance, rule form, solver entailment and outstanding obligations. The scaffold was **designed and directed by a human, and built largely by AI coding agents under that direction**.

Eight novel methodological contributions

“Novel” here identifies the concrete contributions developed in ARA. Comparative priority over other systems is not established.

1. **Novelty relative to each paper, with padding controls.** Joint entailment can count as synthesis even when no individual paper entails the result. R28 rejects Boolean packaging and inconsistent source unions. The corrected real-freeze census found **0 qualifying candidates at a 500-consequence cap**. [R11, R27]
2. **Runtime-owned identity and stage recovery.** Models return judgments or local handles; code binds immutable IDs. Gap1r recovered completed construction and finished **150/150 alignment judgments**, with zero unresolved. R19 applied the same principle to consolidation. [R18, R50]
3. **A diagnosis gate totals missed.** Gap1r contained **141/160 bare atoms and 0 source-shared symbols**. Gap4a's shared-vocabulary prompt also produced zero cross-paper reuse. This evidence motivated accepted-content consolidation; its replacement freeze remained unfinished. [R46, R60]
4. **Packed requests with full-data authority.** Search replay reduced about **733k input tokens to 50k** while gates retained full T. Packing2 then restored complete declared bases in 46/46 maps and 92/92 cuts. Smaller requests alone had omitted essential premises. [R53, R54]
5. **Conservative sampled audit.** An identical prompt produced **44 then 0 flags** without a text repair. Fresh k=3 union auditing retains valid adverse history rather than allowing a favorable sample to erase it. [R20]
6. **Hypotheses with runtime-owned disclosure.** Review exposed **3 X32 defects** involving padding, nonaffirmative disclosure and hidden H. The repaired runtime preserves conditional status and gives H no source-paper credit. Gap3h had no admitted H or derivation. [R52, R61]
7. **Verified carry-over, meaning checks and local repair.** Exact source authority constrains inherited verdicts; changed wording needs unanimous equivalence. T2's structural recovery retained **all 51 prior adverse pairs**. Its final polish still refused. [R57, R66]
8. **A candidate pipeline with a diagnostic stop.** Every eligible route can reach writing while proposal and analysis paths keep their labels. A native pause after **10 empty jobs** requires diagnosis before resume. Two verified original proposal PDFs were delivered. [R33, R56, R64]

Research outcome: zero substantive novel CLOSED-derived contributions established. B/2 and T2 are argued proposals. Formal novelty is weaker than philosophical novelty; the full failures, human actions, credits and route atlas remain in the appendices.

Abstract

ARA is a research scaffold that separates licensed moves (L), a frozen source theory (T), and proposed targets or hypotheses (H). It contributes an enforceable source-to-argument workflow: per-paper novelty with anti-padding checks, runtime-owned identifiers and resumable stages, packed model requests with full-theory validation, disclosed hypothesis premises, conservative union audits and verified sentence carry-over. The findings expose limits that successful gate counts did not reveal. Gap1r froze 141 bare atoms among 160 commitments and had no shared symbols across source papers. A widened forward sweep produced no qualifying lead. Gap3 retained the safety and retraction passages but remained formally disconnected; its packed continuation completed 46 OPEN polarity routes with zero accepted inferences while omitting complete target bases. Offline packing2 replay repaired that visibility defect. **Shared-vocabulary prompting alone also produced no cross-paper reuse in gap4a's completed construction**, supporting a separate consolidation phase without establishing that the unfinished phase succeeds. Identical audit prompts returned 44 then zero flags, and later exports lost their citations; persistent dissent and local source verification addressed separate defects. Two verified original proposal PDFs were delivered, while polished T2 failed its final audit. Gap3h stopped before faithful H formalization. The contribution is an auditable method and a measured diagnosis of this setup; zero substantive novel CLOSED-derived contributions are established. This revision reorganizes the v6 evidence without adding research runs or upgrading pending outcomes.

1. L, T and H: the architecture and responsibility boundary

ARA permits models to propose arguments while keeping authority over an inference in deterministic code. **L** is the versioned bank of licensed moves. **T** is the frozen theory assembled from source commitments. **H** contains proposed targets, cuts and, when explicitly admitted, hypothesis premises. A model cannot create a license by declaring its own step plausible. A deduction must fit a bank form and pass Z3 entailment; a defeasible step must meet its scheme and critical-question obligations. [R3]

The scaffold was **designed and directed by a human, and built largely by AI coding agents under that direction**. The human's research and operational choices are recorded separately from the agents' proposed content. Models used for runtime generation have their own provider receipts; those receipts do not identify every development worker's serving model. [R1–R3, R13]

Rich extraction retains immutable source evidence. Construction selects commitments, checks their formulas and stance, records omissions and freezes T. Search then examines both the target and its negation. A search fork reuses the authenticated freeze. Consolidation or H admission instead requires a new construction-reuse lifecycle before its own single freeze. These distinctions make it possible to inspect what an experiment actually knew. [R3, R18, R50, R52]

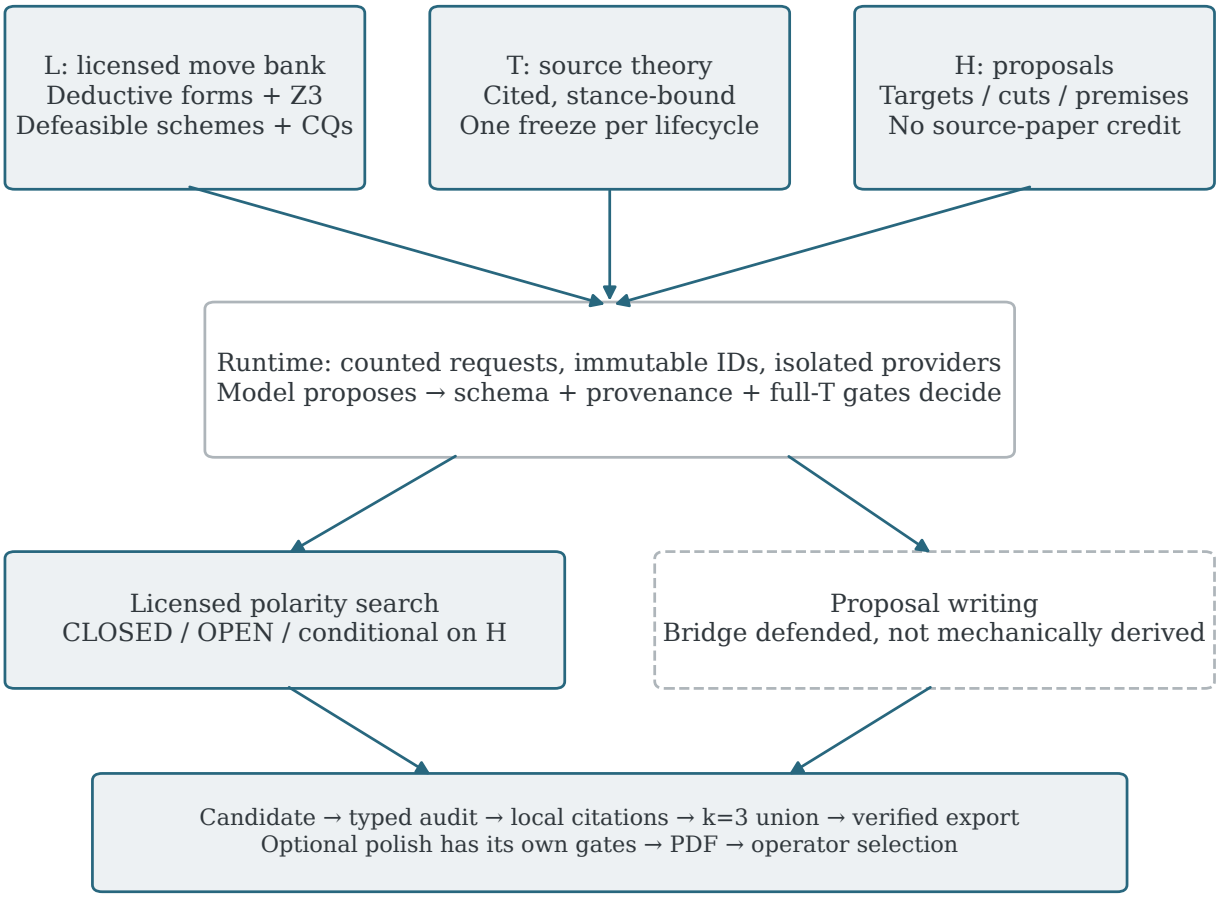


Figure 1. Authority is separated at each boundary. Proposal mode has its own authorization and disclosure; it is not an alternate way to claim a CLOSED route. PDF production follows the status of its input.

CLOSED means the route met that revision's formal obligations. It does not establish source-translation fidelity, premise truth, literature novelty or the quality of the resulting philosophy. OPEN identifies an unclosed obligation. A proposal paper may argue for a bridge principle while its route stays OPEN, and any future derivation using H must remain explicitly conditional on H. The final paper inherits those limits. [R28, R52, R55]

1.1 A contribution with explicit limits

ARA makes source-to-argument responsibility inspectable. A proposed target, a source translation, a model-alignment bridge and a licensed inference have different authority. Its contribution is the machinery that keeps those distinctions visible through construction, search and publication. Each mechanism is backed by a failure receipt or a tested correction. No comparative benchmark or first-ever priority claim follows from that evidence. Appendix A.1 preserves the detailed mechanism table and the abstract-level comparison with The AI Scientist, STORM and SciAgents. [R3, R48–R65]

The decisive distinction is between a result about the machinery and a result in philosophy. A repaired novelty gate can permit genuine synthesis without producing a new philosophical thesis. A faithful proposal paper can defend a bridge that its native route never derived. This report credits each outcome at the level its evidence supports.

2. Constructing T: retrieval, windows, stance and alignment

2.1 Literature enters through recorded evidence

The supplied bibliography contained 53 works; the first acquisition obtained 26 PDFs. Rich extraction produced 5,659 items, quarantined 363 and left 5,296 eligible. The later collection reached 54 PDFs, but those additions did not retroactively enter older freezes. Access limits and the dated modern-literature review remain part of the evidence. A bibliographic entry is not a full text, and a full text is not yet an endorsed premise. [R2, R5, R12]

Extraction preserves premises, conclusions, inference structure, qualifications, source stance and page provenance. Its initial four-item, 400-character summary cap lost too much structure. Rich extraction repaired that design, while validation, grounding and OCR exposed further losses. Appendix A.2 retains the complete counts and corrections, including the different OCR metrics. [R1, R12]

2.2 Construction selects what search can use

A construction window is a counted request over grounded evidence and its dependencies. Its accepted output must satisfy formula, symbol, provenance, stance and capacity checks. Skips remain recorded. Increasing the context allowance does not guarantee valid output; one failed inventory can reject a whole window. [R16, R35]

The coordinator's estimate of 70–90 tokens per commitment was wrong; the pilot measured about 233. Bounded closure and global budgets replaced repeated cap increases. The lesson is operational: a refusal must identify the actual constraint, and recovery must preserve accepted evidence rather than silently dropping difficult dependencies. The incident record retains the original mistaken estimate and all relevant stop counts. [R1]

2.3 Source structure and attribution

A resumed Codex pilot completed **20 searches** but returned only open routes. Diagnosis found **16 of 22** proposed steps with conclusion mismatch, a T of **166 standalone atoms**, and no target-symbol overlap. Structured source inferences, frozen-catalogue guidance, goal binding, joint slot formalization, and reference checks repaired concrete causes of empty search. The next pilot obtained closed routes, but they were depth-zero restatements of individual commitments. [R1, R10]

Own-thesis classification then became entangled with attribution: content definitions saying that an author held a claim made the classifier reject useful own-voice targets as reports. Content-only definitions plus separate stance/provenance repaired that representation. A subsequent lexical filter still rejected ordinary words such as “claims” or “states” in substantive content. Fork 2’s review identified **eight lexical-only exclusions**. The lexical screen became advisory; exact report frames remained hard failures. [R9]

These are representational checks. Endorsed, reported and attacked source positions remain distinct, even when each has a page citation. A model-alignment judgment is not an additional source paper. An atomic formula associated with a paragraph stating a rule is not itself an executable conditional. The later gap2/gap3 anchor results make that distinction concrete. [R28, R35, R36]

Measured bottleneck: paper-local atoms instead of shared predicates

The common explanation for failed cross-paper synthesis is the representation of T, not just the search or context budget. Direct inspection of `runs/codex-gap-1r/frozen-t.json` verifies **141 of 160 commitments as bare propositional atoms**, all with paper-prefixed names, for example `beddorgoldstein2021_info_sensitive_safety`. The remaining **19** have non-atomic syntax: 18 top-level implications and one conjunction. “Structured” here describes the formula tree, not proof that a paragraph's internal quantification, relations or normative force were faithfully formalized. All **147 catalogue symbols are zero-arity atoms**, not a shared predicate vocabulary. [R46]

Counting the papers attached to each symbol's formula occurrences through commitment provenance, **only two of 147 symbols span at least two papers**, and **both gain that status through the single alignment bridge**. Excluding the bridge, **zero symbols are reused across source papers**. The two are the MacFarlane assessment-sensitivity and Schaffer truth-perspectivalism atoms; their bridge remains a model judgment, not source endorsement. Thus the 2/147 count must not be read as two independently established equivalences. The completed alignment examined **150 candidate pairs: 149 unrelated and one directional entailment**; no catalogue symbol mapping changed. [R18, R46]

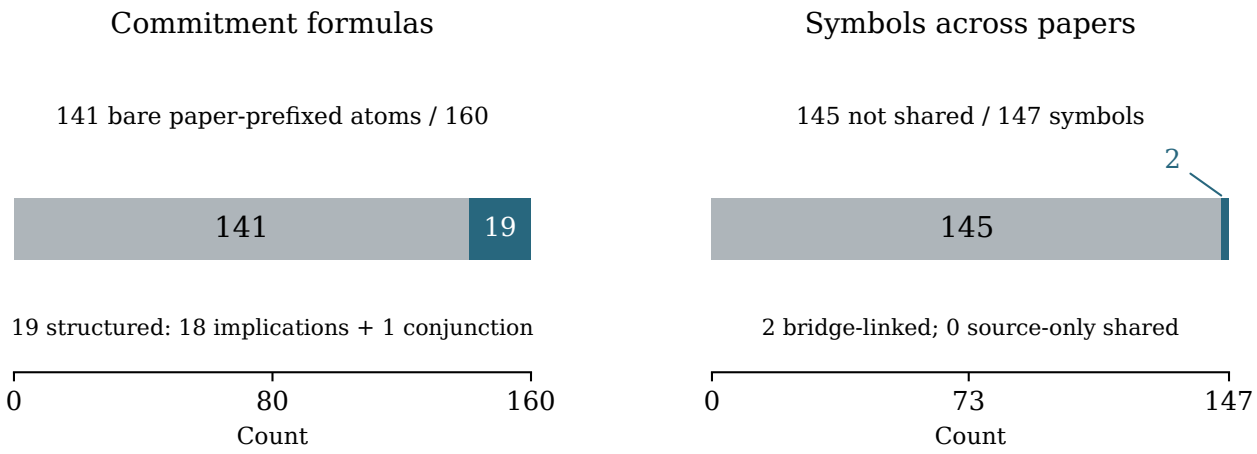


Figure 2. The same gap1r freeze underlies A/B/C and the forward sweep. Structured means non-atomic syntax, not established faithful formalization. Every catalogue symbol is a zero-arity atom. Cross-paper sharing counts formula occurrences using their source provenance; only the two symbols in the alignment bridge span papers. Excluding that bridge, source-only sharing is zero. Alignment tested 150 pairs (149 unrelated, one directional entailment) and merged no symbols. This quantifies the missing common formal vocabulary; it is not an impossibility theorem.

The bank cannot infer the internal content of an opaque proposition from its English name. In particular, an atom naming information-sensitive safety does not expose a reusable safety predicate, its arguments, or a conditional connecting the same assertion and contexts to retraction. The 19 non-atomic formulas and one disputed bridge did not supply the required cross-paper chain. Boolean conjunction/disjunction rules can still fire across otherwise disconnected atoms—as the forward census shows—but this packages premises rather than producing the intended substantive bridge. **No substantive cross-paper derivation was established in the recorded tests.** This is an evidence-based explanation of those outcomes, not a theorem that every possible bank application or future representation must fail. [R28, R44, R46]

The r17 prompt and wider alignment mechanism are now implemented and reviewed; successful source sharing is not established. Free replay on gap1r leaves source sharing at zero. A fresh gap4a construction later delivers the shared-vocabulary prompt but again produces zero actual cross-source reuse. The response is a separate accepted-content consolidation lifecycle, whose r18–r20 attempts are described in §2.7; r21 remains pending. These stages must not be described as a completed before/after improvement. [R49, R50, R60]

2.4 Runtime owns immutable identity

Gap1 completed 74 construction windows and retained 159 commitments from 22/26 papers. Alignment stopped at candidate 36/150 after three replies altered an immutable symbol ID. The validator correctly refused them. The design failure was requiring the model to echo identity, allowing one item to abort the whole alignment, and providing no continuation from finished construction. [R16]

R12 moved identity binding into deterministic code, requested relation/reason only, isolated unresolved items and added alignment continuation. Gap1r reused all completed construction plus 35 accepted judgments, completed the remaining 115, and finished with **150/150 accepted judgments and zero unresolved**. One alignment bridge yielded frozen T=160. That bridge remained semantically disputed; a complete inventory does not establish that its judgment is justified. R19 later applied runtime-owned binding to consolidation slots and evidence handles. [R18, R21, R50]

2.5 Larger windows can lose more evidence

Gap2 accepted only 3/13 windows and retained 57 commitments from three papers, then stopped before freeze. Its ten skipped windows were validation/formalization failures, with zero context/budget/provider refusals. The larger batches changed the unit of failure and lost the safety anchors even while recovering MacFarlane's Retraction Rule. Raising the limit again would not address that defect. Appendix A.3 retains the exact-ID comparison, source losses and refusal sequence. [R35]

2.6 Gap3: the right passages survived, but their formal structure did not

Gap3 completed all **77 terminal construction windows**, freezing **243 source commitments across 23/26 papers**. Stances were **201 endorsed, 14 reported, 28 attacked**. All five Beddor–Goldstein anchors and six MacFarlane anchors survived; the two required safety anchors and the Retraction Rule were endorsed. Alignment judged **148 unrelated, one contrary and one right-to-left entailment**. Applying the judged set refused consistency, so no alignment bridge was added to the frozen source theory. Preserving a refusal is part of the result. [R48, R50]

The freeze contained **212/243 bare atoms (87.24%)**, 212 distinct symbols and **zero symbols shared by at least two source papers**. Source coverage improved over gap1r, but the retained prose rule remained an opaque whole-claim atom. Textual possession of the safety anchor and Retraction Rule was necessary for a grounded attempt; it was insufficient for executable safety-to-retraction reasoning. Gap3's target/search outcomes appear in §4.4. [R48]

Sigma r16's dependency-safe partial acceptance, bounded split retry and more precise attribution screen precede the shared-vocabulary work. The report records a whole-response complexity bypass found and fixed during development. Safe parallel windows remain deferred: snapshot ordering, reservations, symbol collisions and crash/resume evidence are unresolved requirements. Primary-window and retry-entry counts must stay separate; gap4a's 79 primary windows generated 237 terminal entry receipts. No throughput benefit from concurrent construction is claimed. [R45, R60]

The combined r16/r17 delivery at `de4fff8` completed **1,846/1,850 tests** (four skips) and **158/158 selected production-wire controls**; the final independent S17 review reported 382 checks and ten controls with no blockers. This supersedes the earlier r16 proof-pending checkpoint. It establishes tested implementation at that head, while parallel windows remain deferred and live source sharing remains zero. [R45, R49]

2.7 Prompting for shared vocabulary did not create reuse

Gap4a tested the r17 shared-vocabulary prompt in fresh construction. It produced **308 commitments from 21 papers**, including **263 bare atoms**, using **292 symbols**. **Zero symbols were shared across source papers**, and there were zero exact-antecedent cross-paper modus-ponens opportunities. A recorded request contained the new prompt and 290 catalogue symbols. Only 75/263 atoms still had their own source prefix: the naming policy changed, but actual reuse did not. This finding supports an explicit consolidation step rather than treating shared-vocabulary instructions as a completed repair. [R49, R60]

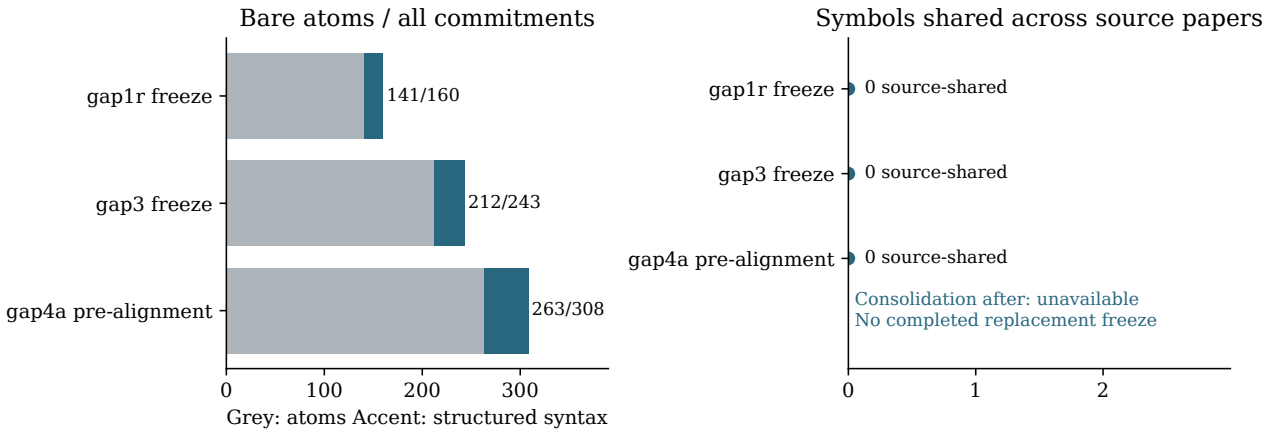


Figure 3. No before/after gain is established. Gap1r has two bridge-linked symbols but zero source-only sharing. Gap4a is completed construction before alignment, not a frozen after-treatment. Structured syntax and source-independent names do not establish faithful shared semantics.

Consolidation re-formalizes accepted source content against a shared catalogue in a new lifecycle. Source IDs, text, citations, stance and provenance remain fixed; old and proposed formulas are both retained. New alignment follows because old judgments bind old formulas. The runtime records model judgments and freezes the new lifecycle once. It neither invents source endorsement nor treats shared symbols as proof of faithful translation. [R50]

The live consolidation attempts remain unfinished at the retained cutoff. R18 stopped on identity mismatches; r19's runtime binding worked but definition screening refused; r20 accepted 22/30 batches, covering 181 commitments, then stopped when a corrected reply regressed from eight required formulas

to seven. R21's cardinality and failure-isolation response was pending. There is **no completed replacement freeze and no after-sharing improvement** to plot. Gap4a itself was paused before alignment, with a separate hold on its pinned search code. Appendix A.4 retains these distinct stopping points. [R50, R51, R60]

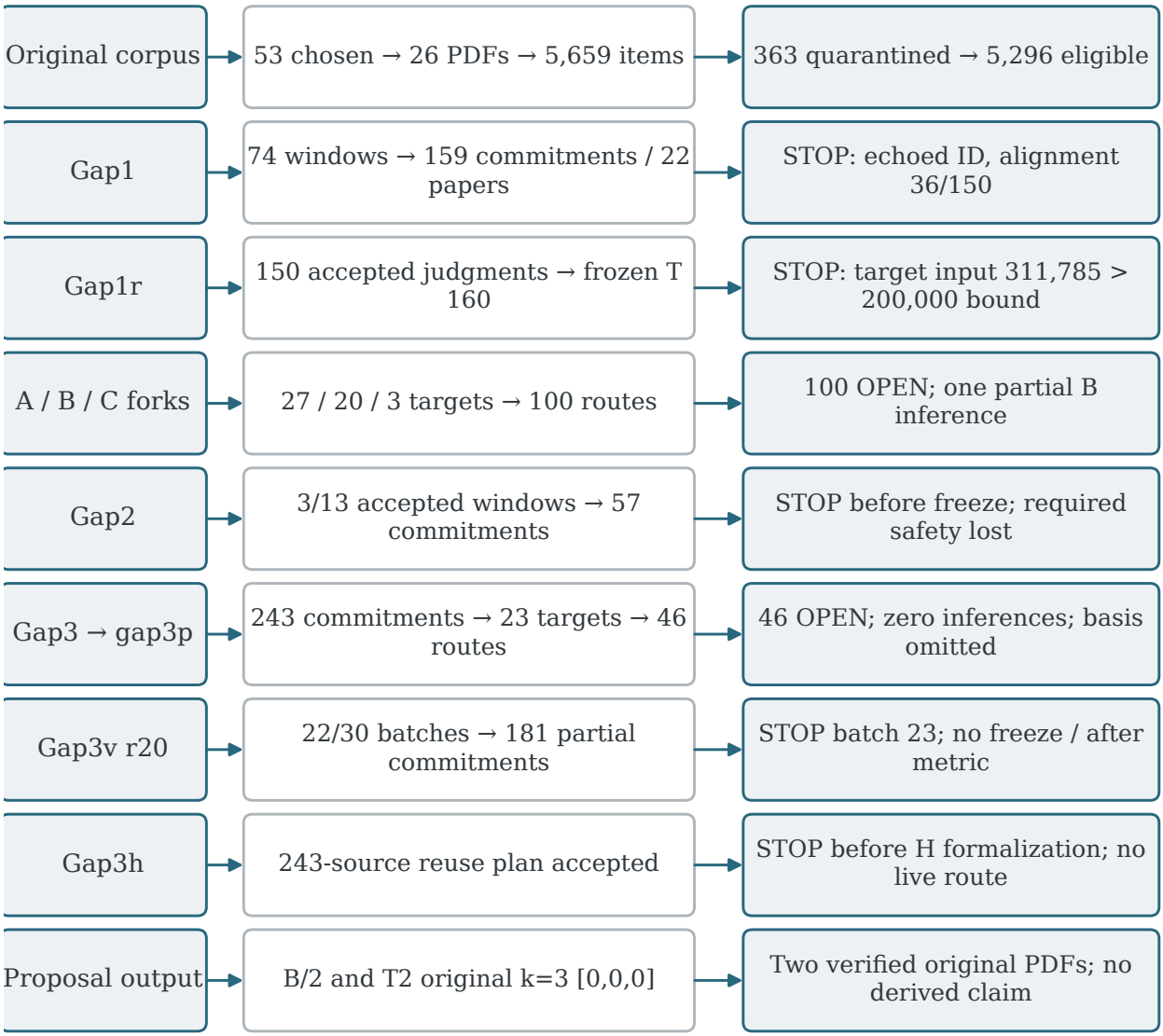


Figure 4. Branched gate funnel: recorded progression ends at an explicit stop or output. Rows are distinct lineages, not successive denominators. Proposal PDFs are not CLOSED gap derivations.

The funnel records distinct experiments and their stops. The proposal PDFs belong to a separate writing path; none is counted as the endpoint of a CLOSED gap derivation.

3. One freeze, controlled forks, and context admission

3.1 Freezing a theory is an evidence boundary

A lifecycle constructs and aligns T, then freezes once. Fork-from-freeze reuses that authenticated theory and its provenance; it does not import improved evidence retrospectively or make the experiments independent. Gap1r reused completed construction and accepted alignment judgments. A/B/C then reused its byte-identical freeze. Gap2's unaccepted rows never entered such a freeze; gap3 starts a new construction lifecycle. [R18, R28, R35, R36]

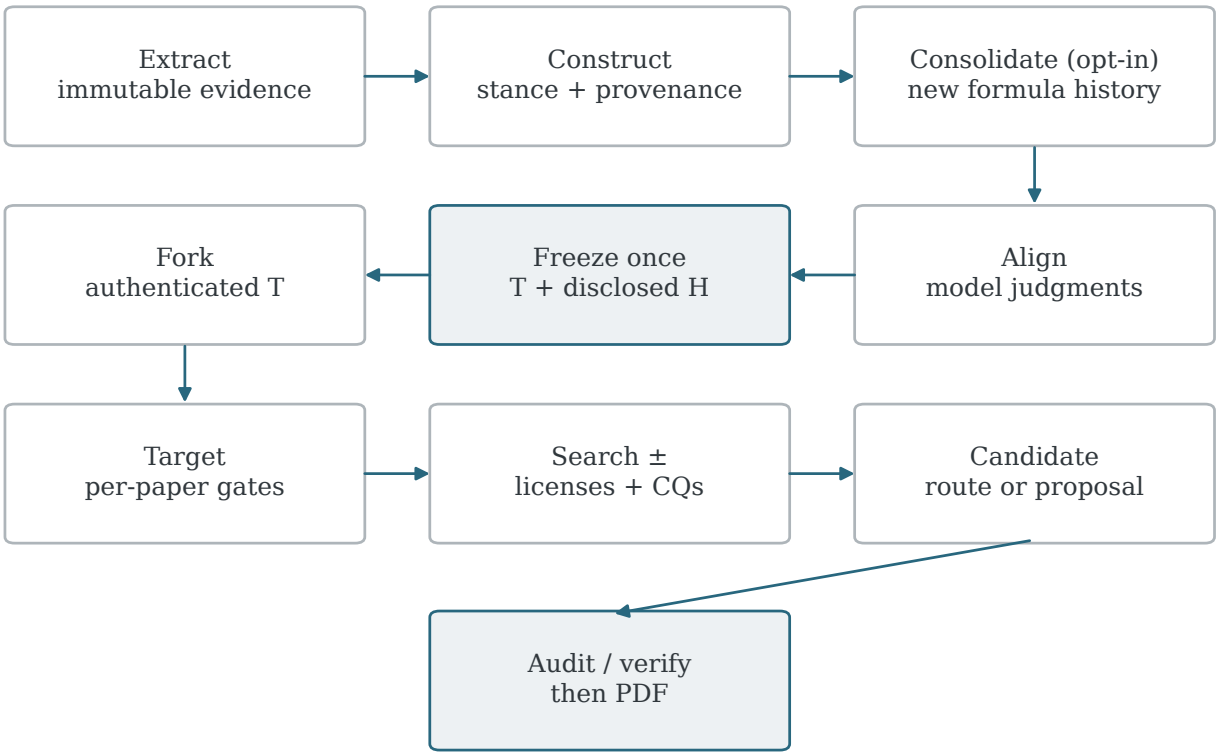


Figure 5. Consolidation or H admission reuses completed construction in a new lifecycle before its own freeze. A search fork instead preserves the existing freeze. Valid completed stages survive later failure; neither path silently edits the source experiment.

A fork-from-freeze preserves T for a new target/search experiment. A construction-reuse lifecycle that consolidates formulas or adds H is different: it creates a new theory and freezes that new lifecycle once. The previous freeze remains intact. Reusing accepted stage receipts saves work without rewriting what a historical experiment knew. [R18, R50, R52]

3.2 Admission must count the request that will be sent

Gap1r froze T but stopped before its first L3 target call: 311,785 estimated input tokens plus 16,384 output allowance exceeded the 200,000 bound. A smaller readiness estimate had not represented the full request. Pinned client metadata advertised 272,000 default and 872,000 maximum context windows, but the adapter initially failed to forward `model_context_window`. Those settings do not establish actual service capacity. [R18, R19]

After the operator authorized loosening the limit, r13 forwarded the setting and packed the gap context. R14 corrected a separate stage-window mismatch by normalizing admission and provider configuration before cache lookup. This repaired the request contract; it did not supply missing premises. [R24, R30]

3.3 A/B/C preserve the freeze while changing the experiment

Fork A's packed target request estimated 122,106 tokens including output allowance; B's full-provenance request estimated 328,468. Both retained the same frozen formulas and stance. Their actual first-call input usage was 60,702 and 171,899 tokens respectively, so conservative admission estimates cannot be read as backend token measurements. The forks also differed in semantic classifications and bounds. They do not provide a controlled estimate of packing's effect on target quality. C supplied exact external T2 probes. All three outcomes are reported in §4.3; full request and accounting details remain in Appendix A.5. [R28]

3.4 Packing must preserve the evidence that the model needs

Gap3 exposed a later request boundary: **42/46 jobs failed before search model dispatch**, with roughly **720k input versus a 567k native bound** and a large novelty payload. Gap3s raised the configured window to 872k and cleared admission, but **67 CLI exits** exhausted reservation headroom: **0/46 completed, 36 failed and ten cancelled**. Their cause remained unknown in those receipts. It is not legitimate to infer the backend's actual context ceiling from these unknown exits. [R48]

The first search-packing change reduced replay estimates from **730,729–733,092 to 47,195–49,558 input tokens**, with a 16,384 output allowance and 239,232 native bound. Gap3p's actual map estimates were **46,940–49,970** and cut estimates **58,243–61,287**. Full frozen-T licensing, novelty, stance and alignment gates remained intact. Yet every root request selected the same ten commitments: **34/46 omitted every basis row, 12 retained one, and zero retained a complete basis**. Both L3 polarities omitted all six declared safety/retraction anchors in mapping and cuts. An ID in a compact index does not supply the full source premise. The same L3 omission existed in gap3s; the evidence does not establish that packing introduced it. [R53, R54]

Packing2 makes the declared basis mandatory through both polarities, child leaves, corrections and shrink attempts. It retains formal/alignment neighborhoods and full vocabulary for arbitrary open slots. Unknown IDs or an oversized mandatory payload refuse explicitly; mandatory evidence is not dropped to obtain an admission pass. On the **same 46 maps and 92 cuts**, offline replay retained the complete basis in every request, with **12 distinct root sets** and all six anchors in both L3 jobs. Estimates were **49,118–58,524** for maps and **60,421–69,841** for cuts, plus 16,384 output. These are request-replay results with zero generation, not proof that the model would produce a useful inference. [R54]

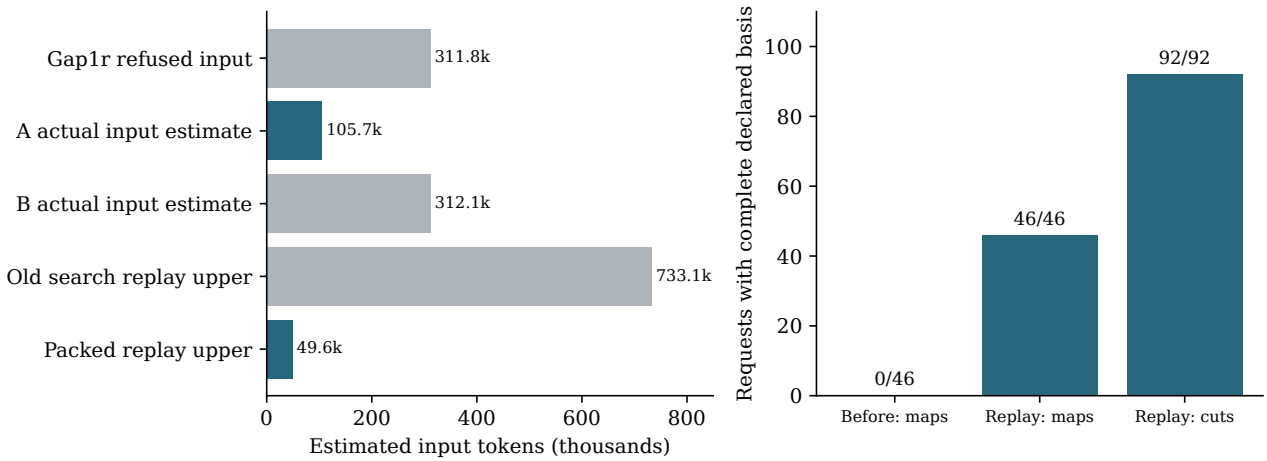


Figure 6. Token reductions and evidence coverage are different achievements. Left: input estimates exclude the 16,384 output allowance, with distinct experiments explicitly named; old-versus-packed search is a replay comparison. Right: packing2 offline replay repairs full basis visibility. No later live derivation is inferred.

The deployment proof also needed correction. A reported 920-control pass was retracted when PACK-F3 found one mutation failing through unrelated AttributeError rather than its intended stance boundary. The repaired owning test at `1878759` detected the intended error. Main `bd60d6a` merged that repair, with 239 affected tests and **920 accepted inherited-plus-repaired controls**, not a fresh 920-control rerun. Later reviewed main `dcbc4b3` included full-basis packing, nested consolidation/provenance readers and native pause diagnosis; its merged affected gate was **274/275, one skip**. Historical live results stay attached to their original heads. [R54, R64]

4. L licenses moves; polarity search exposes open obligations

4.1 A formal label is not a substantive result

The acceptable pilot required source coverage of at least 20/26 papers and at least three genuinely derived routes using distinct supports from two papers. Early CLOSED labels failed that substantive standard: they included single-premise restatements and conjunction packaging. Later fork-3 routes supported writing experiments but used one paper. Appendix A.6 preserves the rejection chart and route census, including overlapping rejection reasons and the distinction between polarity jobs and published routes. [R6, R9–R11]

A licensed step must instantiate a bank entry. Deductive steps need its shape checks and solver entailment; defeasible steps need the scheme and its critical-question obligations. A root can have a licensed inference and still remain OPEN. The mechanism reports which obligation is missing instead of treating fluent plausibility as a license. [R3, R28]

4.2 Novelty gates must allow synthesis without allowing padding

The whole-T novelty gate was another substantive design error. Requiring that the union of T not entail a target prevents deductive synthesis across papers, even when no single paper entails it. The corrected local policy allows union entailment and tests each paper’s endorsed closure separately, while retaining anti-padding, stance, fidelity, and support checks. The placeholder regression combines P from one paper

with P → Q from another and derives Q. This demonstrates the corrected mechanism, not a real literature result. The final bounded real-freeze census, at depth **two** and a **500-consequence cap**, found **zero qualifying candidates**. An earlier six-candidate census was superseded after a packaging defect was fixed. Cross-paper symbol alignment was still a separate integration boundary. [R11]

The R28 repairs distinguish a genuinely joint consequence from Boolean packaging and reject inconsistent endorsed-source unions rather than deriving arbitrary novelty from contradiction. Kappa4's later xi29 review closed R28-F1/F2 at the integrated head. Gap1r's previously observed packaging leads were rejected under that repair; its bounded formula union had been SAT, so the inconsistent-union trigger was absent in that freeze. A passing gate is still weaker than literature novelty or faithful source interpretation. [R18, R27]

4.3 Why A/B/C all remained OPEN

All three forks reached native `awaiting_selection`. No route had been selected at their handoff checkpoint. A/B ran on historical `c99c6b3`; C ran on reviewed `893985a`. All retained the same authenticated freeze: **160 commitments = 159 source entries from 22/26 papers + one model-alignment bridge**. The source-only stance criterion remains **159/160, FAIL**; paper coverage passes. The disputed bridge is not source endorsement. Construction accepted **53/74 windows** and skipped **21**: 16 closure-bound failures, four formalization failures, and one without topical roots. Larger later contexts did not restore those omissions. [R28]

Fork	Targets	OPEN routes	CLOSED-derived	Demonstrated substantive contributions	Terminal argument content
A: packed, two hops	27	54	0	0	All 54 have a single OPEN root.
B: full provenance	20	40	0	0	39 single roots; one three-node partial sign argument, non-L3.
C: external T2 and directions	3	6	0	0	All six have a single OPEN root; no accepted inference, cut, support or CQ answer.

B's partial argument retains a sign inference and two source leaves; CQ1 is supported and CQ2 unsupported. One source's underlying inference is omitted. It remains OPEN and does not establish L3, but it does contain material that can be drafted and criticized. The all-candidates queue contains `gap1pu-B-sign`. C's exact biconditional and its two directions were disclosed external hypotheses from eps2's dossier, not ARA discoveries. Their formulas were whole-statement atoms; the biconditional and contextual quantification were not decomposed and the direction atoms were not formally linked. These are bounded unsuccessful tests, not countermodels or impossibility theorems. [R28, R33]

No closed safety–retraction argument follows from those experiments. B's partial sign argument is useful enough to attempt writing, but its unsupported CQ and omitted underlying source inference remain obligations. C is the focused test of externally supplied T2; its six root-only routes contain no licensed step. ARA's near-T2 proposals, the exact external formulation and the partial sign argument must not be collapsed into one result. [R28]

The sign scheme joins Egan's response to the communicative objection with MacFarlane's criticism of Solipsistic Contextualism's treatment of genuine disputes. It licenses a defeasible step toward the comparative target, but CQ2 remains unsupported. The two source leaves do not constitute a CLOSED conclusion. This is the route used for the refused sign drafts; it is distinct from B/1 and B/2's safety–retraction proposals. [R28, R42]

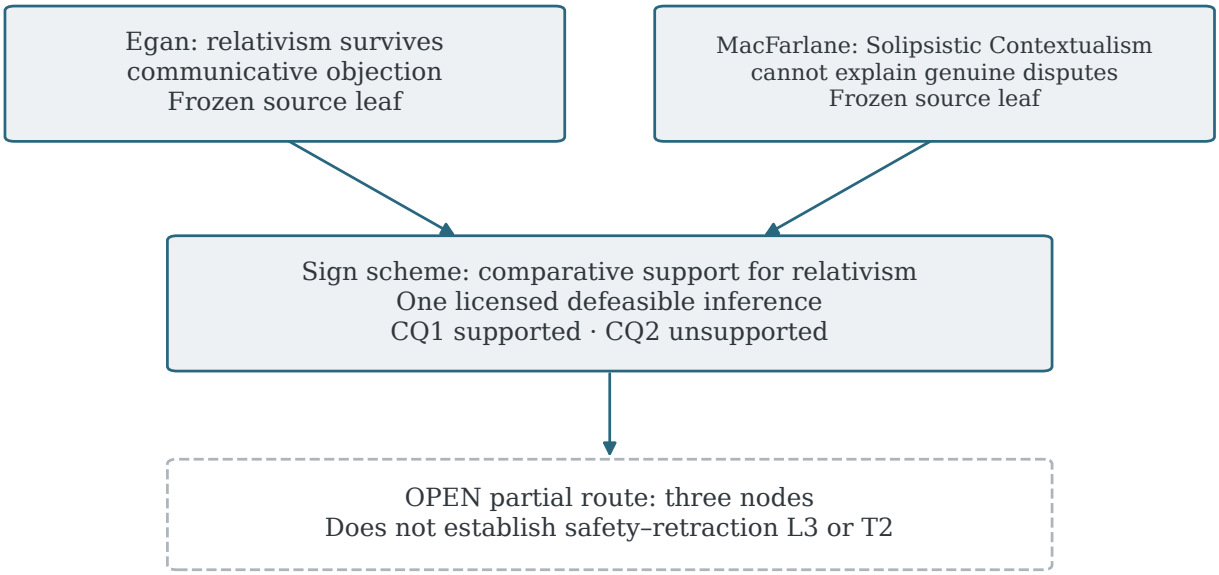


Figure 7. The sole inference-bearing B route is a comparative sign argument. Its unanswered critical question keeps it OPEN. Its attempted sign essays were refused; it is not the B/1 or B/2 proposal route.

4.4 Gap3p: a completed unsuccessful search is not an impossibility result

Gap3 retained **23 authenticated targets**. The exact T2 iff, if and only-if probes each had **zero admitted targets**; their refusals were preserved in later continuations rather than force-admitted. Gap3p reused the 23 retained targets and completed **46/46 polarity jobs**, publishing **46 OPEN routes**, **zero accepted inferences**, **zero CLOSED routes** and **zero eligible queue entries**. All 46 map replies were OPEN; all 92 cuts proposed nothing. Slot formalization and critical-question stages were never reached. No model-response rejection or sanitized CLI error explains this completed run. [R53]

The retained L3 target challenges a sufficiency bridge and maps most closely to partially anticipated T1. A review flags a possible reversal of MacFarlane's sufficient retraction trigger into a necessary condition. Another admitted target combines a consistency-testing claim and a higher-order information claim in a fresh whole-claim atom; its reviewer labels it semantic packaging, although it is not the literal conjunction-padding pattern. Gate admission and distinct target IDs do not establish distinct substantive theses. Appendix E depicts every retained A/B/C and gap3p target with both polarity outcomes; its data retains full texts and route IDs. [R53]

4.5 Pause after empty search, then diagnose the actual request

The operator's **19:52:39 UTC** rule requires stopping after about ten completed jobs with no accepted inference. Gap3p was already terminal, so its first-ten diagnosis is retrospective: **zero licenses**, **zero complete bases in 30 map/cut requests**, and **ten root requests containing none of their declared basis**. The source premise sets also had zero cross-paper symbol sharing. Ten maps returned OPEN and twenty cuts empty proposals; these were schema-valid model responses. The diagnosis identified missing visible premises plus isolated vocabulary, not a transport error or proof of falsity. [R53]

The reviewed native implementation defaults to **ten completed jobs**, counts bank-licensed inferences rather than accepted JSON, preserves completed jobs and emits a pause receipt with exit status 2. Diagnosis records actual packed basis coverage, shared symbols, model outcomes and sanitized rejection categories. Explicit resume starts the next preserved window; automatic reruns would defeat the rule. This mechanism is a resource and investigation policy, not a new logical license. The separate gap4a external guard is disclosed as such and has no live ten-job trigger to report. [R60, R64]

5. Finding gaps backward and forward

5.1 Backward investigation identifies a missing connection

The first three source-level open-problem flags did not survive as three certified open research questions. Currency review found one addressed and two partially addressed; none was certified still open. A later 60-pair conflict pass found three contradictions and seven tensions, but source review again certified no untouched problem. These are bounded retrieval results. [R5, R14]

The safety–retraction question L3 emerged from a more specific missing connection: can worldly-information safety support an assessment-sensitive retraction obligation? The closest read literature partially anticipates that connection. A safety condition, a knowledge claim and an obligation concerning an earlier assertion are different claims; acquiring all relevant passages does not prove a rule linking them. Appendix A.7 retains the currency review, citation-index limits and acquired-source history. [R5, R27]

5.2 Literature comparison is separate from formal novelty

Eps2's nine-thesis dossier classified **four anticipated, four partially anticipated and one not found within scope**. Exact T2, the necessity-and-sufficiency connection, was the not-found thesis. That label is bounded negative evidence, not proof of truth, priority or philosophical novelty. Its sufficiency direction alone is closest to partially anticipated T1. [R27]

ARA's B/1 and B/2 partially match T2's necessity direction while changing the subject or normative force; A/3 is thematically related to T5. Exact T2 came from eps2 and was supplied externally. Neither a near match nor a negation route inherits a proof from a literature label. The full thesis-by-thesis map remains in Appendix A.7. [R28, R38]

5.3 Forward: what premise sets jointly imply

Backward investigation begins with a conflict, failed connection or proposed target and asks which premises would support it. Forward synthesis begins with available premise sets and asks what licensed consequences follow that no individual paper supplies. The original whole-T non-entailment gate obstructed precisely that synthesis; the repaired per-paper gate allows joint entailment while rejecting restatement, inconsistent support and packaging. [R11, R27]

The old native forward pass on gap1r was bounded at depth two / 500 consequences and returned two packaging leads, with **zero substantive contributions**. The answer to whether every premise combination and route had been searched is therefore no. The bounds and available bank schemes delimit coverage. [R18]

The completed native gap1r census widened the bound to **depth six / 20,000 consequences**, producing **zero surviving candidates**, **zero recorded support conflicts**, and **truncated=true** in 493.733 seconds. Its retained consequences were **9,180 conjunction introductions, two conjunction eliminations and 10,818 disjunction introductions**. One further disjunction reached the insertion cap. Only 11 non-scoped deductive entries had fresh forward instantiation; 23 defeasible and two scoped deductive entries needed existing closed-route witnesses. The cap was spent on first-layer Boolean enumeration; configured depth is not observed search depth. The final fwd handoff credits zero candidates and no substantive cross-paper derivation. No exhaustive census or successful 50,000-consequence result is claimed. [R44, R59]

This is the forward signature of the construction defect: the engine can package opaque atoms with conjunction/disjunction but cannot extract their internal safety, context and obligation relations from their names. All A/B/C tests share this same theory and finish **100 OPEN / zero CLOSED-derived**. The initial two padding leads under a smaller bound were not new theses. R32's scheduling repair excludes standalone packaging enumeration and records rule/work coverage; its real gap4a census still awaits freeze. The formal representation, retrieval selection, bank coverage and stopping bounds are jointly limiting. The measurements support a unifying representation diagnosis without proving that every empty model proposal has one exclusive cause. [R28, R52, R59, R60]

Forward gap detection thus asks a demanding question: is there a licensed, non-padded consequence requiring premises from different papers that no one paper already yields? Adding model calls or removing paper prefixes alone does not answer it. The useful next measurement is actual shared formal structure followed by complete request-basis visibility and a licensed consequence, with a separate literature comparison. [R49, R53, R60]

6. H as a proposed bridge: conditional derivation versus argued proposal

6.1 The missing work is substantive

The missing sufficiency connection links worldly-information safety failure to an obligation concerning the same earlier assertion with fixed use and specified assessment contexts. A necessary condition on knowledge does not itself create that obligation. The necessity direction additionally needs an exhaustive or excluding premise covering the relevant grounds of obligatory retraction; a sufficient trigger cannot

simply be reversed. Permission to reject, actual withdrawal, a knowledge claim and an obligation to retract are distinct. Gaining MacFarlane's prose rule is relevant, but neither its atomic encoding nor a generic alignment judgment supplies all those connections. [R27, R28, R35, R36]

6.2 Hypothesis premises and runtime-owned disclosure

Xi r32 admits explicitly proposed H in a new construction-reuse lifecycle before its single freeze. Hypotheses carry `kind=hypothesis`, `stance=proposed`, origin, rationale and linked target; normal formula/signature and consistency checks apply. H receives **no source-paper credit**, no source evidence role and no automatic critical-question support. A qualifying cross-paper route must still use source premises from at least two papers in addition to H. The proposed bridge must not merely restate the target with irrelevant source padding. [R52]

Review found three concrete violations. **X32-F1** closed an H restatement padded with source content; **X32-F2** accepted negated or quoted conditional language as disclosure; **X32-F3** omitted H in summaries/receipts. The first repair closed F1/F3 but left typography/negation bypasses. R32c (`c42fcd5`) moved affirmative conditional disclosure into runtime-owned output, and the final review closed all three. Its full suite reported **1,914/1,918**, with four skips. The review validates those enforcement seams, not the soundness of an actual philosophical H. [R52]

The mathematical form is T plus disclosed H licensing C. The philosophical work of defending H remains separate. A source-stated safety condition cannot be silently promoted into the hypothesis's new normative implication; H also cannot count as a second paper. If a future native route closes, the paper must say which conclusion follows only on which H. It must still examine whether translating the source passages as those formulas is faithful. [R52, R61]

6.3 Proposal-mode papers argue for the bridge

Proposal mode now has actual outputs. B/1, B/2 and A/3 retain **“ARA-proposed thesis; bridge principle argued, not mechanically derived”**; exact T2 retains **“eps2-proposed thesis (tested in ARA); bridge principle argued, not mechanically derived.”** These labels distinguish native target generation from the external dossier. A proposal paper can defend a bridge with reasons and objections even when the corresponding route is OPEN. This is a generated research proposal, not a proof that the bridge is true, new, or successfully derived by ARA. [R55]

The final four-rule writing contract requires an origin label and one abstract disclosure; the bridge stated and defended under Proposal without false source attribution; cited faithful source claims; and no unsupported claim presented as established. Natural supporting reasoning may appear throughout. Repeated “this paper argues” prefixes are not required. B/2 and T2 passed original writer k=3 gates and have verified original PDFs; B/1 and A/3 remain refused. A/3 also exposed conflicting generic-versus-proposal role instructions: six union flags treated proposals as root argument steps, but three unsupported-claim findings still independently blocked it. The subsequent prompt clarification has no demonstrated live acceptance for that refused candidate. [R55, R56]

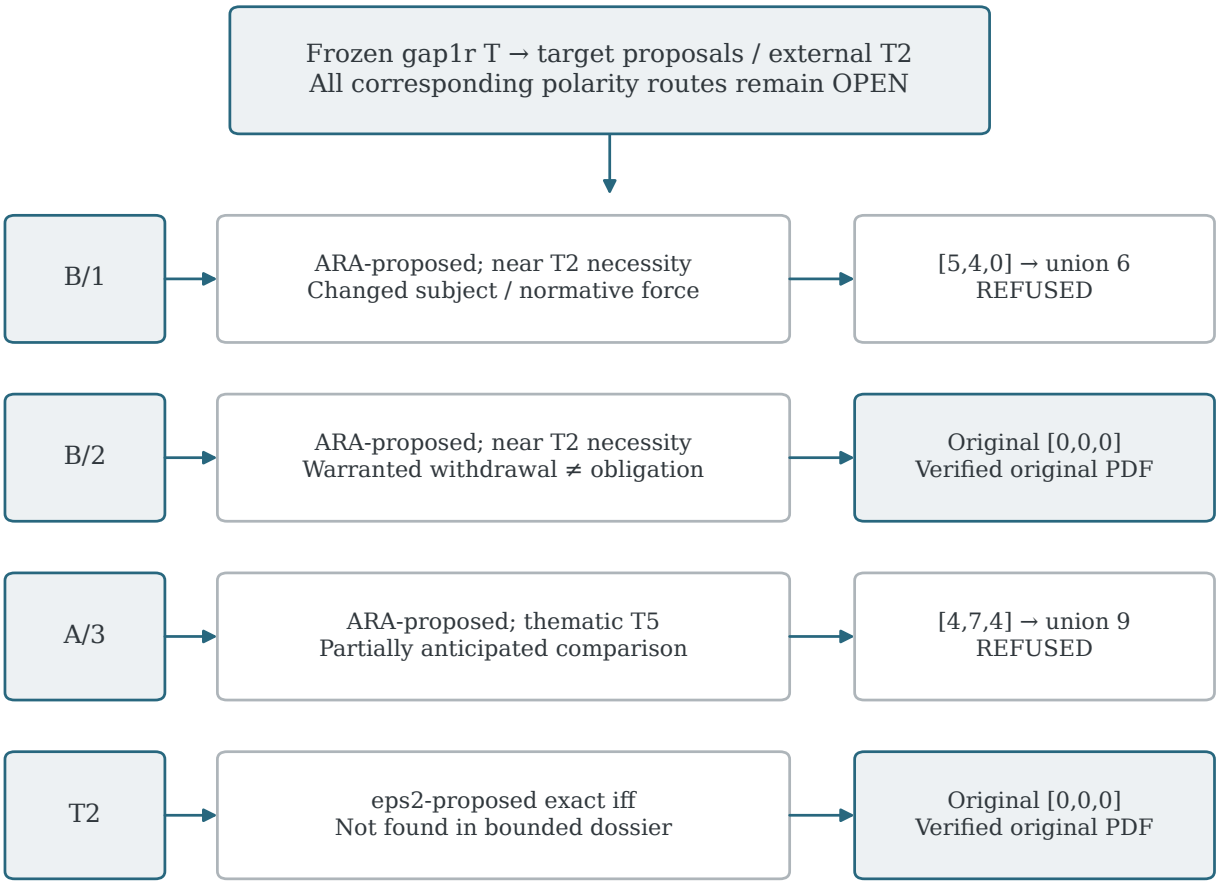


Figure 8. Four proposal candidates retain distinct origins and bounded literature comparisons. B/1 and B/2 only partially match exact T2; A/3 is thematic T5. The T2 sufficiency direction alone is closest to partially anticipated T1. Clean original writer gates do not imply polished acceptance or mechanical derivation.

6.4 Featured combined gap3–T2 route: a conditional design with a real stopping point

The intended combined paper would take gap3's retained Beddor–Goldstein safety commitment and MacFarlane Retraction Rule, disclose H_suff and H_nec, and present a mechanically licensed conditional T2 derivation if one exists. H_suff proposes that safety failure at later assessment, with use fixed, satisfies the Retraction Rule antecedent; the source rule is then separately needed for obligation. H_nec supplies the necessity/exhaustiveness direction. The proposed paper must defend those premises separately. Retraction being a sufficient trigger does not make it the only ground of retraction. [R61]

At 22:29:29 UTC on 30 October, no combined derivation exists. The latest generation-free native reuse plan on reviewed `dcbc4b3` passed: **77 terminal windows, 243 commitments, 150 reusable accepted alignment judgments, zero remaining pairs**. Zeta's semantic preflight then stopped before formalization: the exact available symbols denote whole source claims, not individual safety failures and retraction obligations. A faithful expression of the two H premises over those fixed atoms has not been established. Xi2 confirms that the reviewed code has no separate pre-freeze semantic H-fidelity gate. Well-formed, consistent formulas alone would not certify the intended meaning. [R61]

The receipt records **zero live calls, zero admitted hypotheses, no new freeze, zero generated targets, zero derived or CLOSED routes**, and no destination lifecycle created. This is a worker preflight block, **not a native solver/model refusal**. There is also a disk constraint: free space was 6,629,629,952 bytes; importing the preserved prefix plus one export would leave an estimated lower bound of 5,852,271,144, below the 6 GB search floor. Neither constraint authorizes changing source meanings or reporting a result early. [R61]

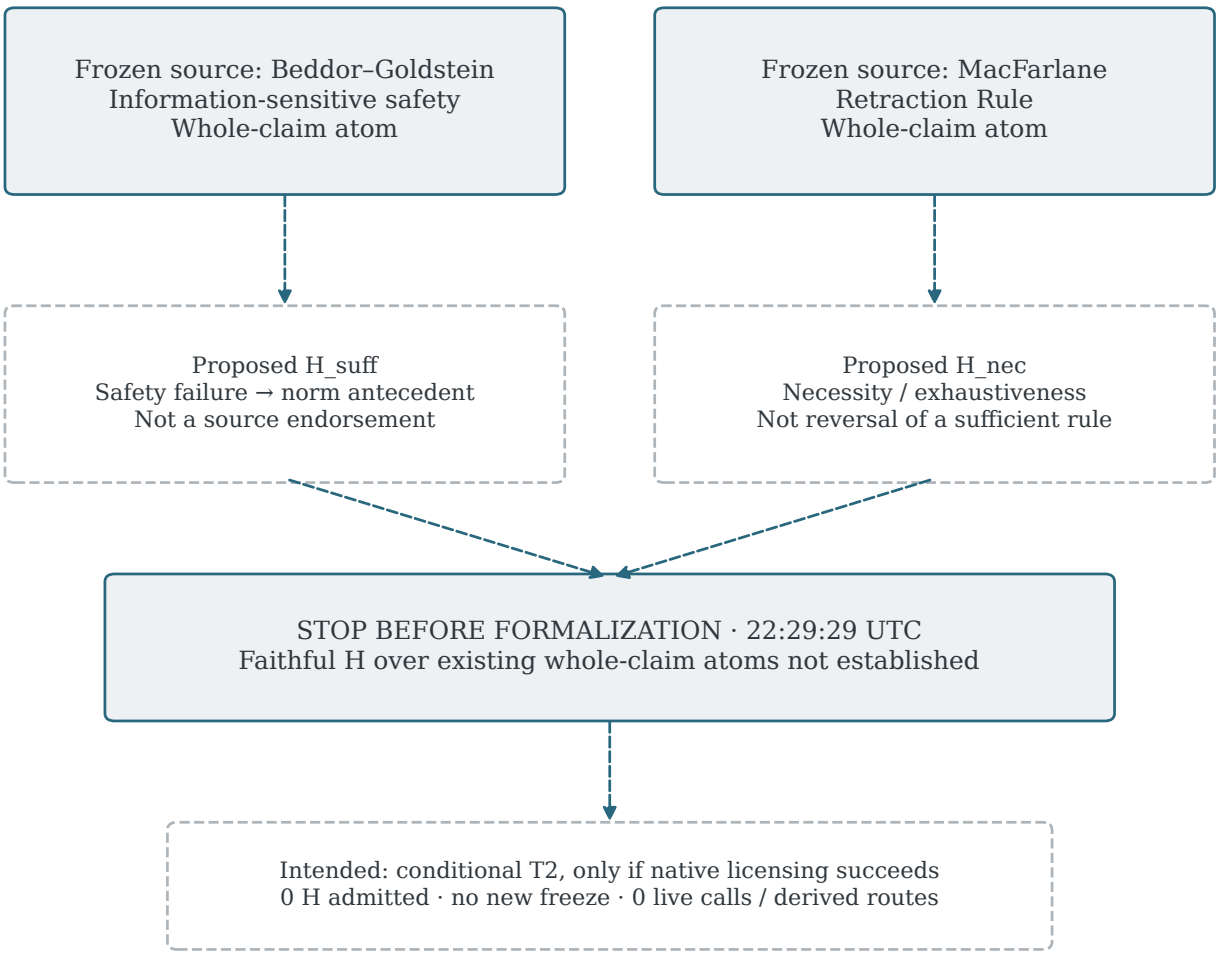


Figure 9. Featured gap3 + T2 conditional design, not an executed derivation. Source reuse is verified; the worker semantic preflight blocks H before model or solver execution. Dashed edges are intended dependencies, not bank licenses. H must be defended and source meanings preserved. A completed route was unavailable at the inspected checkpoint.

The diagram's dashed edges denote the intended research dependency, not accepted bank steps. The completed **22:33 UTC handoff** adds one **held fallback metadata item**, `gap3h-T2-combined`, with route ID null and dispatch hold `awaiting_native_grounding_admission`. The current watcher cannot consume that state. Thus “queued = 1” means **one grounding packet and zero licensed routes**, not a dispatched essay. Nu4 must implement and review truthful route-free grounding admission before it can enter writing; no fake OPEN route or accepted inference was manufactured. The packet binds all 11 anchors and two local PDFs, with 18 physical pages checked for bounds/readability. Those checks do not

replace sentence-level local citation verification. No combined PDF is credited. [R61] A later result must replace this pending state with its exact H formulas, licensed steps, source-use proof, route ID and audit/export receipt.

7. Writing and verification are gates on output

7.1 Typed audit separates claims with different support requirements

A source claim must be faithful to cited evidence. A route step must match its licensed role. Framing, objections and an explicitly argued proposal have different requirements. Nu4's typed audit made those distinctions explicit after binary auditing rejected large amounts of exposition. Sectioned writing also replaced whole-draft length retries. Its first length-successful draft reached 4,642 words, but length alone did not establish support. Appendix A.8 preserves the round-4 typed-audit table and all earlier writer outcomes. [R7]

7.2 Fresh union audits preserve disagreement

A byte-identical cleaned-input prompt received **44 flags and then zero** without a text repair. A clean sample could therefore replace adverse evidence unless the runtime retained history. The k=3 union gate makes fresh uncached requests and keeps every valid current or relevant prior adverse flag binding. It is not majority voting and does not assume independent samples. In the first route-1 batch, samples [0,0,36] plus prior dissent produced a 47-flag union. [R20]

The gate is conservative about disagreement, not a guarantee of source fidelity. Round-6 exports passed their sampled audits yet ended with zero bibliography entries and zero citation markers. Citation stripping was a separate pipeline defect. Both the audit variation and the citation-zero exports remain in the incident record. [R20, R23]

7.3 Citation repair must not repair away the source record

R7 restored local frozen-T citations and retained partial adverse audit history. Both earlier candidates still refused on unsupported or miscited prose; neither reached final consensus. Their two bibliography entries represented commitments from one paper, not cross-paper synthesis. Local PDF verification, valid citation coverage and the final audit must each pass. A clean model judgment cannot substitute for any of them. [R29]

7.4 All candidates, with final selection reserved

The operator authorized drafting every eligible candidate; the operator retains final submission selection. CLOSED routes and OPEN routes with an accepted inference enter the native queue. The later proposal-mode assignment separately covers selected root-only hypotheses under explicit labels; it is not permission to relabel them CLOSED-derived. [R33, R41]

Candidate	Status and time	Evidence and stopping point
gap-T2-analysis	PDF; rendered at the earlier v5 checkpoint	3,879 complete-Markdown words; 4,064 PDF-inclusive words; 17 references; 10 pages / two figures. Analysis of ARA output. No native typed audit, export or Opus pass. Its gap2-pending statement is historical; §2.5 supersedes it here.
gap1pu-B-sign	Refused, 03:14:26 UTC	4,790 words, five bibliography entries; successive audit flags 92 → 101 → 32 → 18. Final 15 miscited + three unsupported claims; missing node n1; three repairs used. Final k=3, citation/originality verification and polish not reached.
gap1pu-B-sign-short	Refused, 03:42:33 UTC	Writer failed the 2,500-word minimum. Attempt counts 1,412 / 1,373 / 0 / 1,203; 1,203 is the last rejected response, not a validated draft. Typed audit, final consensus, verification and polish not run.
B/1 proposal	Refused under four-rule standard	3,171 native words; final samples [5,4,0], two historical adverse receipts, six union findings; three repairs spent.
A/3 proposal	Refused under four-rule standard	2,992 native words; [4,7,4], three historical adverse receipts, nine union flags; three substantive findings remain beyond role-policy conflict.
B/2 proposal	Verified original PDF delivered, 19:36:05 UTC correspondence	Clean original [0,0,0]; 3,168 reading words including method note, excluding references; seven pages. Polished revision refused.
Exact T2 proposal	Verified original PDF delivered, 19:36:05 UTC correspondence	Clean original [0,0,0]; 3,427 reading words on the same basis; eight pages. Polished revision initially failed title quotation fidelity; title-only recovery later passed citations but refused final audit [9,6,0], union 14. [R66]
Combined gap3–T2	Held fallback metadata, handoff 22:33 UTC	One queued grounding packet, zero licensed routes; native writer admission pending; no H admission or combined PDF.

The short retry's policy was an **orchestrator decision**, corrected in correspondence at **03:19:32 UTC**, not a separate human instruction or a competition minimum. For its three-node route with two direct source leaves from two papers, the policy used a 2,500 floor, 2,700 target, 3,000 requested limit and 6,000 runtime ceiling. Audit samples, repair limit and citation policy stayed unchanged. Its failure does not establish that short essays are impossible; it records this bounded writer attempt. Neither refused candidate was enqueued for polish. [R42]

7.5 Adverse history survives retries

Review found several routes by which valid adverse labels could disappear: incomplete inventories on resume and rejected schema retries followed by clean corrections. The repairs union valid dissent before verdict and preserve semantic repair budgets. Citation-only restoration changes markers, not prose; missing local evidence is not positive mismatch evidence. Appendix A.9 retains the finding IDs, reviewed heads and original refusal sequence. [R23, R29, R31]

7.6 Carry-over is bound to verified source text

Carry-over anchors each current sentence to an exact source draft with a complete clean native k≥3 gate. Source hashes, outline, evidence, authorization, sentence offsets, citation markers and bibliography remain bound. Deterministic normalization permits limited prefix removal, typography and whitespace changes. It does not erase arbitrary words, logical symbols or citations. [R57]

Other aligned edits receive three fresh meaning samples. Only unanimous equivalence inherits the source verdict. Different or uncertain votes, including valid adverse votes in malformed retries, leave the current sentence on ordinary typed audit. Decisions persist to prevent favorable rerolls. Unaligned sentences and unrelated adverse history retain normal treatment. The small live seven-pair probe exercised rewording and scope changes; it is not an accuracy estimate. [R57]

Inline repair addresses unsupported elaborations locally using verified source spans and current occurrence markers. The reviewed native seam preserved accepted prefix removal and deleted an unaligned unsupported sentence, with one inline pass and zero whole-paper rounds. That review used fake providers; the actual T2 result is reported in §7.8. B/2's renewed inline outcome remained pending at the retained cutoff. [R58, R66]

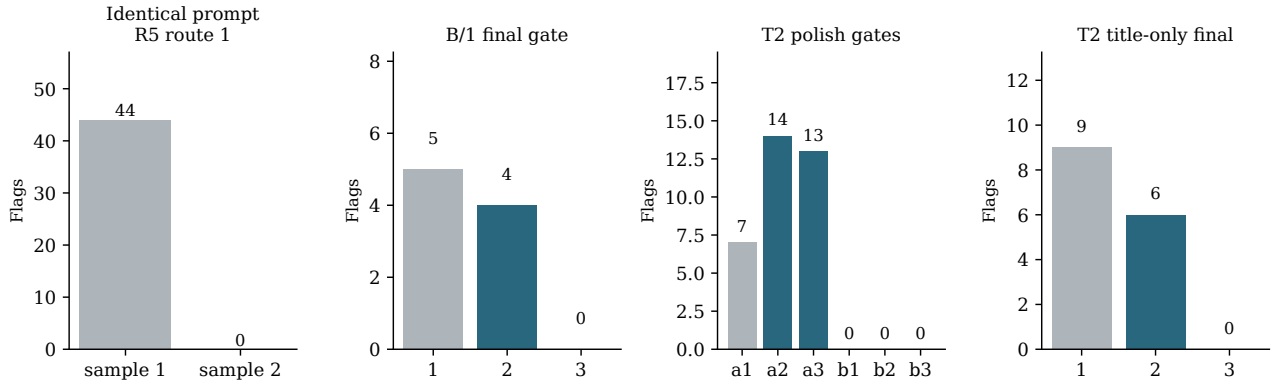


Figure 10. These are separate observed audits, not a quality trend or independent trials. R5 changed 44→0 without text repair. B/1 still refused with six union findings despite a clean sample. T2 had a clean earlier polished gate but failed quotation fidelity; after exact title repair its final [9,6,0] union refused again. The delivered PDF remains its verified original.

7.7 The candidate pipeline preserves result kinds

A graph with a licensed inference can enter route writing, including an explicitly partial OPEN graph. Root-only hypotheses need separately authorized proposal mode. The fallback analysis describes the run and missing premises without claiming a native route. The success-only polish queue carries verified exports; refusals remain in status records. The operator chooses submission candidates after seeing these distinctions. A PDF renderer only lays out the supplied text; it cannot upgrade an OPEN route, an unaudited analysis or a refused polish revision. [R33, R55, R56]

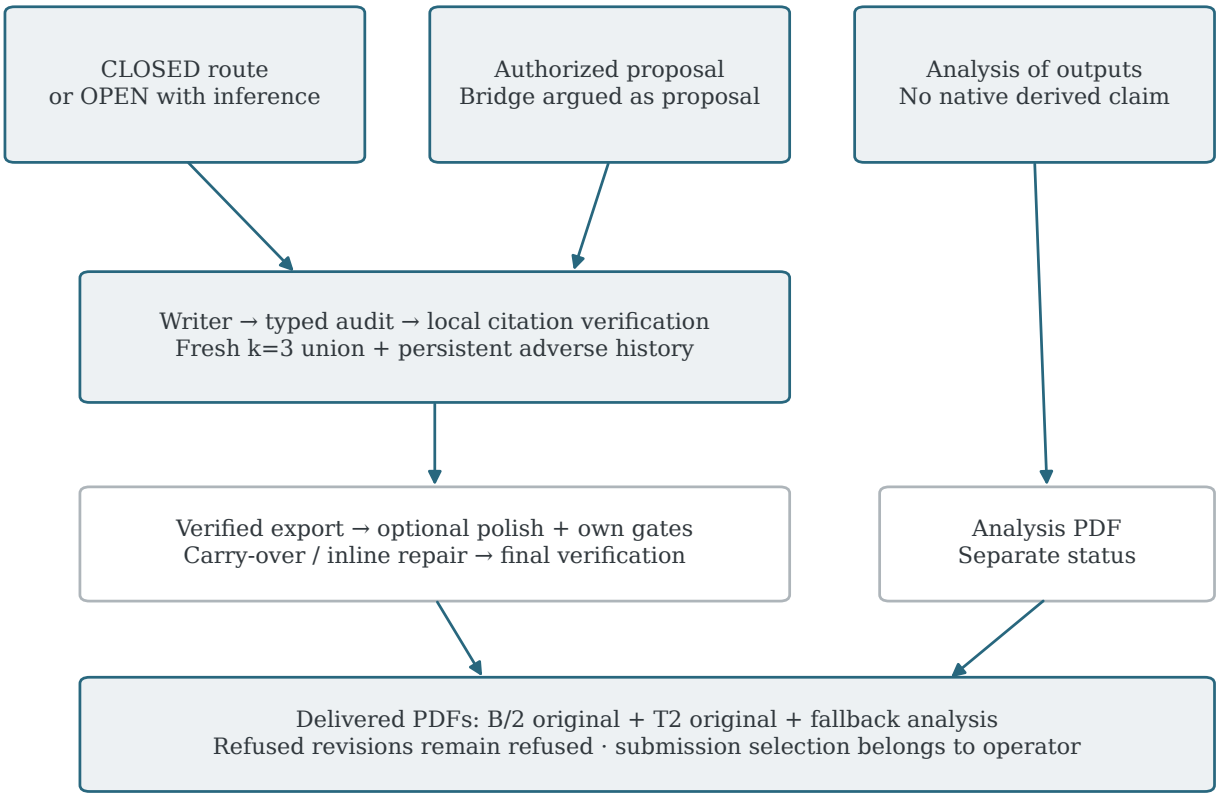


Figure 11. Distinct entry paths and output claims. The current two proposal PDFs reproduce verified originals. Native generation/audit was not claimed for meta's fallback analysis. A refused polished revision is never replaced in the ledger with a rendering pass.

7.8 T2's final refusal shows the boundaries working separately

T2's title-only correction changed four quotation characters to asterisks and passed all six citation checks. A malformed meaning response then returned 98/101 pairs, including 51 adverse votes. The reviewed bounded shape correction preserved every one of those adverse pairs; none inherited a clean source verdict. Of 101 pairs overall, 40 inherited and 61 remained on ordinary audit. [R66]

The final fresh publication samples were **[9,6,0]**, producing **14 union flags over seven sentences**. Polished T2 remained refused despite the clean third sample. No semantic editor ran; repair counts stayed one whole-paper round and zero inline passes. The verified original PDF remained delivered and unchanged. The attempted draft, automatic fallback and delivered original have different word-count scopes, all preserved in Appendix A.10. [R66]

8. Runtime, accounting and human responsibility

8.1 Provider isolation has a measured scope

The campaign moved from API calls to Claude Code and then to Codex after the operator clarified the requested provider. Codex became the ordinary generation path; Opus was separately authorized for polish. Isolation canaries and private transport settings tested specific boundaries. They do not prove the

absence of pretrained knowledge, undisclosed vendor internals or developer-context influence. Requested model names are distinguished from verified serving versions. Appendix A.11 retains the complete provider/version table, failed canaries, calibration and transport history. [R13]

8.2 Prompts and model decisions remain evidence

Recorded prompts instruct extraction, bounded construction, target proposal, licensed search, typed writing and publication. Raw receipts preserve full prompts, responses, correction histories and rejected calls. The accompanying prompt supplement contains inspected historical examples; it is not a complete replay package for later consolidation, hypothesis and carry-over stages. Stage rules are distinct from task-specific source content. [R3, R15, R57]

8.3 Accounting is scoped to its actual ledger

The fork-6 ledger deduplicates receipts across 67 lineage runs. It records **113,530,631 measured token components**, **$131.785270 measured API USD**, and **$1.844820 unknown-at-reservation USD**. These are lineage totals, not a campaign invoice. Subscription tokens do not have an inferred API-dollar equivalent. Imported evidence must not be charged again. The detailed provider/stage table, charts and later supplements remain in Appendix A.11. [R4]

Gap3p separately records 4,213,242 input, 36,104 output and 254,976 cache-read tokens. No reconciled total covers all authoring agents, reviewers, consolidation and publication. Unknown response usage retains its explicit reservation treatment. The report makes no new generation calls and does not manufacture a global cost from incompatible receipt scopes. [R53]

8.4 Human decisions and coordination are part of the method

The operator designed and directed the scaffold, chose the topic and bibliography, approved budgets and provider changes, set research priorities and retained final submission selection. The orchestrator and workers implemented and operated it. Appendix C preserves all 59 timestamped research-action entries plus separately disclosed untimed instructions; Appendix B credits every recorded worker. [R1, R26]

Coordination failures remain disclosed: the 28 October idle period lasted 4 hours 15 minutes 40 seconds, and a later approximately four-hour stall went unnoticed while disk, canary and cleanup decisions blocked delivery. Reports also required repairs to size, mobile overflow and event-origin attribution. These affected elapsed work and reporting, not the logical status of the routes. Appendix A.11 preserves the exact incidents, corrections and cleanup authority. [R1, R4, R63]

9. Novelty, reviewed mechanisms and the remaining limits

9.1 Honest novelty status

Zero substantive CLOSED-derived contributions have been established so far. The measured paper-local atomic representation of T is the unifying explanation for the failed cross-paper synthesis (§§2.3, 5.3); r17 prompting alone did not establish sharing, and r18–r21 consolidation has no completed replacement freeze (§2.7). ARA did propose near-T2 hypotheses. That is a proposal-generation result, not

a licensed argument, an exact match to the dossier's only not-found thesis, or priority in the literature. The exact T2 is eps2-proposed and externally tested in ARA. All A/B/C routes remained OPEN; the available B sign argument remained partial. The fallback analysis is a rendered analysis candidate; both native sign drafts refused. B/2 and T2 are now verified original proposal PDFs. Gap3p adds 46 OPEN root-only routes, and gap3h is blocked before H formalization. These updates do not change the zero substantive novel CLOSED-derived count. [R27, R28, R38, R42]

The formal novelty gate is weaker than philosophical novelty. A target may pass source-relative checks without being a useful, faithful, defended or literature-new contribution. Conversely, a hypothesis may be interesting while lacking a missing bridge premise; accepting that premise as H would establish at most the disclosed conditional derivation, if actually licensed. Proposal-mode prose must argue for that premise. No pending feature or larger search is counted as its own success. [R11, R34, R39–R41]

9.2 Review requires the intended failure to be tested

Independent review caught defects in padding, source unions, context configuration, hypothesis disclosure, evidence packing and audit history. PACK-F3 also invalidated a reported 920-control pass: one control failed through an unrelated error instead of its intended stance boundary. The pass was retracted and the owning test repaired. N4-F1 similarly required a corrected no-reroll test whose fixture did not merely exhaust its votes. [R54, R57]

Reviewed code, a passing replay and a live research result are different evidence. Each verdict belongs to its exact revision and exercised boundary. Appendix A.12 keeps the complete review-finding tables and deterministic iteration figures, including failed proof and inherited-control qualifications. [R21–R24, R49, R52, R54, R58, R62]

9.3 Remaining limits and current pending register

Source selection remains incomplete and access-bounded; literal stance and source-fidelity requirements are separate from coverage. Atomic rules and a disputed alignment bridge remain material limits. Intake's prohibition on human philosophical content also has an unresolved literal tension with bibliographic titles and supplied filing labels; four such labels were not four selected theses. Corpus neutrality was not independently verified. [R2, R18, R28, R35, R36]

Privacy's early API refusal did not change visibility; the later operator report and coordinator check did. Anonymity is checked on these deliverables, not inferred for raw logs. Source acquisition, exact model serving identity, statistical independence of audits, full campaign cost and competition readiness are not certified. [R1, R13]

Pending item	Last explicit evidence checkpoint, UTC on 30 October	What remains unproven
Consolidation r21 / replacement gap3v freeze	21:05:16 assignment	Stable cardinality and failure isolation; completed freeze, faithful formulas and after-sharing metrics.
Gap4a alignment / C2 / C3	20:10:10	First alignment dispatch blocked by disk policy; search also held on pinned basis-omission defect; no frozen or searched result.
Combined gap3h–T2	22:29:29 preflight	Faithful H over existing atoms; adequate disk headroom; H admission, freeze, conditional route and paper.
B/2 inline publication / combined grounding admission	Receipts collected at 22:46 UTC	B/2 final native verdict; gap3h route-free packet consumer. T2 title-only recovery is completed and refused (§7.8).
Construction concurrency	R16 report, collected for v6	Ordered snapshots, reservations, collisions and crash/resume correctness; no parallel-window benefit claimed.
Philosophical novelty / submission choice	This revision's cutoff	Independent substantive evaluation, priority and operator's final selection.

A pending item receives no completed result bar or invented success count. The repaired search software now exists, while the next real shared-vocabulary freeze and successful cross-paper route do not. This paper's rendering checks are separate from native paper audit, source fidelity and competition eligibility. [R51, R53, R57, R60, R61, R64]

9.4 What discovery this record actually supports

The work generated hypotheses, located missing-premise obligations and discovered engineering failures that explain why apparently promising routes were empty. Those are useful research outputs with different warrant. A near-T2 target is a proposal; a mapped literature absence is bounded evidence; a conditional derivation would be a formal achievement relative to disclosed H; a substantive novel contribution requires a defensible claim beyond those ingredients. None of these categories should be collapsed into the number of PDFs. The strongest present contribution is an auditable method and a concrete negative result about this construction/search setup, with reproducible repairs and unresolved semantic work.

Appendix A. Incident record and detailed receipts

This appendix preserves the detailed v6 incidents, quantities and qualifications behind the shorter main account. Original section numbers identify the source passage; references to those sections point to the corresponding concept in the main text. Evidence and pending states retain the v6 cutoff.

A.1. Contribution boundaries and comparison with neighboring systems

Original §1: architecture and responsibility boundary

The human chose the research area, supplied the bibliography, directed the scaffold, approved budgets and provider changes, and set acceptance and submission priorities. Coding agents implemented, reviewed and operated it. ARA generation used separate provider calls; isolation of those calls does not establish isolation of development workers. The operating rule prohibited the human from supplying original philosophical arguments, premises, distinctions or a personal view. Recording actions does not independently certify competition compliance. The human-action ledger now contains **59 timestamped entries**, plus earlier untimed design and bibliography preparation. Worker use of a native `HUMAN` input boundary does not count as a distinct human intervention. [R1–R3, R34]

Every later effort is an instance of the same process: represent admissible evidence in T; propose something in H; try to license a move through L; record what remains open; and gate any written account on its actual support. Proposal-mode writing and proposed hypothesis premises extend this process in different ways. Neither licenses quietly converting a new idea into source evidence.

ARA separates **L**, the versioned bank of permitted move forms; **T**, the working table of source-linked commitments; and **H**, model-proposed cuts or hypotheses. A model is not allowed to create a license by asserting that a step is plausible. A step must instantiate an existing bank entry. Deductive entries require both shape checks and Z3 entailment. Defeasible entries require the relevant scheme and its critical questions; a failed or unsupported critical question leaves an obligation open. [R3]

Rich extraction records are immutable source evidence. A construction lifecycle selects a bounded working subset, formalizes it, records source stance and omitted support, and then freezes T. Search uses that freeze. A verified fork copies the frozen prefix and provenance, proposes new targets, and uses a separate cache. It reuses premises, not statistical independence. Reported or attacked source positions are not silently converted into endorsed premises. [R3, R4]

For each target, ARA searches both the target and its negation. The engine records proposed, licensed, discarded, and unresolved moves. A closed route means the engine’s formal obligations were met under that version’s policy. It does not establish that a source was faithfully translated, its premises are true, the result is new to the literature, or its prose is good philosophy.

Originality was pursued through several independent checks: avoiding direct restatement of T, testing single-support entailment and tautologies, classifying whether the target states an own thesis, checking the relation between its text and formal definition, requiring distinct source supports, and later comparing

with retrieved literature. The gate design changed substantially. Its defects are part of the method's history, not evidence of an invariant guarantee. [R9–R11]

This revision extends the concept-based account with completed construction/search evidence, proposal papers and the latest reviewed mechanisms. Completed results are tied to their run or reviewed head. A live checkpoint is not a final result, a code review is not a philosophical validation, and a rendered PDF is not a native export. The companion data preserves earlier snapshots and records this revision's sources separately. V1–v5 remain unchanged. [R34]

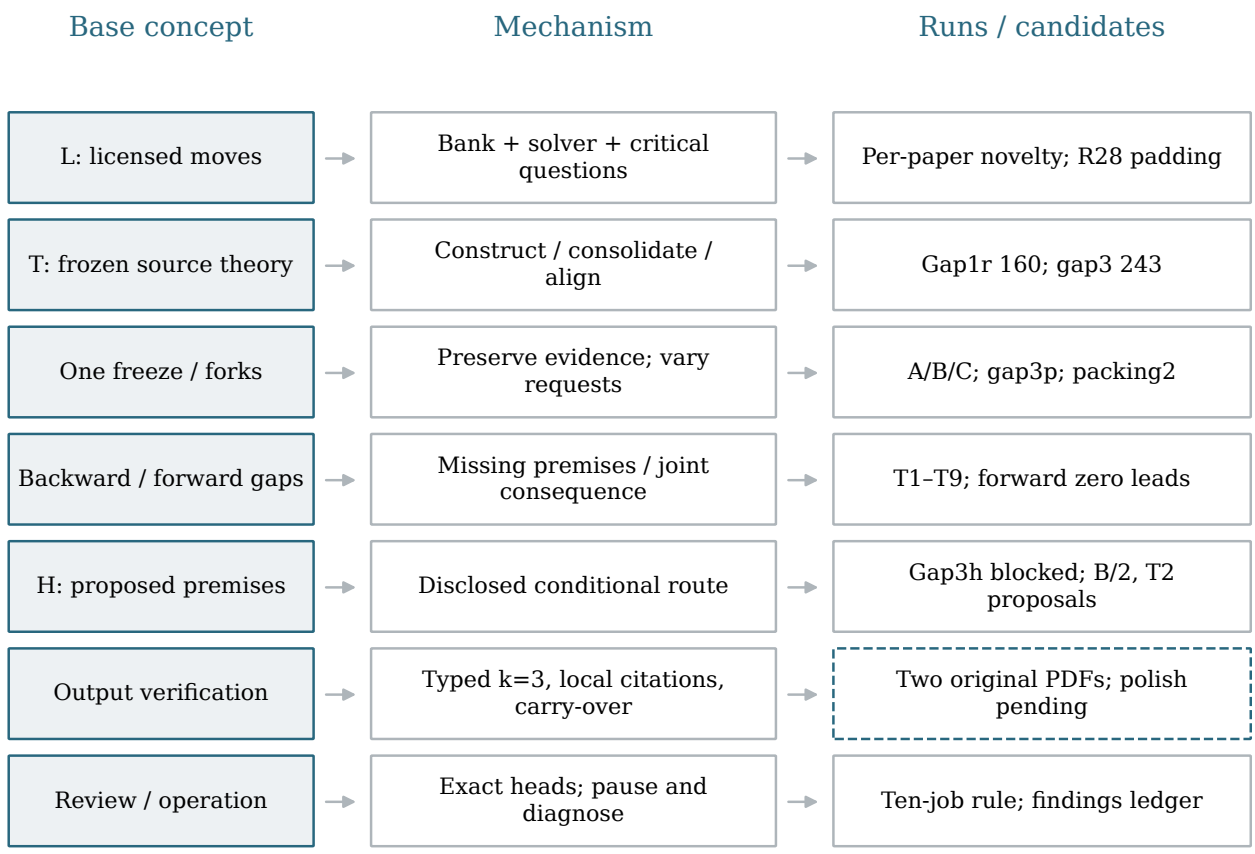


Figure 12. One architecture organizes the experiments and extensions. Source construction determines T; L licenses search; H includes targets and disclosed proposed premises under the reviewed extension. Arguing for a bridge principle differs from deriving a result conditional on it. Pending mechanisms receive no success count.

Original §1.1 What is distinctive, and what is not yet a novelty claim

The system's unit of accountability is a licensed, source-bound move. A language model can suggest a target, a translation, a bridge or an explanation; none supplies its own formal authority. This differs in emphasis from a research-writing workflow whose principal artifact is a paper: here a paper must retain the status of the route beneath it. An OPEN graph remains OPEN after fluent exposition. A proposal remains proposed after publication. A formula proved from an H premise remains conditional on that premise. These are enforceable design commitments, not evidence that the underlying philosophy is correct. [R3, R52, R57]

The AI Scientist describes idea generation, code execution, experiments, paper writing and automated review; STORM describes retrieval and multi-perspective pre-writing for grounded articles; SciAgents describes graph-based hypothesis generation and critique. Those are relevant neighboring aims. Their inspected abstracts do not supply enough implementation evidence to certify that any ARA feature is absent from every competing system. Our comparison therefore identifies ARA's implemented responsibilities and failure evidence, without an unsupported first-of-its-kind claim or a head-to-head benchmark. [R65]

Methodological contribution in this system	Concrete mechanism	Executed evidence and boundary
Source-relative synthesis	Per-paper endorsed-closure novelty; allow joint entailment, reject padding and inconsistent unions	R28 fixes and placeholder cross-paper derivation; no substantive real-source novelty yet (§4.2).
Stable evidence identity	Runtime binds immutable IDs; models return semantic decisions or local handles	Gap1 recovery completed 150 judgments; consolidation reused the lesson after r19 (§2.4, §2.7).
Representation diagnosis	Count atoms and actual cross-source formula reuse separately from name prefixes	141/160 atoms in gap1r; 212/243 in gap3; gap4a changed names without creating sharing (§2.7).
Context without reduced authority	Pack model-visible evidence, retain full T for licensing; mandatory declared basis in packing2	46 maps and 92 cuts regain complete basis in offline replay (§3.4).
Conservative output evidence	Fresh k=3 union with persistent adverse history; local frozen-T citation checks	44→0 nondeterminism and bibliography-zero failures remain recorded (§7).
Disclosed conditional research	H has no source-paper credit; runtime owns affirmative conditional disclosure	X32 review repairs; gap3h has no admitted H or completed derivation (§6.2–6.4).
Resource-aware investigation	Pause after ten empty jobs; diagnose basis, sharing and responses before explicit resume	Reviewed native mechanism; historical gap3p diagnosis is retrospective (§4.5).
Candidate accountability	Separate derived, partial, proposal and analysis paths; preserve refusal status	Two verified original proposal PDFs and explicit failed alternatives (§7.4).

The empirical lessons are strongest where a concrete failure was reproduced and a targeted repair was tested. Their general usefulness is plausible, but comparative superiority and philosophical novelty require further evidence. [R48–R64]

A.2. Extraction, source coverage and construction budgets

Original §2.1 Literature and extracted evidence

The initial bibliography contained **53 works**, dated **1967–2024**. Only **four** were from **2020 onward**, and none was later than **2024**. The first acquisition pass obtained **26 PDFs**, leaving **27** originals requiring access elsewhere. The accessible subset covered **1996–2024**, with **three** works from **2020 onward**. Intake accepted all bibliographic entries, but bibliographic intake was not full-text extraction. [R2, R5]

The first small-summary design was rejected because a **four-item, 400-character** cap could not preserve argument structure. Rich extraction instead retained page-linked premises, intermediate conclusions, inferences, distinctions, qualifications, and reported or attacked positions. The complete original-corpus pass produced **5,659 items** in **26 nonempty graphs**. Of those items, **363** were quarantined: **312 directly** and **51 because of dependencies**. **5,296** remained eligible. **Nineteen** graphs were classified as extracted and **seven** as partial; **57** snapshot-scoped extraction gaps remained. These gaps concern source processing and must not be confused with open research problems. [R1, R12]

The later acquisition and currency audit added **28 full texts**: **25 modern selections** and **three** missing originals. The local collection therefore reached **54 PDFs**. The modern search recorded **44 candidates**; **19** modern selections and **24** originals remained unavailable, yielding **43 acquisition requests**. Access failure is a limit on the review, not evidence that a problem is open. Newly obtained texts were not part of the frozen pilot evidence at the supplement's checkpoint. [R5]

The small-summary cap was a design mistake against the stated aim of preserving arguments. Rich extraction removed that cap, but strict output contracts then rejected useful material. A cheaper Sonnet extraction produced malformed inference structure; a subsequent retry reduced a partial paper to **nine items**, compared with **108** from the Opus extraction. The cheaper path did not pass the richness test. The first complete-corpus attempts also stopped on reversed page ranges, missing inference dispositions, duplicate correction IDs, and reconciliation failures. [R1]

The repairs separated deterministic wire repair from invented source content, added bounded corrections, allowed disclosed reconstruction or a no-inference disposition, and retained accepted ranges through best-effort reconciliation. Shape validation alone had admitted placeholder-like junk: a later grounding pass quarantined **245 of 2,333** items in an intermediate corpus. Union publication repaired source loss across resumed snapshots. The final quarantine inventory remained visible rather than being deleted to improve a success count. [R1, R12]

OCR was tried on **82 affected pages**. Successive diagnostics reported **22** recovered pages under the first metric, then **15** under revised whole-page rules. Span-level repair recovered **131 of 677** damaged spans, with only **nine of 82** pages clean. These are different metrics, not a monotone improvement curve. Review found confidence dilution and required a further fix. OCR availability did not establish complete source recovery. [R1]

The later T1–T9 dossier obtained seven further lawful full texts and inspected the two modern comparisons that had previously been inaccessible. This is a dated acquisition update, not a retroactive expansion of any earlier freeze. Its substantive comparisons are discussed in §5. [R27]

Original §2.2 Bounded selection and what a skipped window loses

The first rich build dropped **40 dependencies**, so no freeze was accepted. Later construction could not fit whole-paper provenance into a window: one singleton input was approximately **86,000 tokens** against a **70,000-token** batch limit. Filtering metadata to the selected window repaired that particular cause, but further refusals exposed the absence of a global plan. [R1]

The coordinator then justified a lower symbol-growth allowance using a claimed **70–90 tokens per commitment**. That claim was wrong. The pilot worker corrected the observed symbol cost to approximately **233 tokens per commitment**. The report retains the error and its correction. Bounded closure, disclosed support cuts, global working/catalogue budgets, and soft per-window shares replaced successive cap increases. An intermediate run had stopped at **802 tokens against an 800-token share**, despite remaining global capacity; the corrected policy allowed borrowing within the global bound. [R1]

A construction window is a counted request over selected grounded evidence, with dependency closure, formula and symbol checks, stance, page provenance and bounded corrections. A larger context allowance does not guarantee a usable reply. All-or-nothing window acceptance can discard many valid-looking items when one required inventory or definition fails. Skipped material must remain visible as an omission, not silently disappear from the account of T. [R16, R35]

A.3. Immutable identity, alignment recovery and the gap2 stop

Original §2.4 Immutable identity and resumable alignment

The fresh gap pilot ran at detached revision `0a31afb`. It completed **74 construction windows: 53 accepted and 21 skipped**, retaining **159 working commitments from 22 of 26 papers**. These were pre-alignment artifacts. **There was no frozen T**, so the frozen-coverage criterion was not met even though the working set represented enough papers numerically. The supervisor exited with status **2** at **13:36:58 UTC on 29 October**, after **16,340.66 seconds**. [R16]

Alignment stopped at candidate **36 of 150**. It had accepted **35 judgments**: **34 unrelated** and **one right-to-left entailment**. The next candidate received **three invalid replies**, each changing the immutable symbol-ID fragment `r27_39` to `r27-39`. The responses called the relation unrelated, but their changed reference violated the exact-reference contract. **The validator correctly refused them.** The model text was not silently normalized and the working set was not promoted into a freeze. No bridges or canonical rewrites were applied; the final consistency phase was never reached. [R16]

The free `resume-construction --plan-only` check also refused: “construction resume requires a terminal window refusal.” Construction had already finished, and that revision had no alignment-resume entrypoint. Starting a new run with retained cache was the documented alternative; it would replay construction and could make new calls. That alternative was not executed as an in-place recovery. **Targets, forward synthesis, routes, writing, and an L3 resolution were never reached.** A rendering pass over an empty route inventory is not evidence of route-summary quality. [R16]

Coverage and semantic limits survived the completed construction stage. Skipped windows included capacity/empty-closure losses, a no-root selection, lexical attribution false positives, and provenance/schema refusals. The sole accepted entailment judgment was disputed on source review and involved an attacked description, not an endorsed safety–retraction bridge. Several original L3 anchors did not survive construction; a surviving retraction-related statement could not silently replace the full retraction rule. Four conflict inputs also lacked bindings because their required source contributed no commitment. These were qualified draft inputs, never admitted frozen gaps. [R16]

Independent review of this exact revision also retained blocking **INTEG-F1**: irrelevant fresh-atom wrappers could package existing cross-paper facts and receive a native derived label. This defect was independent of the earlier identifier stop. The pilot generated no target or route on which to test it; an empty padding-review inventory does not clear the defect. L3 therefore remains partially anticipated, with no resolution established by this pilot. [R16]

The identifier validator was right to refuse gap1's altered reference. The avoidable design error was asking a model to echo an immutable ID, letting one item's bounded failure abort the run, and offering no continuation from finished construction. Sigma r12 moved identity binding into the host-owned candidate record, requested relation/reason only, recorded per-item unresolved outcomes under a declared threshold, and added alignment continuation. Gap1r exercised that continuation with zero unresolved outcomes. Recovery preserves stage receipts; it does not certify the semantics of the retained bridge. [R17, R18, R21]

At `4cefe35`, recovery restored all **74 construction windows**, **159 source commitments**, and **35 accepted legacy judgments**. It completed **115 pending pairs** without replaying construction: **150/150 accepted judgments**, comprising **149 unrelated** and **one right-to-left entailment**, with **zero unresolved**. Applying the one model-alignment bridge produced **frozen T = 160 (159 source + 1 bridge)** across **22/26 papers**, with no canonical rewrites. This is executed continuation evidence, not merely a repair brief. The source inventory retained **134 endorsed, 10 reported, and 15 attacked** commitments. The inherited bridge remains semantically disputed; accepting its receipt did not establish a safety–retraction inference. [R18]

Sigma r12 supplied ID-free alignment replies and resume-alignment; later review closed its integration fixture issue. The design lesson is to bind immutable IDs in deterministic code, isolate a bad item's failure, preserve accepted judgments, and resume from completed stages. The recovered freeze has **160 commitments: 159 source entries from 22/26 papers and one disputed model-alignment bridge**. Zero unresolved alignments does not establish that the bridge is substantively justified. The literal all-source-stance criterion is **159/160, FAIL**, while paper coverage passes. Construction retained **53/74 windows**; its **21 skips** comprise 16 closure-bound failures, four formalization failures and one no-root window. [R18, R21, R28]

Original §2.5 Gap2: more admitted context, less retained evidence

At the **30 October 03:51:27 UTC** pre-freeze check, gap2 had accepted **3/13 windows**, skipped **10/13**, and retained **57 source commitments from three papers**: MacFarlane 42, Schaffer 10 and Veltman five. Its stances were 46 endorsed, eight attacked and three reported. The correspondence records the stop decision at **04:04:01**; later gap3 status confirms **gap2 stopped without a freeze**. The pre-freeze receipt itself still records a pending stop question, so the later stop must not be attributed to that earlier receipt. [R1, R35, R36]

All 26 papers remained in the counted plan. No selected closure was cut, and there were **zero context/budget/provider-refused windows**. The ten skips were terminal validation/formalization failures, not an extra category to add to the ten skipped windows. Increasing batch limits changed partitioning from 74 to 13 windows; copied cache entries could no longer provide exact-input reuse.

Whole-window failures then lost more material. This observed granularity effect does not establish a controlled causal claim about model quality. Raising limits again would not address the actual validation failures. [R35]

Exact-ID comparison found **five retained, 52 added and 154 lost source rows** relative to the old 159. Only eight of 48 evidence IDs in one formerly skipped window were recovered; no old skipped window was fully recovered. Gap2 gained MacFarlane's general Retraction Rule and four qualifications, but represented them as five atoms. Both Beddor–Goldstein windows failed, so the earlier safety/worldly-information premises disappeared. Eight other connected conditionals did survive, but none supplied the missing safety-to-retraction or necessity/exhaustiveness bridge. Thus gap2 was worse for the intended L3 test without being a strict subset or worse in every respect. [R35]

One rejected definition described Joey as maintaining a modal belief; this was a possible false author-attribution hit. Another diagnostic normalized nullable wire fields before applying correction patches: that repaired a schema issue but still left **94/95** required commitments. Neither diagnostic justified admitting the rejected window. Dinges and several newer comparison texts remained absent from native extraction; acquisition or an anchor nomination cannot manufacture their premises. [R35]

A.4. Shared vocabulary and consolidation attempts

Original §2.7 Shared vocabulary, consolidation and the repeated identity lesson

R17 (`de4fff8`) made shared-vocabulary prompting and wider alignment independently opt-in. The prompt asks for source-supported predicates, roles and terms against the growing catalogue; it no longer prefers paper-prefixed names. The wider pool uses signature compatibility and weighted lexical similarity with top-k neighbors and an explicit global cap. It is bounded lexical retrieval, not a semantic oracle. Defaults preserve the older path. Faithful atoms remain permitted; the runtime neither forces equivalence nor invents an ontology. Free replay changed no source formulas and demonstrated no improvement in source sharing. [R49]

Gap4a is the live test of that prompt. At its completed construction checkpoint, **308 commitments from 21 papers** comprised **263 atoms, 38 implications, four conjunctions and three universals**, using **292 symbols**. Actual source sharing and exact-antecedent cross-paper modus-ponens opportunities were both **zero**. Only **75/263** atoms still began with their own source prefix: removing a prefix did not make two formulas reuse the same symbol. A recorded request contained the shared-vocabulary contract and 290 catalogue symbols, so absence of the new instruction does not explain this outcome. [R60]

Gap4a's 2,000 selected alignment pairs came from a 3,338-pair top-k pool and 30,955 signature-compatible pairs. At the **20:10:10 UTC** checkpoint, disk policy had paused the run before the first alignment dispatch. No freeze, forward C3 sweep or searched route existed. Search was separately held after an offline counterexample showed its pinned code could omit a declared target basis. These are two distinct holds. The later packing2 repair does not hot-swap the already-running process. [R60]

R18 introduced **accepted-content consolidation** in a fresh lifecycle. It imports verified terminal construction, re-formalizes accepted source commitments against a shared catalogue, runs new alignment, optionally admits H, and freezes once. Source IDs, text, citations, stance and provenance remain unchanged; old formulas and new formulas are both retained. Old alignment judgments bind old formulas and therefore cannot be silently carried into the new alignment. The nested consolidation receipt and provenance-v5 permit replay validation and disclosed model judgments. The same text can have a different proposed formal representation without becoming new source evidence. [R50]

Attempt	What executed	Stopping point	What cannot be claimed
Gap3v / r18	Fresh consolidation attempted	First batch refused reuse/evidence-ID mismatches; zero accepted batches	No consolidated freeze or sharing result.
Gap3v / r19 (`f11fc18`)	Runtime-owned slot/handle binding worked in live calls	Legacy attribution lexicon rejected consolidation definitions	Identity repair is not semantic success.
Gap3v / r20 (`cdec8ff`)	22/30 batches accepted, representing 181 commitments; 41 new model receipts: two canaries, 22 accepted and 17 rejected consolidation replies	Batch 23: seven formulas for eight slots, then eight with definition refusals, then seven again; bounded budget exhausted	Partial prefix is not a final consolidated theory. Zero alignment, target or search calls.
R21	Schema cardinality plus per-batch failure isolation assigned at 21:05:16 UTC	Pending at this revision's source cutoff	No freeze or after-sharing metric credited.

R19 generalizes gap1's lesson: do not ask a probabilistic model to echo immutable IDs. Runtime-owned handles bind formulas and evidence back to immutable source rows. R20 adds precise declaration-specific feedback; successful corrections demonstrate useful feedback, but the terminal batch shows that a correction can fix one constraint and regress another. R21's proposed response is structural cardinality plus failure isolation, not unlimited retries or manual invention of model text. R20's exact-head offline result was **1,987/1,991 tests** with four skips and **39 selected controls**; a real fork test remained unexercised because no new freeze existed. [R50, R51]

The vocabulary-after figure appears in the main text.

The funnel figure appears in the main text.

The figure separates construction outputs, failed gates and distinct writing lineages. None of the proposal PDFs is counted as the endpoint of a CLOSED gap route.

A.5. Client windows, admission bounds and packed/unpacked forks

Original §3.2 Client windows and the refused target request

The first L3 target request was then refused **before dispatch**: **311,785 estimated input + 16,384 output allowance = 328,169**, against a **200,000** bound. Thus “312k > 200k” describes the input estimate alone; admission counted the larger total. **No target call, target, route, selection, or L3 resolution occurred.** A prior readiness check estimated only **51,998 input tokens** and did not represent the full target request.

The full prompt was **627,750 UTF-8 bytes**, including **332,316 bytes** of provenance. The stopped request incurred no generation spend. Recovery's supervisor elapsed time was **13,773.98 seconds**, ending with status **2**. [R18]

Pinned Codex client metadata advertised a **272,000 default / 872,000 maximum** context window. These are client settings, not verified service capacity. The adapter initially failed to pass `model_context_window`. After the operator authorized loosening the token limit, sigma r13 forwarded it explicitly and added gap-scoped packing. The offline one-hop plan retained three full and 157 compact rows and estimated **122,108** total tokens. It was a plan, distinct from the subsequent actual two-hop fork A. [R19]

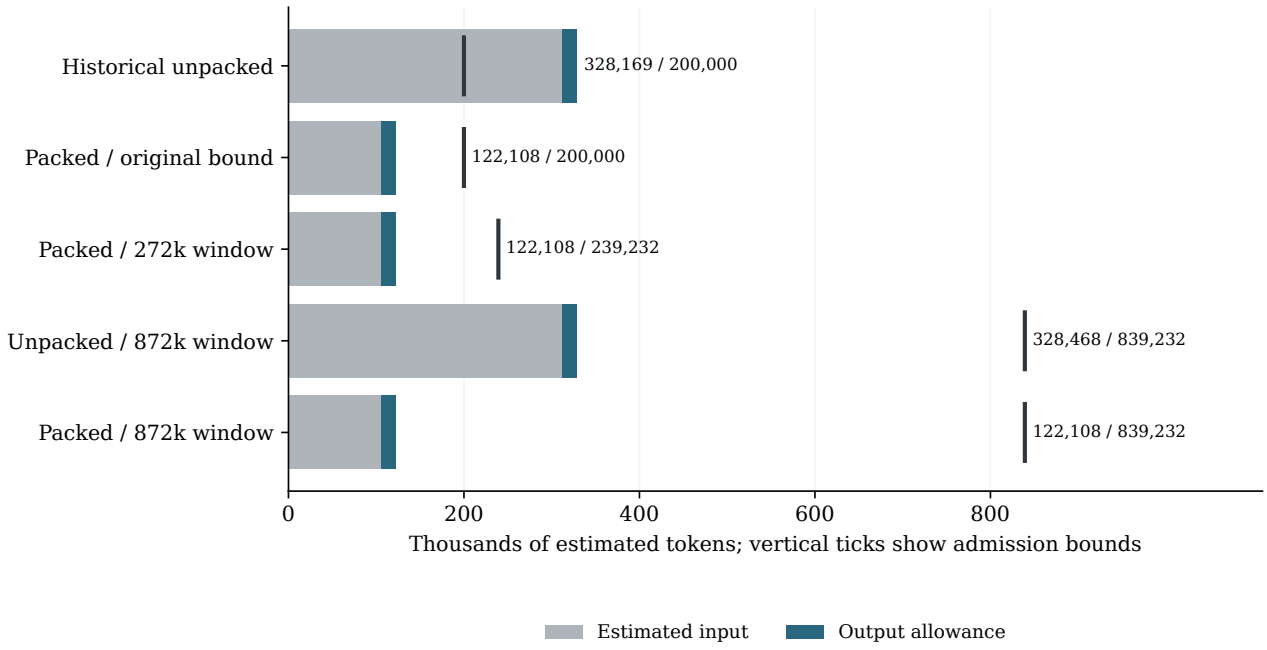


Figure 13. Full L3 request totals include output allowance. Only the first row is a historical refusal; the remaining rows are free offline plans. Packing uses one hop (3 full + 157 compact commitments); the later actual two-hop A results are reported separately in §3.3. This historical unpacked template adds 299 tokens. The later stage-window defect was closed by r14 review. These plan estimates do not prove service capacity.

Sigma r13's stage-specific configuration split could admit 587,839 estimated tokens under an effective 272,000 window because a larger inherited bound remained active. Sigma r14 normalized admission and provider configuration before cache lookup and explicitly refused null clearing. Kappa4 closed S13-F1/F2 at `7d19ccc`; the fix was merged into `893985a`. A/B/C used run-level windows only and avoided the historical trigger. These are offline enforcement findings, not proof of a model's maximum serving window. [R24, R30]

Original §3.3 Actual packed and unpacked requests

A's actual two-hop L3 context retained **three full and 157 compact commitments**, including every frozen formula and stance. Its first live estimate was **105,722 input + 16,384 output allowance = 122,106** tokens; offline replay was 122,108. B retained full provenance for all 160 commitments:

312,084 + 16,384 = 328,468 estimated tokens. A used a run-level 272,000 window and 239,232 bound; B/C used a run-level 600,000 window and 567,232 bound. No stage override triggered the historical stage-window defect. [R28]

Actual first-call usage was **60,702 input / 958 output / 3,968 cache-read** for A and **171,899 / 699 / 0** for B. The larger limit admitted a conservatively estimated request. These receipts do **not** show that the provider needed more than 200,000 actual input tokens or establish its maximum serving capacity. A/B also differed in ten semantic classifications before packing, despite identical pair inputs; token bounds differed, and there was only one arm of each type. Target-count differences cannot be causally attributed to packing. Their deterministic preparation was interrupted and explicitly recovered, with original receipts and freezes preserved; these were not uninterrupted runs. [R28]

The completed fork outcomes are in §4. Raising an admission limit admitted a conservatively estimated request; it did not supply missing premises, improve the freeze or establish a causal quality benefit. Fresh usage is reported separately in §8. [R28]

A.6. Historical pilot gates and route classifications

Original §4.1 Formal search and the outcome census

The operator-approved acceptable pilot required a clean native selection boundary, source coverage of at least **20 of 26 papers**, at least **three** genuinely derived routes, and readable summaries/reporting. Each qualifying route had to state a novel own thesis, avoid single-commitment entailment, use at least **two distinct supports from two papers**, respect stance, and contain a licensed inference. Pilot 5 met the clean-run, coverage, and rendering requirements but failed the substantive-route requirement. A native label alone did not satisfy the criterion. [R1, R6]

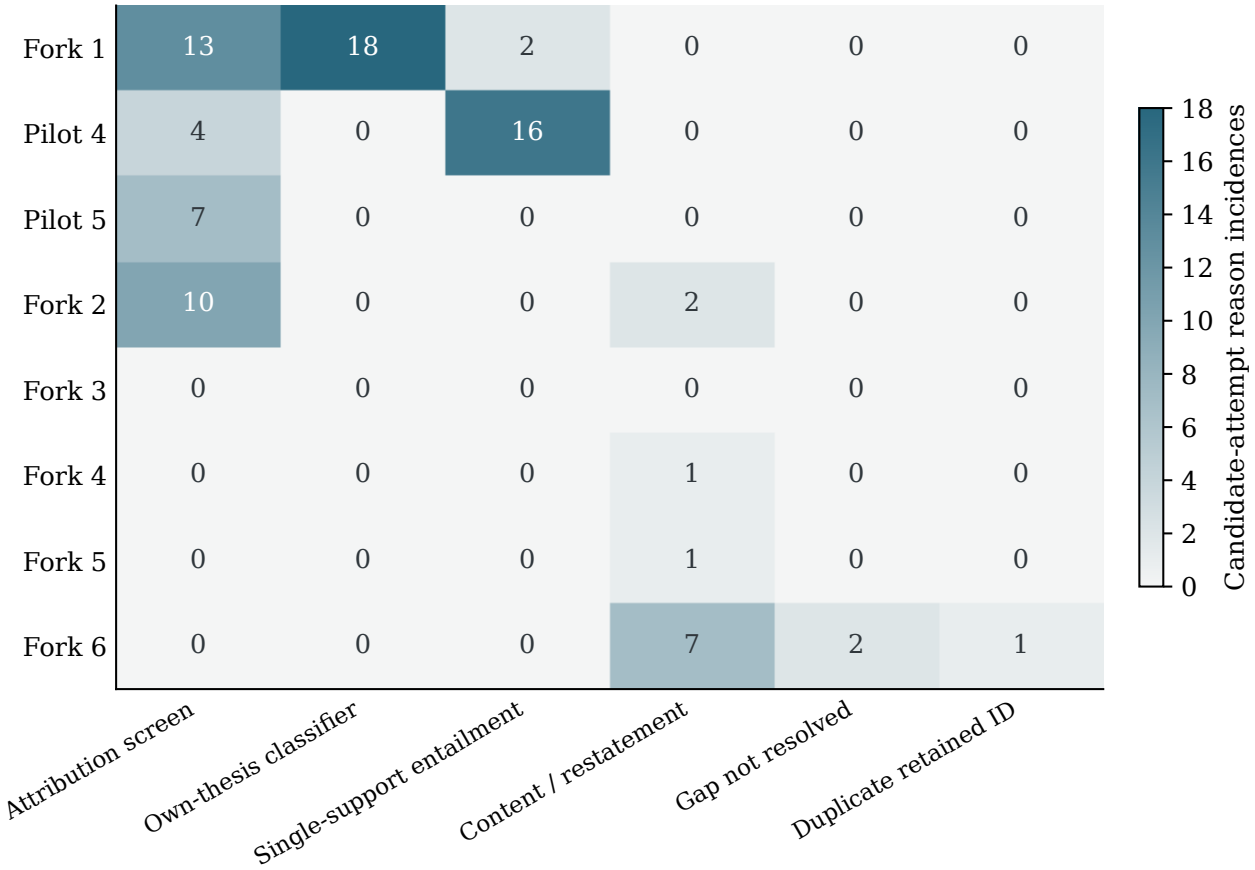


Figure 14. Native rejection reasons across pilot generations. The same candidate-attempt can occur in more than one column; columns must not be summed into a rejection total. “Attribution screen” includes lexical false positives, especially before that screen became advisory. Repeated proposal attempts count separately. Zero means no recorded rejection under that reason, not proof of semantic novelty.

The chart counts candidate-attempt incidences under the recorded gate reason. Reasons can overlap. In fork 1, an attribution screen and the own-thesis classifier could both reject a candidate; adding the columns would overcount rejected candidates. Later pool receipts record an explicit retained/rejected disposition. The figure preserves this difference rather than treating all generations as a controlled, identical experiment. Exact rows and event hashes are in the data supplement.

The gates caught several real defects: single-premise restatements, unused-antecedent conditionals, Boolean packaging, renamed assertions, ornamental proof steps, and unsupported gap-resolution labels. They also created false positives and exclusions: attribution vocabulary, an order-dependent paper count, stance-blind support slicing, a duplicate target identifier, and the outer target cap. Passing all gates was necessary under a given policy but never sufficient for literature novelty. [R4, R9–R11]

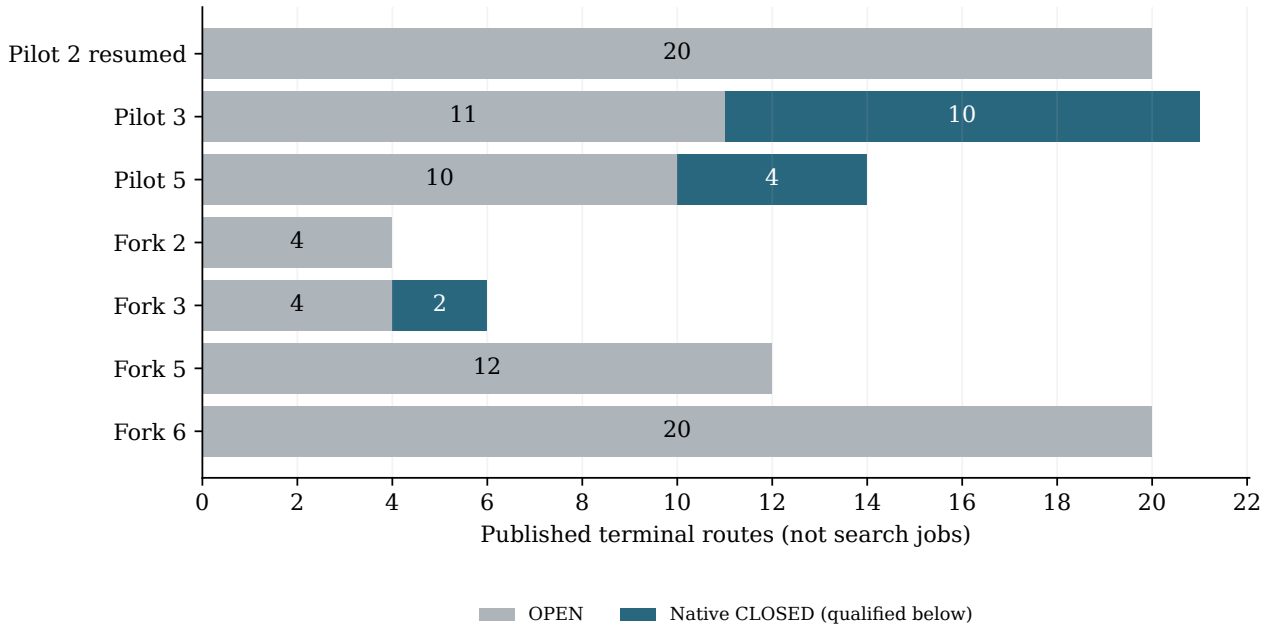


Figure 15. Closed is a formal outcome. Pilot 3's ten closed routes were trivial restatements; pilot 5's four were conjunction packaging; fork 3's two derived routes used a single paper. None met the complete accepted cross-paper criterion. Pilot 3 published twenty-one routes from twenty polarity searches.

The operator selected the two available fork-3 derived routes only after their single-paper limitation was reported. Those routes supplied writing experiments. The later L3 priority was a separate research direction, not a retroactive claim that the selected routes had already resolved that gap. No successful cross-paper pilot is established by the evidence covered here.

A.7. Backward gaps, literature currency and the T1–T9 dossier

Original §5.1 Backward: conflicts, missing connections, and literature currency

The initial gap detector found **three source-level open-problem flags**. It did not establish that those problems were still open in the current literature. The follow-up checked the source context, later literature, and proposed resolution theses. Its verdict was **one addressed, two partially addressed, and zero certified still open**. [R4, R5]

Flag	What the system treated as unresolved	What the currency check found
G1	Whether disagreement motivates a relativist account and where the effect belongs	Later theories and experiments address the comparison. Remaining disagreement does not certify an untouched problem.
G2	Nested probability and the interpretation of a particular reply pattern	Later formal/pragmatic models and experiments address the mechanism. The exact remaining scope was only partially settled by the accessible material.
G3	An older absence-of-model or absence-of-diagnostic claim	MacFarlane's later models and fixed-use/retraction diagnostics address that claim. Relevant works were already in the original accessible corpus.

L3 emerged from the failed search for connecting premises and a targeted bridge-literature investigation: could worldly-information safety support an assessment-sensitive retraction obligation? The initial acquisition report classified the link as unknown. The deeper check revised it to **partial**. The safety anchor already discusses retraction and assessor information; Dinges's later work connects estimates of epistemic qualities, including safety, to retraction while allowing a rival semantic interpretation. These are substantive anticipations. The accessible texts did not establish the specific safety-to-retraction derivation sought by ARA. [R5]

The focused search covered **2018–2026** and retrieved indexed first-generation citation sets, including **11** works citing the safety anchor, **116** citing the earlier assessment-sensitive account, and **449** citing the later book. Those counts describe retrieved index records, not papers all read in full. The index omitted an actual book citation found in a read PDF, demonstrating that absence from a citation graph is unreliable negative evidence. At that earlier checkpoint two especially relevant modern comparisons remained inaccessible; the later dossier obtained and inspected both. [R27] [R5]

A separate live semantic-conflict pass classified **60 pairs**: **three contradictory**, **seven in tension**, **35 compatible**, and **15 unrelated**. Source/currency review of the ten flagged pairs yielded **nine partial** and **one unknown**, with none certified still open. L3 was omitted by that bounded top-pair retrieval and was investigated separately. The operator approved the restrained framing **"not found within stated search scope, partially anticipated."** That framing is retained here. It is not a claim that ARA resolved L3 or established priority. [R1, R14]

Original §5.2 The dossier is a comparison map, not a proof

Eps2's completed dossier classifies **nine analyst-generated theses: four anticipated, four partially anticipated, one not found within scope**. Its only not-found thesis is the exact T2 necessity-and-sufficiency connection. The verdict does not establish truth, priority, feasibility or an exhaustive current gap. Seven acquired full texts support this analysis without becoming premises in the old freeze. [R27]

Dossier thesis	Comparison subject	Bounded verdict
T1	Safety failure sufficient for obligatory retraction	Partially anticipated
T2	Safety failure necessary and sufficient for obligatory retraction	Not found within scope
T3	Asymmetric loss without relativism	Anticipated
T4	Stable acceptance and withdrawal	Partially anticipated
T5	Assessment-relative worldly-information selector	Partially anticipated
T6	Flexible truth assessment, current-context retraction	Anticipated
T7	Retraction, cancellation and uptake	Anticipated
T8	Later responses help determine earlier content	Anticipated
T9	Fragmented/topic-sensitive knowledge with a retraction norm	Partially anticipated

ARA did propose near-T2 targets. Kept B/1 and B/2 partially match T2's necessity direction, while changing the subject or normative force: knowledge claims, warranted retraction, or an assessor requiring retraction are not the exact fixed-use, original-speaker obligation. Kept A/3 is only thematically related to T5. None is exact T2 or either exact direction. C's exact T2 came from eps2 and was supplied as a disclosed external test. Its sufficiency probe is closest to partially anticipated T1; the global T2 not-found label cannot override that closest-thesis comparison. Negation routes inherit only a comparison label, not a positive thesis's literature support. [R28, R38]

A.8. Writer length, typed audits, union batches and citation loss

Original §7.1 Typed claims and the early writing results

Repeated whole-draft retries did not supply actionable length feedback. Section budgets and append-only expansion preserved accepted material while filling deficits. Citation splitting, identity screening on resumed drafts, and revision-marker handling also required repair. Typed auditing improved the distinction between exposition and unsupported argument, but audit judgments varied markedly across runs. A zero-flag model judgment is evidence about that invocation, not an independent semantic proof. The retained adverse judgments and the final retrieval failures remain part of the record. [R7]

The initial whole-essay writer produced **eight attempts** across the selected routes, all below the then-configured **4,500-word** minimum; their range was **1,590–4,376 words**. No audit was reached. A sectioned writer then used an outline with section budgets, deterministic assembly, and targeted expansion. Its first length-successful draft reached **4,642 words**, but failed audit on **four sentences**. A repaired **4,604-word** draft passed a subsequent native audit, then stopped before external evidence retrieval. [R1, R7]

Later evidence changes the interpretation of that pass. A new binary audit flagged **101** sentences in the same previously accepted draft. Typed auditing distinguished source claims, argument steps, framing, objections, and unsupported claims. In the later recorded comparison, route 1 began with **75/245** flags and route 2 with **25/245**. Repairs yielded **4,714-word** and **4,619-word** drafts with zero flags in their final pre-verification audits. Both native runs nevertheless stopped at `search_evidence` with `cli_tool_event`. An isolated web diagnostic succeeded, but neither native lifecycle produced accepted retrieval evidence, citation/originality classification, the final audit, or an essay export. **Verified native exports remained zero at the round-4 checkpoint.** [R7]

Table 1. Round-4 typed pre-verification audit, reproduced from the writer’s receipt. “Audit passed” does not mean that retrieval, citation/originality verification, or export passed. [R7]

Route / stage	Native words	Flags / sentences	Outcome
Route 1 imported draft	4,604	75/245	Revision required
Route 1 accepted repair 1	4,714	0/245	Pre-verification audit passed
Route 2 imported draft	4,638	25/245	Revision required
Route 2 accepted repair 1	4,525	74/244	Revision required
Route 2 accepted repair 2	4,619	0/244	Pre-verification audit passed

The increase from **25 to 74 flags** on route 2 shows why these labels should not be read as a stable quality score. Native word counts include machine markers. Excluding both node and citation markers, the final drafts contain **4,578** and **4,498 words** respectively; route 2 is two words below the then-configured **4,500-word** floor under that stricter count. Neither round-4 draft was a verified export. Nu4 r5 was pending at the version-2 checkpoint; its later completion and round-6 limitations are reported below. This historical table is unchanged. [R7, R17, R20]

A separate publication branch polished the preserved earlier route-1 draft through isolated Claude Opus calls. It produced a **4,743-word**, **12-page** PDF with **two deterministic figures**. Its native audit covered all route nodes and its exact T-citation membership check passed. External verification remained blocked on that branch. This polished artifact is evidence of successful prose and rendering work on one selected route; it is not the L3 gap paper and is not proof of a verified competition entry. [R8]

Original §7.2 Audit nondeterminism, union gates and citation loss

Round 5 (`0368a17`) identified the refused event as `item.completed → web_search → action.type=other` and permitted that exact native web event only in `search_evidence`. Both routes then completed native export with originality labelled `possible_overlap`. That label is not a novelty pass. After quotation cleanup, route 1 received **44 flags**, then **zero** on a byte-identical cleaned-input prompt without a text repair. Route 2 changed from **eight to six flags** before substantive repair. Final audits could replay an accepted cache sample without reconciling the adverse verdict. These observations show audit nondeterminism, not measured error rates or evidence that the clean sample was correct. [R20, R22]

Round 6 (`d9ac619`) introduced a **k=3 union gate**: fresh requests bypass the response cache, and any current or valid prior same-prompt adverse flag remains binding. It is neither majority voting nor a claim of statistically independent samples. Repairs address the union and return through verification and fresh audit. The exact final-stage batches were: [R20]

Route / batch	Flags per sample	Prior adverse	Union	Unanimous / sentences	Verdict
1 / 1	0, 0, 36	44	47	209/245	reject
1 / 2	41, 52, 42	none	53	234/246	reject
1 / 3	0, 0, 0	none	0	245/245	accept
2 / 1	30, 0, 0	none	30	214/244	reject
2 / 2	2, 0, 5	none	6	238/244	reject
2 / 3	1, 15, 1	none	15	230/244	reject
2 / 4	0, 0, 0	none	0	244/244	accept

Unanimity counts agreement in binary votes; unanimous adverse votes still reject. Route 1 used **10 actual fresh audit calls** for **nine valid final-batch samples**; route 2 used **13 calls** for **12 samples**. Initial audits and bounded schema correction explain the difference. The first route-1 batch retained the earlier 44-flag result despite two clean samples. Complete valid adverse history therefore worked; the incomplete-history loophole later found by review required the repairs recorded in §7.5. [R20, R23]

Both round-6 lifecycles exported, but **each final bibliography had zero entries and each body had zero citation markers**. Native word counts were **5,387 / 4,413**, or **5,336 / 4,385** without machine markers. Both originality verdicts remained `possible_overlap`. External verification had removed the local source citations; subsequent clean audits did not restore them. These are native exports, not submission-ready essays or proof of source fidelity. Route 1 ran before a later cleanup-length/resume-budget correction; route 2 used the corrected source. The report preserves that source-version distinction rather than claiming both ran at the final committed head. [R20]

At the R6 checkpoint, citation repair was pending in R7; §7.3 gives the completed outcome. The requested fix verifies frozen-T references against local full text and page evidence first, uses web search as supplementary evidence, and refuses an export whose source claims lack citation coverage. It also must preserve valid adverse evidence from incomplete audit inventories. Publication's separate default floor is **3,000 words**, distinct from the historical writer construction target of **4,500** and the **6,000-word** ceiling. Native counters and marker-free counts must not be interchanged. [R20, R22, R23]

Original §7.3 Citation retention repaired; R7 still refused

Nu4 R7 at `72c11e7` repaired citation retention and partial adverse-history handling. Each earlier route uses **two support commitments from one paper**, `gsv1996`; two bibliography entries are not two papers or evidence of a 26-paper synthesis. Citation-only restoration preserved the prose and restored two entries and two markers per candidate; the retained source spans were locally grounded at physical PDF page 18. Both candidates nevertheless refused. [R29]

Earlier route	Recorded sentence audit	Candidate bibliography / markers	Native words / marker-free words	Native export
1	141/245 unsupported; after the remaining repair, 173/246 unsupported	2 / 2	5,199 / 5,148	Refused
2	123/244 flags: 121 unsupported and two miscited	2 / 2	4,413 / 4,385	Refused

Successful R7 native exports: 0/2. Final k=3 samples: not run for either candidate. Both stopped before native citation/originality verification and final consensus. The local verifier diagnostic retained both entries, but did not create a native acceptance event. No favorable rerolls were used and stale exports were removed. R7 also corrects the earlier word-counter description: native words include title/body and markers, **exclude bibliography**; the comparison removes markers from that same scope. Restoring citations fixes a gate defect without making unsupported prose source-faithful. [R29]

A.9. Audit history, polish failures and verified sentence carry-over

Original §7.5 Audit history and citation verification must survive retries

R2-F1 exposed dropped partial adverse history on resume; R2-F2 exposed citation stripping. DLV-F1 then showed that valid adverse labels inside rejected schema retries of the current fresh batch could disappear when a corrected response was clean. Pi3's `3c9c8a4`, included in delivery round30b `6c6cf84`, unions those labels before verdict. The reviewed pipeline verifies frozen-T source evidence locally, restores only citation markers without changing prose, preserves semantic repair budgets, and independently checks support coverage before export. Missing local evidence is not positive mismatch evidence. Clean sampled judgments still do not prove source fidelity or statistical independence. [R23, R29, R31]

Original §7.6 Polish failures, carry-over and meaning-preserving edits

Publication has its own gates. B/2 initially failed the isolation canary's string-format reporting; the operator approved a Claude-only list format while retaining strict empty-field semantics. The reviewed canary then passed. A subsequent context refusal was addressed with a publication-only counted bound, leaving Codex admission separate. Neither transport repair accepted the prose. B/2 used three semantic repairs; its last complete gate was **[0,1,1]**, and its final operator-stopped gate had only **[4,10]**. There is no third sample or clean final gate to infer. [R56]

T2 polish moved from **[7,14,13] to [0,0,0]**, but native verification refused two quoted article titles. All six citation identity/locator checks passed; originality remained unverified/advisory and was not the refusal trigger. Formatting those exact titles in italics is a narrowly scoped correction, not permission to change substantive claims or bypass verification. On the operator's instruction, the delivered B/2 and T2 PDFs reproduce the verified pre-polish originals, retaining the original body/title hashes, origin labels and clean gate receipts. Their small status notes and deterministic reference/layout formatting are disclosed. The existing refused polished PDF is not a delivered success. [R56]

The audit-flags figure appears in the main text.

Nu4's carry-over mechanism anchors each current sentence to a source draft with a complete clean native k≥3 gate, exact source hashes, matching outline/evidence/authorization and retained sentence offsets. Deterministic normalization permits ownership-prefix removal and limited typography/whitespace changes; it does not erase citations, logical symbols or arbitrary words. Adding or changing an ownership prefix requires meaning assessment. Citation markers and bibliography records must remain unchanged. Source roles remain effective, including inherited roles, without requiring incidental role labels to agree in every original sample. [R57]

Other aligned edits receive **three fresh meaning samples**. Only unanimous equivalence inherits the source verdict. A different/uncertain vote, including a valid adverse vote in a malformed retry, leaves ordinary typed auditing in force. Failed equivalence does not itself refuse publication; the current text can still pass its ordinary audit. Decisions persist, preventing favorable rerolls. Unaligned sentences and unrelated adverse history retain ordinary treatment. This is a disciplined reuse of verified evidence, not a claim that meaning equivalence is mechanically decided. [R57]

The native live probe used requested `gpt-6-luna`, with serving version unverified. Seven pair expectations passed: two rewordings received three equivalent votes each; quantifier/scope, attribution, negation, modality and claims-versus-sources scope changes received three different votes each. This is a small classifier/schema probe, not an accuracy estimate. The latest source-retry fix has **27 owning tests** and independent review at `757bdda`; the initial publication continuation used reviewed combined `3a365a5`; the later inventory repair advanced its pin to `6ddcaf1` (§7.8). [R57, R58]

Pi3's inline repair uses verified source spans and current occurrence markers to repair unsupported elaborations locally. The combined reviewer exercised the native seam: accepted prefix removal survived raw model dissent, one unaligned unsupported sentence was deleted, and the final audit passed with one inline pass and zero whole-paper rounds. Fake providers and stubbed verification transport were used in that review, so it is not a live candidate success. **T2 subsequently refused the final audit after passing title/citation verification (§7.8); B/2 inline-repair outcome remains pending.** [R58]

A.10. T2 title-only continuation and retained meaning dissent

Original §7.8 Late T2 result: repaired formatting, preserved dissent, refused polish

The first native title-only continuation replaced exactly four quotation characters with asterisks, leaving every other Markdown character unchanged. All six citation items verified. It then stopped before publication k=3 because the meaning stage returned **98 of 101 requested pairs**; its **47 equivalent / 51 different** votes were retained. A malformed inventory is not permission to erase its valid adverse evidence. [R66]

A bounded shape correction was reviewed on integrated `6ddcaf1`: exactly two attempts per meaning sample, precise missing/extra/duplicate feedback and retained dissent, without changing prose or semantic repair budgets. Four owning regressions and twelve independent combined checks passed; four

distinguishing controls were restored. The renewed native stage corrected a 99/101 inventory within its single correction. **All 51 previously adverse pairs remained adverse and none inherited**; overall, **40 of 101 pairs inherited** the source verdict and 61 stayed on ordinary audit. [R66]

The final fresh publication samples were **[9,6,0]**, with **14 union flags over seven sentences**: seven unsupported-claim labels, five unsupported-elaboration labels and two original-draft-fidelity labels. A clean third sample did not override dissent. **Polished T2 remains refused.** No semantic editor ran; whole-paper repairs remain one and inline passes zero. The attempted polished draft is **3,786 reading words**. The automatic fallback metadata reports **3,385 words**, a different artifact; the delivered original PDF with its method note remains **3,427 words**. Its original file and audit evidence were not overwritten. [R66]

This result narrows the earlier diagnosis: the title formatting defect is fixed, while final sentence-level support/fidelity still fails. It also supplies live evidence that malformed-response dissent survives the structural recovery. B/2 has started its separate same-draft inline continuation on the reviewed head; no final acceptance or replacement PDF is yet credited. [R66]

A.11. Providers, prompts, accounting and human coordination

Original §8. Runtime, prompts, human time and accounting

Original §8.1 Providers and isolation

Role in the recorded work	Model / version evidence	Limit
API extraction and early construction	Receipts identify `claude-opus-5`; early cheaper-model experiments requested `claude-sonnet-5-5`	API ledger scope below excludes some early experiments. No underlying weight revision beyond reported identifiers is available.
Subscription construction/search	`claude-opus-5-5`, Claude Code `2.1.306`	Vendor runtime context remains; effective prompts differ from direct API prompts.
Codex pilots and writing	Requested `gpt-6-luna`, `codex-cli 0.163.0`	Serving identity was recorded as `unverified` when omitted by CLI output. A requested pin is not a serving receipt.
Publication-only polish	`claude-opus-5-5`, Claude Code `2.1.306`	Separately authorized polish, not authorization to resume Claude pilots.
Scaffold coding, orchestration, this report	Agent roles recorded in briefs/correspondence	A complete per-worker model/version and token ledger is unavailable in the supplied sources. No exact model identity is inferred from a role name.

The API-to-Claude switch followed the credit failure and the human request for an agent route. The Claude-to-Codex switch followed the operator's clarification that "chat" meant ChatGPT/Codex; the active Claude pilot was halted. Later, the operator explicitly prohibited further API credit use and authorized ordinary agent calls by default, with Claude Opus reserved for the requested final polish. These were human choices, not automatic provider fallback. [R1, R13]

The Claude canary and review established the recorded configuration's suppression of operator customizations and identity output. They did not establish absence of vendor built-ins or byte-identical API context. Subsequent reviews caught escaped-output leakage and unsafe diagnostic text; allowlisted diagnostics replaced free-text retention. Codex used fresh processes, temporary configuration, isolated host homes/process state, disabled project instructions, skills, memory, and external tools, plus a context canary. Generic built-in tool definitions could still be visible; actual tool events rejected the response. Strict-schema projection required its own preflight. Isolation evidence is specific to the tested transport/version and threat model, not proof against all contamination. It does not erase pretrained model knowledge or establish that every generated idea came only from the supplied papers. [R13]

The later web-evidence transport deliberately differed from the earlier tool-free transport. Round 4 rejected an unidentified native event; round 5 diagnosed the specific web-search event and narrowly permitted it for `search_evidence`. Subsequent native exports are recorded in §7.2, together with their citation loss. That repair did not authorize other tool events. [R7, R20]

API construction stopped with opaque request failures, then a count-endpoint diagnostic identified insufficient credit. The earlier request failure was suspected to have the same cause, but that retrospective attribution was not proven. A subscription route was then implemented. Its first strict isolation probe failed because built-in context, hooks, and identity-related material remained. The revised contract excluded operator content and customizations while disclosing unavoidable vendor context. Output and decoded-string scans, fresh sessions, bounded retries, and canaries were reviewed repeatedly. [R1, R13]

Subscription capacity was also misestimated. A calibration comparison of aggregate usage with initial context inflated a factor from **0.553 to 1.424**, leaving **58 of 70** windows skipped in the first Claude pilot. Like-for-like accounting and a local rather than inherited token cap repaired the comparison. A later **401** after host authentication refresh exposed stale in-memory credentials; per-call credential reads and one bounded re-read retry addressed it. These repairs did not prove that future provider behavior would remain stable. [R1, R13]

Original §8.2 Prompt disclosure

Stage prompts are topic-neutral templates combined at runtime with source text, the frozen catalogue, target/route state, explicit schemas, and correction feedback. Full prompt and response receipts were retained. Worker briefs separately specified implementation tasks, budgets, permitted providers, stop conditions, and expected evidence. They directed scaffold development; they were not all passed verbatim to philosophical generation calls. [R2, R3, R15]

Representative recorded instructions included:

- Extraction: "Preserve the paper's argument structure" and "Do not reduce this to a fixed number of central claims."
- Construction: a selected source inference must become a content conditional using the exact endpoint formulas, rather than a new standalone atom.

- Target proposal: assert a thesis in the system's own voice; preserve frozen meanings; do not invent support to fill the requested pool.
- Search: instantiate the supplied bank and answer critical questions from available evidence. The engine retains licensing authority.
- Writing: compose from the selected route and source context, preserve citations and node coverage, and meet explicit length bounds.
- Publication: improve presentation without adding claims, preserve citation/marker sequences, then run the native audit again.

The quoted extraction sentences are short instruction excerpts, not literature quotations. The other entries are functional descriptions. `prompts.md` supplies the exact recorded instruction prefixes and their digests, including successive variants. Current source templates alone would be insufficient to reproduce earlier runs because gates, correction messages, schemas, effective provider instructions, and runtime inputs changed. Exact raw prompts remain privately recoverable through the event hashes. [R15]

Original §8.3 Scoped accounting

The principal ledger is `reports/overnight/fork6/accounting.json`. Its report covers **67 lineage runs** and deduplicates identical event receipts across copied prefixes and imports. We reaggregated its MODEL receipts by provider and stage and checked equality with its measured, unknown-at-reservation, and token totals. Per-run totals must not be added: many include the same extraction and freeze history. [R4]

This is a measured lineage subtotal, **not total campaign cost**. It excludes some early smoke/development work, later writer/publication branches, coding and review-agent conversations, the orchestration conversation, and production of this metapaper. The early smoke receipt separately reports **$7.059330**. The initial development record also retained unknown usage reservations. No reconciled global invoice or subscription fee allocation is present, so no all-in dollar total is asserted. [R1, R4]

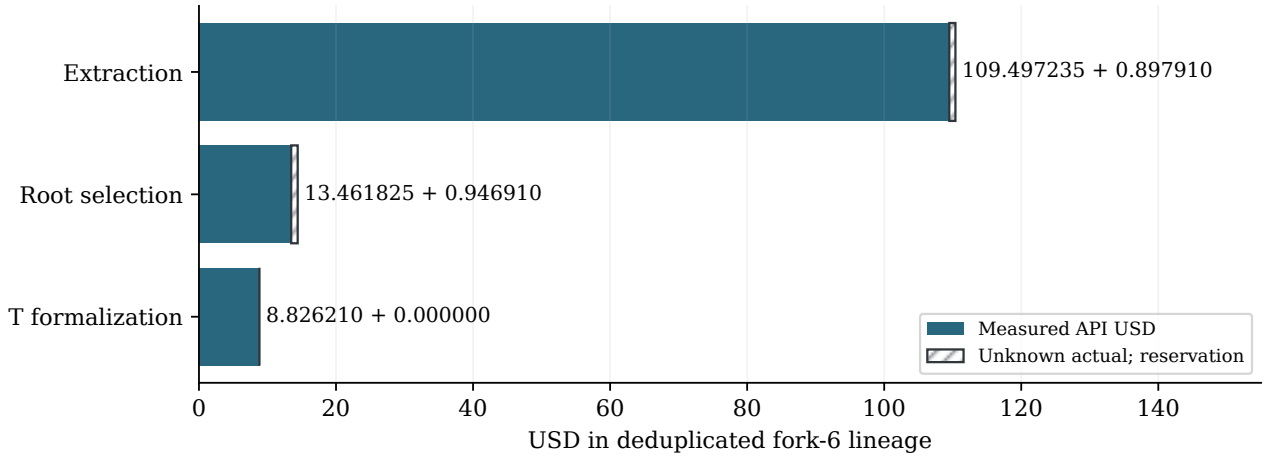


Figure 16. Measured API cost is $131.785270; unknown-at-reservation is $1.844820. The sum, $133.630090, is a budget-accounting figure, not an invoice. Subscription calls contribute no API USD here. Earlier development/smoke work and later writing/polish are outside this report's lineage.

The ledger contains **4,534 measured MODEL receipts** and **12 unknown-at-reservation receipts**. MODEL receipts include cache replays and validation/preflight work; these counts are not unique uncached generation calls. Recorded measured usage totals **113,530,631 tokens**, including the separate cache components. Unknown usage contributes **1,700,600 reserved tokens**, not measured consumption. API measured USD is **$131.785270**; the **$1.844820** unknown reservation makes the budget-accounted sum **$133.630090**. A failed response with no usage receipt does not become measured because it was budgeted. [R4]

Provider scope	Input	Output	Cache read	Cache write	Unknown reserved tokens
Anthropic API	11,314,239	3,008,563	0	0	175,424
Claude Code subscription	4,012	1,972,083	5,762,812	49,414,935	661,810
Codex CLI subscription	39,728,717	1,087,909	1,228,160	0	863,366
Legacy: provider absent	8,373	828	0	0	0

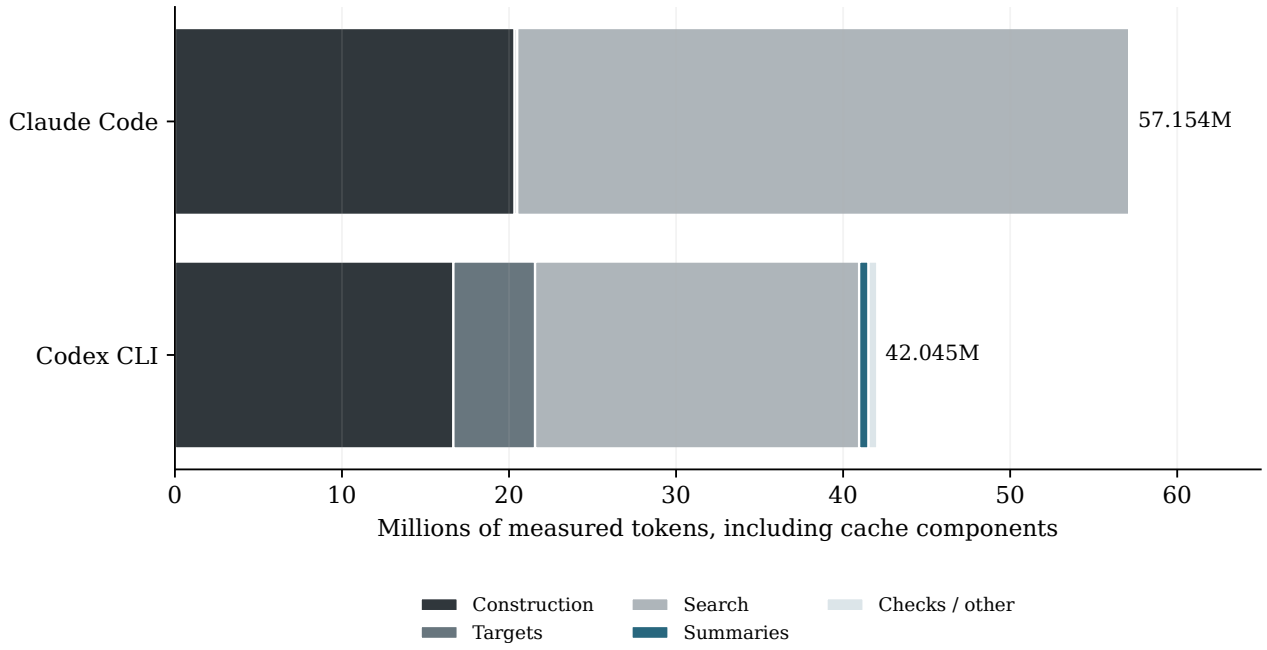


Figure 17. Measured subscription usage in the same deduplicated lineage, grouped by native stage. Cache reads and writes are retained; these are accounting components, not unique prose tokens. Unknown reservations and an unclassified-provider legacy receipt are shown in the accompanying table, not these bars. Coding-agent and orchestration conversations are outside this ledger.

A small legacy receipt requests `gpt-6-luna` but lacks an explicit provider field. Its **9,201 measured tokens** remain separately unattributed instead of being silently assigned to an API or subscription provider. Subscription list-price equivalents, where available, are not API payments. The data retains every native stage and token component; the figure groups stages only for readability.

Separate supplements and missing totals

The writer’s round-2 repair/audit supplement records **143,996 input**, **5,888 cache-read**, and **35,228 output tokens** across **eight actual responses**, all requesting Codex. These numbers exclude its preflight and earlier writing attempts. The publication receipt records **155,926 input**, **49,281 cache-read**, **208,242 cache-write**, and **183,617 output tokens**. Its **37 MODEL receipts** include cache replay; **19 dispatches** produced **18 measured responses** and one timeout with unknown usage retained at reservation. Thus its token aggregate is mixed measured/reserved, not wholly measured. These supplements are not added to the lineage subtotal. [R7, R8]

Gap pilot 1 has a separate local ledger: **242 fresh Codex calls**, comprising **221 main-run calls**, **20 schema-preflight calls**, and **one standalone canary**. Measured fresh usage is **4,769,322 input + 235,926 output + 266,752 cache-read = 5,272,000 tokens**; cache writes and server-tool uses are zero. New API spend is **$0**, while subscription dollar-equivalent is unavailable. Requested model identity remains `gpt-6-luna`; served identities are unverified. The pilot report’s inherited extraction charges are not new spend and are not added again to the fork-6 ledger. Its free recovery inspection made no new model calls. [R16]

Gap1r adds **197 fresh Codex calls**: **177 main-run calls** (two canaries, 115 alignment, 60 semantic classification) and **20 preflight calls**. Its measured components are **724,135 input + 36,474 output + 279,680 cache-read = 1,040,289 tokens**, with **$0 new API spend**. Together gap1 and gap1r used **439 calls / 6,312,289 measured token components**. Inherited extraction charges must not be added again. Gap1f and sigma r13's offline plans made no new generation calls. At that earlier accounting checkpoint the enlarged-window A/B runs were pending; their later fresh usage appears below, and this report does not manufacture an R5–R6 global token total from their audit-call counts. [R18–R20]

Summed measured model wall time in the lineage is **99,802.960 seconds**, approximately **27.723 hours**; unknown-at-reservation receipts add **747.604 seconds** of recorded wall time. Calls can overlap, so neither sum is elapsed campaign duration or human labor. The extended correspondence span is **8 days, 21 hours, 7 minutes, 22 seconds**; the principal accounting and research checkpoints are unchanged. No complete operator-reported labor duration is available. Zero-minute fields on worker-generated command receipts cannot establish zero human effort. [R1, R4]

Fork	Fresh local Codex calls	Input tokens	Output tokens	Cache-read tokens
A	319	10,110,944	141,719	732,032
B	272	12,934,452	136,006	455,296
C	29	4,918,072	13,864	31,232

These components include local runtime canaries and reporting summaries, and exclude inherited extraction/construction and separate preflights. Shared A/B preflight and C preflight each used **20 calls**. Input/cache overlap is not assumed, and inherited charges must not be counted again. New API spend was **$0**; subscription dollar cost remains unknown. The earlier cost figures retain their fork-6 scope. [R28]

No complete campaign total for gap2, gap3, the wider forward sweep, new writer work or this report is inferred from the older lineage. Partial live counters are not final invoices. New API spend of zero in a Codex run does not mean free subscription usage or zero authoring cost. [R28, R35, R36]

Gap3p accounting: local continuation only

The gap3p handoff reports **4,213,242 input tokens, 36,104 output tokens and 254,976 cache-read tokens**, with no cache writes or server tools in that local accounting. The 4,504,322 sum is a sum of receipt categories, not a new global token definition or subscription dollar estimate. Inherited gap3 construction must not be charged again. The broader campaign still lacks a reconciled authoring-agent, reviewer, consolidation and publication total. Failed/unknown CLI responses can retain reservation upper bounds; those are distinct from measured usage. [R53]

R20 consolidation records 41 new accounted model receipts; imported unknown-usage receipts retain their reservation treatment. This report does not add call counts to token counts or assign subscription dollars from API list prices. New pipeline stages also mean that the historical prompt supplement is no longer a complete current replay package. [R51]

Original §8.4 Coordination and reporting errors

The coordinating agent acknowledged an idle period from **12:33:41 to 16:49:21 UTC on 28 October**, or **4 hours 15 minutes 40 seconds**. It waited for a go/no-go decision instead of starting an already available alternative in parallel. Earlier, its instruction to stop at “any cap” caused a worker to interrupt an extraction pass after a best-effort reconciliation refusal; that was a briefing error, not evidence that the corpus could not be processed. [R1]

Reporting itself needed repair. A **3.7 GB** HTML report crashed Chromium; a later sidecar-based report was approximately **2.45 MB**. Fork 6’s native report rendered on desktop but retained **19 pixels** of mobile overflow. Its event-origin index also mislabelled **15,663** inherited-prefix occurrences; a separate overlay corrected attribution without altering native report bytes. The main accounting deduplicates receipts, so imported lineage must not be summed as new spend. These defects concern the original run reports; this metapaper has its own rendering checks. [R1, R4]

The original all-candidates short-length attribution was corrected rather than counted as an operator action. Old report directories were archived only after verification according to the correspondence; source hashes in this paper identify snapshots, not a guarantee that every old source path remains unpacked. The operator's recent scheduling and parallel-work instructions are disclosed in Appendix C. [R1, R42]

A later orchestration failure was recorded at **15:50:34 UTC on 30 October**: no new paper had arrived during an approximately four-hour stall while disk reservations, a canary decision and idle cleanup work blocked progress. The orchestrator acknowledged that it had not noticed the stall and reassigned investigation. At 08:34:20 it had also recorded that correction messages ended workers' turns without

continuation. These coordination failures are not model-search results; they affected elapsed delivery time. The operator subsequently required the orchestrator to limit its own investigation, rotated xi/kappa4 to xi2/kappa5, and adopted the ten-empty-job diagnostic rule. [R1]

Evidence cleanup is itself an operational intervention. Janitor archived 55 directories with upload/download hash verification, freeing a reported 1.35 GB; active records and uncertain items were excluded. Later triage classified the screened hits as zero real secrets, 246 fake fixtures and zero unknown, without printing values. This does not certify all campaign logs secret-free. Worker-owned shared-memory cleanup was authorized separately, restricted to finished scratch with no process users. Meta deleted none in this revision. [R1, R63]

A.12. Independent review, corrected controls and the iteration history

Original §9.2 Review findings ledger: mechanism proof has a scope

Finding or reviewed change	Disposition at the named head	Limit
S12-F1: ID-free integration fixture drift	Closed on later integrated sigma12/xi line	Fixture repair does not certify source-alignment meaning.
NP-F1–3: imported screening, title audit/cache, duplicate occurrence permissions	Closed for publication at pi3 `a6a2f9b`	No promotion of a failed polish fallback to a successful edit.
R28-F1/F2: padding and inconsistent union	Closed by xi29 `0c4d472`, independently re-reviewed	Placeholder regressions and gate closure are not a new philosophical result.
S13-F1/F2: merged control, stage-window mismatch	Closed at sigma14 `7d19ccc`; merged `893985a`	Actual client capacity and semantic quality not proved.
R2-F1/F2: partial adverse history, citation stripping	Closed in reviewed delivery `6c6cf84`	R7 actual exports remained 0/2.
DLV-F1: fresh schema-retry adverse labels lost	Old `d24288b` blocked; pi3 `3c9c8a4` included in round30b closes it	Clean retries cannot erase valid adverse labels.
Sigma15 anchors and context-derived budgets	Kappa4 merge verdict; 1,460/1,464 full tests, 77 Sigma controls and 25 synthetic-merge controls	No faithful new T or live pilot success follows.
Xi31 integration at `f76c1b3`	1,673/1,677 full tests; all 820 controls; no fixes needed	Four opt-in tests skipped; offline compatibility, not live or submission readiness.

The delivery round30b review separately ran 253/254 tests, 170 controls and 33 probes. Do not aggregate tests across heads into a claimed full-suite result. Sigma15's later full review closes the pending label from the earlier v5 checkpoint; xi31 checked the actual merge and retained both decision histories. The later X32, packing, consolidation and output-gate findings below distinguish completed code repairs from pending live results. [R21–R24, R27, R30, R31, R43]

Later review finding	Corrective mechanism / disposition	Evidence boundary
X32-F1/F2/F3	H-restatement padding, nonaffirmative disclosure and hidden H; closed by r32c runtime-owned disclosure	Mechanism reviewed; gap3h has no actual admitted H.
PACK-F1	Open slots lost frozen vocabulary; packing2 retains full catalogue where dependency inventory is incomplete	Gap3p never reached slot formalization.
PACK-F2	Declared target basis omitted; mandatory basis now survives shrinking, polarity and child leaves	Offline replay restores 46/46 maps and 92/92 cuts; no useful live inference proved.
PACK-F3	Mutation failed for unrelated error; earlier 920 pass retracted and owning test repaired	Repaired plus inherited accepted controls, not a fresh 920 rerun.
S17-F1/F2	Compatibility/control issues closed on final r17	Prompt delivery does not prove actual shared vocabulary.
N4-F1	No-reroll control exhausted fixture votes instead of testing persistence	Corrected control detects intended six-versus-three call failure; raw failed proof retained.
Combined audit-inline	Exact reviewed union at 3a365a5, six independent seam checks, zero blockers	Fake-provider seam proof; candidate verification remains separate.

Boundary	Failure found	Disposition / limit
Identity	S12: fixture drift; gap1 echoed IDs	Runtime binding / resume; semantics separate
Output evidence	NP-F1–3; R2-F1/F2; DLV-F1	Screening, titles, occurrence IDs, citations and retry dissent retained
Novelty	R28-F1/F2: padding / inconsistent union	Closed in reviewed xi29; real novelty still unproved
Context	S13-F1/F2: stage/provider mismatch	Normalized before cache/admission
H disclosure	X32-F1–3: padding / negation / hidden H	Runtime disclosure; gap3h not derived
Retrieval	PACK-F1/F2: missing vocabulary / bases	Packing2 offline basis coverage restored
Proof quality	PACK-F3; N4-F1: unrelated failure	Pass retracted; causal controls repaired
Inline output	Combined 3a365a5: 6/6 seam tests	Reviewed; live publication verdict pending

Figure 18. Review findings are attached to exact revisions and exercised boundaries. “Closed” refers to a software finding, not to an argument route or philosophical question.

The review architecture separates author implementation, a non-author adversarial review, exact-head integration and live candidate evidence. A mutation count alone is insufficient: the failed assertion must exercise the intended boundary. Reviews also preserve stale failures and explicit retractions rather than replacing history with the latest green number. [R49, R52, R54, R57, R58, R62]

Concept / iteratio	Observed failure	Response / outcome
21–25 Oct: extraction	Small summaries; malformed ranges/inferences	Rich evidence, grounding, disclosed omissions
Construction bounds	86k singleton / 70k; false 70–90-token claim	Window metadata, bounded closure; ~233-token correction
29 Oct: gap1 / r12	74 windows; alignment ID mismatch at 36/150	ID-free recovery → frozen T160; target admission then stops
Novelty / R28	Whole-T exclusion; padding / inconsistent union	Per-paper gate; zero qualifying real-source census
29–30 Oct: forks	141/160 atoms; zero source sharing	100 OPEN, only one partial B sign inference
30 Oct: gap3 / packing	243 commitments; admission / unknown CLI exits	Gap3p fits but omits bases; 46 OPEN
30 Oct: r17 / gap4a	Prompt delivered; 0/292 source-shared symbols	Consolidation added; alignment still pending
30 Oct: r18–r21	Echoed IDs, attribution, cardinality regression	r19 binding; r20 22/30 batches; r21 pending
Writing / audits	44→0; bibliography zero; refused polish	k=3 union, local citations, verified originals
Review / operation	Basis omission, unrelated mutation failure, empty jobs	Packing2, corrected proof, native ten-job pause

Figure 19. Iterations grouped by the core mechanism they changed. The alignment refusal, cardinality stop, invalid pass retraction and pending repairs remain visible; this is not a monotonic success timeline.

Appendix B. Credits

The detailed historical rows below retain their recorded rounds and evidence. Their in-progress descriptions are dated to v4; the current updates in Appendix B.6 and the concept sections supersede them. The roster now includes fwd and the three later workers janitor, kappa5 and xi2: **43 workers plus one orchestrator**. Agent IDs are not human names.

B.1 People, agent identifiers, and model evidence

The operator designed and directed the scaffold; AI coding agents built most of it under that direction. The operator chose the topic and literature, approved budgets and providers, set acceptance criteria and research priorities, selected writing routes, and requested publication and this report. Appendix C records the human actions. All names below are **agent identifiers, not people**. There are **43 distinct workers plus one orchestrator**, or **44 credited agent roles**. The count excludes the operator. [R1, R26]

Orchestrator — `brain` in correspondence, `@agent` in the live roster. The recorded harness is Claude and the model family is Claude; an exact model/version pin is unrecorded. From 21–29 October it planned work, wrote briefs, dispatched approved worker contexts, sequenced author/reviewer work, merged branches, coordinated budgets/providers, and reported to the operator. It also arranged output publication. These coordination credits do not transfer human design or research decisions to the agent. Evidence: `correspondence/log.txt`, `briefs/meta-credits.md`, and the read-only roster receipt in `data.json/credits`. Its acknowledged mistakes are retained: the 70–90-token construction estimate (§2.2), the “stop at any cap” briefing error and idle period (§8.4), and the logged delay in submitting the xi r29 prompt on 29 October. [R1, R26]

Each worker row records **Codex harness; exact worker model/version unrecorded**. The credits brief supplies the campaign-level Codex attribution. The initial log explicitly identifies alpha and beta as Codex; the v4 roster snapshot confirmed nine then-current worker contexts. For retired workers, that campaign attribution is not an independently recovered per-session telemetry receipt. No worker is assigned `gpt-6-luna` merely because its code called that model. Model pins for ARA generation are listed separately in §8.1. [R1, R26]

The later Opus authorization concerned the **isolated publication polish pass** operated by pi3; pi3 itself is a Codex worker. “Opus for polish only” describes that later authorization, not the whole history: earlier API extraction and Claude subscription pilots used the models already disclosed in §8.1. No exact orchestrator subtype is inferred from the polish model. [R1, R8, R13, R26]

B.2 Intake, extraction, runs, and reporting foundations

Dates below are UTC calendar dates in October 2026 and identify recorded work/rounds, not continuous labor. A review credit names what the reviewer caught or verified; the corresponding author receives implementation credit. Report paths identify the evidence and its versioned qualifications. [R26]

Worker / harness / model	Recorded scope	Contribution, findings and evidence
alpha · Codex; model unrecorded	21 Oct; smoke series 5	Executed five live smoke runs at 5998c90 and recorded accounting/audits. The planned pilot was held, then reassigned to zeta; no alpha pilot result is credited. Evidence: `reports/alpha-live-runs.md`; `reports/alpha-pilot-section.md`
beta · Codex; model unrecorded	21 Oct; literature intake	Prepared intake for 53 bibliography entries; distinguished plain-text ingestion from full-text extraction. Flagged the committed-content rule tension and the lack of evidence for corpus neutrality. Evidence: `reports/beta-literature.md`
epsilon · Codex; model unrecorded	21 Oct; initial acquisition	Obtained 26 lawful PDFs from the 53-work list and recorded 27 access requests, versions, hashes and text-layer limits. No complete-access claim. Evidence: `reports/epsilon-pdfs.md`
gamma · Codex; model unrecorded	21 Oct; extraction rounds 1–6	Built per-paper extraction, rich graphs, bounded corrections, reconciliation and batched T construction. Changed the small-summary design after review; recorded Sonnet richness failures and repaired D4-B1 gap loss and D5-B1 continuation bounds. Final integration ba3a004. Evidence: `reports/gamma-extraction.md`; `reports/gamma-round6-report-draft.md`
delta · Codex; model unrecorded	21 Oct; initial review + re-reviews 1–6	Reviewed gamma. Identified missing assembled-context protection, cached lineage loss, the inadequate four-item cap, long-source/large-paper failures, reconciliation gap loss (D4-B1), and absent per-paper continuation bound (D5-B1). Cleared the final scoped repair at ba3a004. Evidence: `reports/delta-review.md`; `reports/delta-rereview6-section.md`
theta · Codex; model unrecorded	21 Oct; rounds 1–3	Repaired invalid context requests, added disclosed per-paper failure outcomes and completed-paper resume. Fixed iota T-B1 policy-sensitive reuse and T-S1; final be7468d. Evidence: `reports/theta-context-fix.md`
eta · Codex; model unrecorded	21 Oct; shared-T rounds 1–2	Implemented verified reuse of a single frozen T through --from-frozen, with source integrity and provenance checks. Rebased on theta and fixed E-B1/E-S1–3; final 185a318. Evidence: `reports/eta-shared-frozen-t.md`
iota · Codex; model unrecorded	21 Oct; theta/eta review and two re-reviews	Reproduced changed-policy resume bypass (T-B1), shared-T compatibility/provenance defects (E-B1/E-S1–3), and combined-suite failures. Rechecked both repaired branches before integration. Evidence: `reports/iota-review.md`; `reports/iota-theta-rereview-section.md`; `reports/iota-eta-rereview-section.md`
zeta · Codex; model unrecorded	21–29 Oct; full26, subscription/ Codex pilots, forks 1–6, gap1/r/f; A/B assigned	Operated extraction, pilots, frozen forks, route summaries and writing attempts; preserved failure/accounting receipts and performed source/substantive review. Corrected the coordinator's 70–90-token claim to about 233. Distinguished trivial/single-paper/padded results from contributions; documented gap1's ID stop, gap1r's 160-item freeze and admission stop, and gap1f's non-launch. gap1pu outcomes remain uncredited here. Evidence: `reports/zeta-full26-part7.md`; `reports/zeta-fork6.md`; `reports/zeta-gap1r.md`; `reports/overnight/gap1f/handoff.md`; `briefs/zeta-gap-fork2.md`
kappa · Codex; model unrecorded	21–22 Oct; acceptance rounds 1–3	Built real-response replay and extraction acceptance repairs, provider-schema preflight and bounded corrections. The live probe reached four of six papers, not full acceptance. Fixed lambda F1 diagnostic echo and F2 null-inventory recovery at 0eb8592; retained the deferred qualification. Evidence: `reports/kappa-acceptance.md`
lambda · Codex; model unrecorded	22 Oct; review + re-review	Caught request-content echo in diagnostics (F1) and null-inventory correction failure (F2), despite passing tests. Checked v4 reuse, required re-extraction of an incompatible paper, and confirmed the narrow repairs. Evidence: `reports/lambda-review.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
mu · Codex; model unrecorded	22–23 Oct; rounds 1–7	Built range-union salvage, grounding/quarantine and monotone publication; disclosed placeholder output rather than calling it faithful extraction. Fixed snapshot/gap loss and pi's prerequisite defect, projected construction metadata, accelerated imports at 1d832b2, and added startup verification caching at cdc9753. Evidence: `reports/mu-four-papers.md`
omicron · Codex; model unrecorded	22 Oct; mu review + re-review	Caught accepted-inventory loss (B1), undisclosed legacy gaps (B2), and shared-T lineage incompatibility (B3). Reviewed mu/xi/eta integration and the repaired mu/xi rebase. Evidence: `reports/omicron-review.md`
pi · Codex; model unrecorded	22 Oct; mu round-4 review	Found union isolation omitted applies_to prerequisites (B1). Checked that the current published corpus had no affected instance; that bounded result supported the disclosed temporary integration, while mu repaired the general defect. Evidence: `reports/pi-review.md`
rho · Codex; model unrecorded	23 Oct; mu rounds 5–6 review	Verified prerequisite handling, metadata projection and import optimization. Caught a verification mismatch: the recount used an empty audience instead of the CLI default (S1). Required the pilot command to match the measured plan. Evidence: `reports/rho-review.md`
nu · Codex; model unrecorded	22 Oct; xi report/log review + re-review	Caught full-graph snapshot growth and repeated chain verification (B1), provenance defects (B2), renderer crashes (B3), and a remaining report failure (R1). Drove hashed sidecar delta tracing and honest legacy-ancestry disclosure. Evidence: `reports/nu-review.md`; `reports/nu-rereview-draft.md`

B.3 Construction, providers, search, and their reviewers

Worker / harness / model	Recorded scope	Contribution, findings and evidence
sigma · Codex; model unrecorded	23–29 Oct; construction rounds 1–14	Implemented bounded closure, global borrowing/soft shares, disclosed window skips and construction resume; diagnosed insufficient credit. Added linear formula validation, structured/stance-aware T, cross-paper alignment, ID-free replies and resume-alignment (r12, 4cefe35), then gap packing and explicit window forwarding (r13, c99c6b3). At the v4 checkpoint R14's S13-F1/F2 repair was in progress; §9.2 records its later closure. Evidence: `reports/sigma-bounded-closure.md`; `reports/sigma-round12.md`; `reports/sigma-round13.md`; `reports/sigma-round14.md`
tau · Codex; model unrecorded	23 Oct; bounded-closure review	Reviewed sigma's 5009af1 closure cuts, mandatory dependencies, omission attribution and global budget arithmetic. Confirmed the scoped implementation while retaining model-owned relevance and semantic limits; no blocking defect found. Evidence: `reports/tau-review.md`
upsilon · Codex; model unrecorded	23 Oct; sigma r2 review + r3 re-review	Found that nominally soft shares still vetoed closure admission and could starve windows (B1). Reproduced the repair at 1411e91 while checking global bounds and mandatory dependencies remained enforced. Evidence: `reports/upsilon-review.md`
phi · Codex; model unrecorded	23 Oct; sigma/xi r4 integration review	Verified construction-resume receipts/accounting and the compact report plus sidecars. Checked additive merge conflicts and offline resume; recommended integration without claiming a completed live pilot. Evidence: `reports/phi-review.md`
chi · Codex; model unrecorded	23 Oct; sigma r5 review	Reviewed count-endpoint refusal handling, explicit credit failure and plan-only resume at 0e0a7fc. Cleared the scoped code while preserving the operational credit block; did not infer a successful live recovery. Evidence: `reports/chi-review.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
omega · Codex; model unrecorded	23–29 Oct; provider rounds 1–16	Implemented isolated Claude Code then Codex CLI transports, canaries, credential refresh, safe diagnostics, calibrated counting, local usage caps and strict-schema projection. Fixed psi/kappa2 provider findings. Later switched the configured default to codex_cli and tested new-paper extraction; r16 f7067a1 still failed both richness/structure trials under the repair-fraction gate. Evidence: `reports/omega-subagent-provider.md`
psi · Codex; model unrecorded	23 Oct; review + provider follow-ups/finals	Caught nondeterministic hashed-event ordering (F1), escaped-output leakage (F6), retry bypass (F7), credential transport exposure (F8), aggregate reservation misused as request size (F11), and lost correction-stage model override (F12). Rechecked the integrated repairs before the subscription pilot. Evidence: `reports/psi-review.md`
kappa2 · Codex; model unrecorded	24 Oct; omega/xi review + re-review	Found that sanitized diagnostic free text still retained synthetic credential values and short prompt echoes (F1). Reviewed failure continuation and stable publication order, then rechecked allowlisted diagnostics and integration f9c5a8f. Evidence: `reports/kappa2-review.md`
lambda2 · Codex; model unrecorded	24 Oct; auth/integration review	Checked per-call credential refresh, one bounded re-read retry, sanitization and accounting at omega 9a8e51a / integration 50ddc05. Found no blocking defect within the stated transport threat model. Evidence: `reports/lambda2-review.md`
rho2 · Codex; model unrecorded	25–26 Oct; OCR rounds 1–4	Integrated local PP-OCRv6_tiny, then PaddleOCR-VL-1.6 and span-level repairs. Reported 131/677 repaired spans and nine fully clean pages out of 82; did not claim complete recovery. Fixed tau2's confidence-dilution finding at 9aad013. These OCR models are task dependencies, not rho2's worker identity. Evidence: `reports/rho2-ocr.md`
tau2 · Codex; model unrecorded	26 Oct; provider/OCR/schema/resume reviews	Found OCR confidence dilution (F1) and stale awaiting_selection state that could suppress resumed search results (F2). Reviewed Codex isolation, strict-schema projection, linear formula validation, search continuation and their repairs; distinguished the actual pilot state from the general F2 defect. Evidence: `reports/tau2-review.md`
xi · Codex; model unrecorded	22–29 Oct; report/search/integration line, rounds 1–29; r30 assigned	Built ara report, sidecar traces, parallel search and resume-search; diagnosed goal/formula drift and atomic T. Implemented target/stance/novelty gates, CQ retrieval, semantic gap discovery and forward mining (r25–r29), integrating sigma. At 0c4d472 fixed R28-F1 Boolean padding, R28-F2 inconsistent-union suppression and S12-F1, plus solver lifetime. Delivery-branch r30 is an assignment, not a completed result here. Evidence: `reports/xi-report-tool.md`; `reports/xi-search-diagnosis.md`; `reports/xi-round25.md`; `reports/xi-round29.md`; `correspondence/log.txt`
upsilon2 · Codex; model unrecorded	26 Oct; structured-T/search integration review	Reviewed sigma r8 and xi r12; resolved the scratch integration at 6511ff6 and verified a native placeholder route closes while a flat negative stays open. Disclosed three merge conflicts and the need for a fresh live schema preflight. Evidence: `reports/upsilon2-review.md`
phi2 · Codex; model unrecorded	27 Oct; xi r13–r15 reviews	Reproduced single-commitment/tautology padding (B1) and attributed-report admission (B2), then checked their repairs. Found the single-support disclosure report crash (R1), later cleared at f89cd37. Evidence: `reports/phi2-review.md`
kappa3 · Codex; model unrecorded	27 Oct; sigma r9 / xi r16 integration review	Verified content definitions separated from attribution, stance-bound provenance and own-thesis/content classification. Produced the reviewed integration b48d56f; retained the distinction between offline validation and a useful real-corpus pilot. Evidence: `reports/kappa3-review.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
rho3 · Codex; model unrecorded	27 Oct; sigma r10 / xi r17 integration review	Reviewed per-window construction correction and target-pool changes, exercised native defeasible closure, and committed scratch integration 7c7bdc8 with additive conflicts resolved. No blocking defect found in that scope; no live quality guarantee. Evidence: `reports/rho3-review.md`

B.4 Documentation, publication, literature, and methodology

Worker / harness / model	Recorded scope	Contribution, findings and evidence
omicron3 · Codex; model unrecorded	27 Oct; documentation cleanup + rebase	Reduced the README and organized concern guides, corrected stale/contradictory descriptions, preserved decision history and applied authorized identity redaction. Integrated xi's current fork/target guidance at 5b512de without executable changes. Evidence: `reports/omicron3-docs.md`
chi2 · Codex; model unrecorded	27 Oct; documentation review + re-review	Caught documentation integration conflicts and stale selection/fork guidance (F1). Checked command/link coverage, retained history and anonymity, then cleared the rebased documentation in the recorded re-review. Evidence: `reports/chi2-docs-review.md`
kappa4 · Codex; model unrecorded	27–29 Oct; xi r18–r29, sigma11–13, writer/publication reviews	Independently challenged novelty, stance, isolation and publication gates. Findings include R22-F1, R23-F1, R25-F1/F2, W1, S11-F1–3, INTEG-F1, R28-F1/F2, S12-F1, NP-F1–3, R2-F1/F2 and S13-F1/F2. Recorded closures separately; the later xi29 review rejects both exact gap1r padding leads. §9.2 carries the reviewed dispositions; Appendix B.4 dates the earlier credit snapshot. Evidence: `reports/kappa4-review.md`; `reports/kappa4-writer-review.md`; `reports/kappa4-sigma11-review.md`; `reports/kappa4-r2-review.md`; `reports/kappa4-sigma13-review.md`; `reports/kappa4-xi29-review.md`
nu4 · Codex; model unrecorded	28–29 Oct; writer rounds 1–6; r7 in progress	Built sectioned composition and targeted revision, repaired sentence splitting and imported-draft screening (W1), and added typed audit roles. Fixed the exact web_search Other event in r5 (0368a17), exposed 44→0 flag variation, and implemented k=3 fresh union audits in r6 (d9ac619). Disclosed two citation-free exports; R7 citation/adverse-history repair remains uncredited as complete. Evidence: `reports/nu4-writer.md`; `briefs/nu4-round-7.md`
pi3 · Codex; model unrecorded	28–29 Oct; publication + quality passes 1–5	Built isolated Opus polish, typed post-polish audit/repairs, deterministic argument figures and native PDF rendering. Delivered 0aa6a29 and later quality passes; fixed NP-F1–3 at a6a2f9b. Q5 b6fc454 adds bounded structural retries and sampled publication audits, at the v4 checkpoint final R7 integration/live polish remained pending; a failed polish fallback is not a successful edit. Evidence: `reports/pi3-publication.md`; `reports/pi3-quality3.md`; `reports/pi3-quality5.md`; `reports/kappa4-r2-review.md`
eps2 · Codex; model unrecorded	28–29 Oct; modern literature, L3 and novelty dossier	Acquired lawful bridge/modern full texts, checked older open-problem claims against 2018–2026 literature and qualified L3 as partially anticipated. The later dossier maps nine analyst-generated theses: four anticipated, four partial, one not found within scope; seven further lawful full texts added. This is literature analysis, not an ARA-derived novel result. Evidence: `reports/eps2-bridge-literature.md`; `reports/eps2-L3.md`; `reports/eps2-methodology-supplement.md`; `reports/eps2-L3-novelty.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
meta · Codex; model unrecorded	29 Oct; metapaper v1–v4	Assembled the methodology, human-action ledger, source-linked numerical data, review dispositions and deterministic figures; rendered PDFs and checked accounting, anonymity and layout. V3 preserves the evidence boundary; v4 adds this complete named-worker credit register. Evidence: `briefs/meta-paper.md`; `briefs/meta-credits.md`; `reports/papers/metapaper/v3/verification.json`

Later receipts, dated separately from the research checkpoint. The **20:42:29 UTC** correspondence entry records kappa4's xi29 review and integration of `0c4d472`. Its report closes **R28-F1, R28-F2 and S12-F1** and rejects the exact gap1r padding leads in offline reproduction. This is a later software-gate result; it does not rerun gap1r or create a substantive route. [R1, R27]

At **20:55:52 UTC**, eps2's L3 dossier superseded the earlier "dossier pending" label: **nine analyst-generated candidate theses**, **four anticipated, four partially anticipated, one not found within scope**, plus **seven** further lawful full texts. The not-found thesis is the exact necessity-and-sufficiency claim linking safety failure to obligatory retraction. "Not found" does not establish novelty, viability, priority or a successful ARA derivation. The dossier also reports that both previously inaccessible modern comparison papers were now obtained and inspected; other access limits remain. Its thesis map must not be presented as targets that zeta generated. [R1, R27]

At the v4 credit checkpoint, sigma14, nu4 R7, A/B/C and delivery integration were still pending. Their later completed receipts are now reported under construction/context (§§2–3), search (§4), writing (§7) and review (§9.2). Those historical pending entries are superseded by the executed and explicitly pending results in §§2–7 of this revision. This preserves the difference between an assignment and an executed result. [R26–R43]

B.5 Roster reconciliation and limits

The original 39-worker roster was reconciled against the correspondence prefix and Markdown under both `briefs/` and `reports/`, including nested evidence/handoff directories. Generated metapaper copies were excluded to avoid counting this table as evidence for itself. Large native-run transcripts were hash-inventoried but excluded from worker-name discovery; model-generated text is not worker-authorship evidence. `data.json/credits` preserves the source inventory, hashes, candidate mentions, per-worker references and harness attribution limits. Review-only workers and earlier retired workers receive their own rows. This is a record-based attribution, not a recovered total of their tokens, elapsed work or individual philosophical authorship. [R26]

Two filenames would otherwise double-count workers: `briefs/omicron2-review.md` explicitly names **kappa2** in its title/body and required completion line; `briefs/sigma2-review.md` explicitly names **kappa4**. They are aliases in the record, not evidence of additional omicron2 or sigma2 contexts. Strings such as mu4/mu6, theta2/theta3, sigma5/sigma11/sigma12/sigma13 and xi29 refer to rounds; `eps*` is a family shorthand in the request. The distinct recorded literature workers are **epsilon** and **eps2**. No independent nu2, nu3 or pi2 context was found in the inspected records, so none is invented. [R26]

B.6 Current responsibilities, new worker, and retirement

Worker / harness / model	Recorded scope	Contribution, findings and evidence
fwd · Codex; model unrecorded	30 Oct; native forward sweep	Operator-approved sweep of cross-paper premise combinations on gap1r's authenticated freeze at `f76c1b3`; preserve licenses, SAT, stance and anti-padding gates; send qualifying routes for writing and later eps2 literature checking. Sweep completion not credited; the bounded census and subsequent stop request are recorded in §5.3. Evidence: `briefs/fwd-sweep.md`; `reports/fwd-sweep.md`; correspondence.

Omega retired at 06:27:19 UTC after the recorded provider work. **Kappa4** handed off at 19:46:59 and retired after **kappa5** accepted review ownership at 19:47:48; **xi** handed off at 20:06:45 and retired after **xi2** accepted at 20:07:37. Fresh contexts did not erase earlier findings or inherit unverified completion claims. **Sigma** owns construction/consolidation and gap3v; **zeta** owns gap3 continuations and source/route review; **fwd** owns gap4a and forward diagnostics; **nu4** owns writer/carry-over; **pi3** owns publication/inline repair. Meta writes this report without dispatching native generation or selecting a candidate. [R1, R49–R64]

Worker / harness / model	Dates / scope	Concrete contribution and evidence
janitor · Codex; model unrecorded	30 Oct; approved storage tasks	Preserved evidence manifests, archived and download-verified eligible run records; screened hits with no values exposed. Reports `janitor-evidence.md`, `janitor-records.md`, `janitor-secret-triage.md`. Operational cleanup does not establish research correctness.
kappa5 · Codex; model unrecorded	30 Oct; successor non-author reviewer	Caught PACK-F3's unrelated-failure control and N4-F1's exhausted-vote control; reviewed search-pause/full-basis integration and audit-inline boundaries. Reports `kappa5-packing-review.md`, `kappa5-nu4-audit-inline-review.md`, `kappa5-inline-combined-review.md`.
xi2 · Codex; model unrecorded	30 Oct; successor integration/search	Finished mixed-executor packing controls, retracted the unsupported pass, repaired the intended stance test and merged approved packing/pause/reader changes. Evidence `xi-packing2.md`, `overnight/gap3p/search-ready-checkpoint.md`.

The roster is **43 workers plus one orchestrator**. The credits describe recorded work, not an equal contribution claim or a complete per-worker serving-model log. Earlier exact model pins remain unrecorded unless an actual receipt supplies them. The orchestrator is recorded as a Claude model; under the current provider policy, Opus generation is restricted to the separately disclosed polish path. Earlier API extraction use is disclosed in §8.1. New requested `gpt-6-luna` meaning-probe usage is a runtime-stage receipt, not proof of every worker's model. [R57]

Contribution cluster	Recorded identifiers
Sources / extraction	alpha, beta, epsilon, gamma, delta, theta, eta, iota, mu
Pilots / source review	zeta, kappa, lambda, omicron, rho, tau, upsilon
Construction / search	sigma, xi, xi2, omega, fwd; phi, psi, chi
Integration / independent review	kappa2, kappa3, kappa4, kappa5; rho2, rho3, tau2, upsilon2, phi2, omicron3, chi2, lambda2
Writing / publication / literature	nu, nu4, pi, pi3, eps2, meta
Operations / direction	janitor; orchestrator; anonymised human operator

Figure 20. Compact role map; Appendix B retains individual histories and evidence paths for all 43 workers plus the orchestrator. Agent identifiers are credits, not human identities. Model/version uncertainty is retained.

Appendix C. Complete recorded human-action entries

Timestamps are UTC. Entries are grouped exactly by correspondence record; one entry may contain several instructions or actions. H01–H03 are indirect reports of operator actions. Summaries remove identity and credential locations while preserving the decision. The corresponding anonymised record text and original line number are in `data.json`.

ID / UTC timestamp	Recorded human action
H01 · 2026-10-21 00:02:03	Approved the initial coding workers and spending: five smoke runs capped at $5 each and one pilot capped at $100. This approval is reported by the coordinator.
H02 · 2026-10-21 00:12:18	Chose per-paper logged extraction; the existing pilot was held pending that capability.
H03 · 2026-10-21 00:48:04	Asked the team to collect PDFs; the operational brief restricted collection to lawful open access.
H04 · 2026-10-21 02:03:18	Chose rich extraction rather than the compressed-summary option and said they were obtaining remaining papers.
H05 · 2026-10-21 03:38:41	Approved one additional build-T attempt by increasing its cap from $4 to $6.
H06 · 2026-10-21 03:40:20	Directed extraction to use a cheaper model and asked the team to get through to the pilot.
H07 · 2026-10-21 18:21:02	Pulled and pushed the main branch to the remote repository.
H08 · 2026-10-21 18:25:51	Approved an approximately $600–700 all-in budget, a shared frozen T across candidates, and the start of extraction/pilot work.
H09 · 2026-10-22 03:32:13	Asked for immediate repair of the four-paper failure policy; requested looser workable restrictions and OCR if needed.
H10 · 2026-10-22 05:33:48	Requested a full overnight pass, a $200 cap, and a detailed morning report with search-tree traces; authorized workers and method iteration. The coordinator interpreted the cap as new spend within the earlier campaign budget.
H11 · 2026-10-23 13:17:07	Requested an agent/connector route, a fresh T rebuild, parallel work, and iteration until an acceptable pilot.
H12 · 2026-10-24 15:28:25	Clarified that “use chat” meant ChatGPT/Codex and prohibited Claude-subscription pilots. The active pilot was halted.
H13 · 2026-10-25 22:26:45	Requested integration of the new open-source Baidu OCR family in parallel for future content recovery.
H14 · 2026-10-25 22:53:20	Completed a separate ChatGPT/Codex login. No credential value or identifying location is reproduced.
H15 · 2026-10-27 02:06:18	Requested plain-language documentation and repository organization using an existing internal methodology; the resulting assignment was documentation-only.
H16 · 2026-10-27 15:15:37	Saved a repository access token. The coordinator then pushed and started periodic pushes; no credential value is reproduced.
H17 · 2026-10-27 15:38:31	Approved the complete pilot and entry criteria, including 20/26-paper coverage, three qualifying derived routes, a 4,500–6,000-word audited essay, an anonymised methodology, and the October 31 AoE deadline. Set October 29/30 checkpoints.

ID / UTC timestamp	Recorded human action
H18 · 2026-10-28 16:52:30	Directed “Write C,” asked for at least one resolvable literature gap, and asked to build all routes. The coordinator recorded selection of all available CLOSED-derived routes for writing.
H19 · 2026-10-28 20:59:16	Requested a Claude Opus final pass producing professional papers, visualizations, Markdown, then PDF.
H20 · 2026-10-28 21:02:01	Questioned the lack of novel conclusions; requested bridge-paper discovery/ingestion, creative investigation, consideration of architectural changes, and examination of contradictions.
H21 · 2026-10-28 21:03:17	Required explicit disclosure of missing modern literature, acquisition of that literature, and accurate claims about open questions.
H22 · 2026-10-28 21:03:45	Directed the team to obtain the full texts and put them in the full-text collection.
H23 · 2026-10-28 21:06:14	Requested maximum parallelization of the ongoing work.
H24 · 2026-10-28 21:14:27	Prohibited further credit-based API work and authorized normal agent calls as the default; the coordinator retained Claude only for the separately requested Opus polish.
H25 · 2026-10-29 06:17:05	Gave GO for the safety–retraction gap paper and this visual metapaper; approved “not found within stated search scope, partially anticipated,” directed best effort from available sources, and requested an output repository.
H26 · 2026-10-29 06:20:26	Asked to make the scaffold repository private and place outputs in a folder for later deletion. The visibility API request was refused with HTTP 403 for insufficient administration permission; that action did not change visibility; H28 records the later successful change.
H27 · 2026-10-29 19:09:07	Authorized loosening the token limit. The later adapter change forwards an explicit provider window; raising native admission alone was insufficient.
H28 · 2026-10-29 20:04:19	Reported the repository private; the coordinator verified visibility through the API, copied outputs, and pushed them. This supersedes the earlier failed privacy request.
H29 · 2026-10-29 20:41:14	Questioned idle completed agents and context limits, and asked whether the work was pursuing real novelty. The coordinator requested an L3 novelty dossier, delivery integration, and this report revision.
H30 · 2026-10-29 20:56:46	Requested a repository push. The coordinator recorded main and four work-in-progress branches as pushed; this was a publication action, not a research-result approval.
H31 · 2026-10-29 21:09:25	Requested named credits for every worker in the metapaper. Worker identifiers refer to agents; the human remains the operator.
H32 · 2026-10-30 02:09:23	Authorized drafting every available candidate as soon as it is ready; final submission selection remains with the operator. The coordinator assigned native writing/audit to nu4, polish to pi3, and the separate fallback analysis to meta.
H33 · 2026-10-30 05:47:40	Reported apparently empty pilot process windows. The coordinator explained that ARA logs to files, closed completed A/B windows after retaining results, and confirmed gap3 remained live.
H34 · 2026-10-30 05:51:57	Approved construction improvements while requiring the current run to stay on schedule; the coordinator assigned default-off repairs without touching gap3.
H35 · 2026-10-30 06:22:01	Requested parallel work and the maximum number of papers, and expressed that one output seemed novel. The coordinator distinguished near-T2 proposals from derived results and assigned four proposal-mode candidates plus safe parallel construction work.
H36 · 2026-10-30 06:27:19	Authorized proposal papers and hypothesis-premise work in parallel without stopping ongoing work; asked whether premise combinations and routes were being used to find literature gaps. The coordinator disclosed bounded earlier forward coverage, retired idle omega and assigned the native fwd sweep.

ID / UTC timestamp	Recorded human action
H37 · 2026-10-30 06:28:39	Approved the proposed hypothesis-premise extension; the coordinator assigned xi round32.
H38 · 2026-10-30 06:28:53	Requested an immediate metapaper update unified under ARA base concepts.
H39 · 2026-10-30 08:14:07	Authorized autonomous progress while away and requested skepticism that produces results; no permission to overstate outcomes.
H40 · 2026-10-30 08:15:11	Required the orchestrator to limit its own work; source investigation to owners and merges after independent review.
H41 · 2026-10-30 08:28:44	Restricted cleanup to unnecessarily stored material saved in version control, then directed no changes for space; coordinator stopped cleanup.
H42 · 2026-10-30 08:37:51	Authorized an additional cleanup worker for inactive, restorable checkouts while retaining run/cache evidence.
H43 · 2026-10-30 08:45:03	Asked for reasonable cleanup restrictions; coordinator briefly simplified them.
H44 · 2026-10-30 08:45:45	Clarified that cleanup should stay strict and proposal-publication requirements should be reasonable; four substantive disclosure/source rules retained.
H45 · 2026-10-30 15:50:34	Asked whether vocabulary repair applied to gap3 and whether papers had arrived; coordinator acknowledged no new papers and a prolonged stall, then assigned consolidation.
H46 · 2026-10-30 16:07:08	Approved the canary list-format reporting change under the publication worker’s conditions.
H47 · 2026-10-30 17:07:54	Authorized cleanup of already-recorded old agent records through the cleanup worker, excluding active sessions and retaining manifests.
H48 · 2026-10-30 17:22:10	Allowed two session-infrastructure processes to be excluded from the cleanup attribution blocker.
H49 · 2026-10-30 17:38:27	Requested relevant evidence be saved remotely before local cleanup; coordinator required private release upload/download hash verification and scanning.
H50 · 2026-10-30 18:41:32	Authorized each worker to clean only its own finished shared-memory scratch, with no active users or other workers’ objects touched.
H51 · 2026-10-30 19:29:05	Requested immediate PDFs of the verified B/2 and T2 originals and stopped polish rounds.
H52 · 2026-10-30 19:36:47	Requested cleanup and numbered reorganization of output folders.
H53 · 2026-10-30 19:41:33	Approved fresh successor worker contexts kappa5 and xi2 after handoffs.
H54 · 2026-10-30 19:52:39	Required pause and diagnosis after approximately ten empty search jobs: inspect target bases, shared symbols and responses/rejections before continuing.
H55 · 2026-10-30 19:54:00	Asked why Opus polish failed; coordinator requested failure classification rather than treating all failures alike.
H56 · 2026-10-30 20:28:07	Directed prefix-only carry-over, auditing/inline repair of unsupported elaborations, and exact title formatting plus re-verification for T2.
H57 · 2026-10-30 20:29:47	Required wording changes to be audited but accepted when not meaningfully different; coordinator assigned three-sample unanimous equivalence.

ID / UTC timestamp	Recorded human action
H58 · 2026-10-30 22:25:53	Requested a detailed v6 metapaper centered on core concepts, new mechanisms, routes and updated visualizations; also asked which candidate was promising to await.
H59 · 2026-10-30 22:28:10	Withdrew the requested discovery-framing citation and asked to combine gap3 sources with T2; conditional-derivation paper if licensed, source-grounded proposal fallback otherwise.

Earlier human actions without a timestamp in this log. The human designed and directed the scaffold, chose the topic, and supplied/committed the initial bibliography and filing labels. The intake review found no added argument annotations, but flagged an unresolved literal-rule tension: bibliographic titles and human-provided grouping labels can name positions despite the repository’s blanket prohibition on committed philosophical content. Four position-related filing labels were not four selected theses. The collection’s claimed completeness and neutrality were not independently verified. No selection policy is promoted here into proof that the corpus was balanced. [R2]

Earlier v5 direction without a separate timestamped operator entry. The operator directed meta to make the measured 141/160 atomic commitments, 2/147 cross-paper symbols and 150-pair alignment the central construction/forward-search finding, add a deterministic two-panel figure, and mark sigma r17's shared-catalogue/wider-alignment repair pending. The correspondence records the coordinator's diagnosis and dispatch at 06:46:15–06:46:47; it does not timestamp this operator message separately. This direction is disclosed without inventing an additional timed human-action row.

V7 editorial direction. The operator requested a one-page executive summary, a shorter concept-based main text and preservation of the full evidence in appendices. This is an editorial action after the retained research cutoff, not a new run, result or timestamped research-ledger entry.

Appendix D. Evidence register and reproducibility

Each citation identifies a source snapshot, not a live guarantee that the file has not changed. `data.json/v6_source_snapshots` records this revision's new hashes and collection times; older hash sets retain their original scopes. Numerical statements are indexed in `data.json/numerical_claim_index`. Pending statuses are anchored to the stated times, even if work continues during rendering.

ID	Primary sources and responsibility
R1	`correspondence/log.txt`, frozen through 2026-10-30 22:28:10 UTC: decisions, iterations, human actions, and timing.
R2	`briefs/beta-literature.md`, `reports/beta-literature.md`, `reports/epsilon-pdfs.md`: supplied bibliography, intake scope, rule issues, initial access.
R3	Scaffold `CLAUDE.md`, `PLAN.md`, `SOURCE_MAP.md`, `DECISIONS.md`, and `docs/`: architecture, boundaries, lifecycle, policies and their limitations.
R4	`reports/overnight/fork6/{data,accounting}.json`, its content-addressed event sidecars, and `reports/zeta-fork6.md`: deduplicated accounting, gap/search outcomes and report defects.
R5	`reports/eps2-bridge-literature.md`, `eps2-L3.md`, `eps2-methodology-supplement.md` and their JSON twins: initial sweep, focused currency check, cumulative coverage.
R6	`reports/zeta-codex-pilot-5.md` and `reports/overnight/codex5/`: frozen structure, stance, symbol reuse, targets and routes.
R7	`reports/nu4-writer.md`, round-2 and round-4 native receipts, plus `zeta-writing-test.md` and `zeta-writing-route2.md`: length repair, audit variation, blocked retrieval and export.
R8	`reports/pi3-publication.md` and `pi3/runs/pi3-proof/live-result.json`: preserved input, polish, native audit, usage and rendering evidence.
R9	`reports/zeta-fork1.md` through `zeta-fork6.md`, native target-gate receipts and `kappa4-review.md`: rejection mechanisms, counterexamples, false positives and substantive route review.
R10	`reports/zeta-codex-pilot-2-resumed.md`, `zeta-codex-pilot-3.md`, `xi-search-diagnosis.md`: atomic T, conclusion mismatch and trivial closure.
R11	`briefs/xi-round-27.md`, `reports/xi-round27.md`: per-paper novelty correction, superseded census and final bounded zero-candidate census.
R12	`reports/zeta-full26-part7.md`, `gamma-extraction.md`, `mu-four-papers.md`, `sigma-bounded-closure.md`: extraction inventory, quarantine, retention and construction repairs.
R13	`reports/omega-subagent-provider.md`, provider review reports and scaffold `docs/providers.md`: API/CLI versions, isolation scope, calibration and diagnostics.
R14	`reports/zeta-conflicts.md`: bounded semantic-conflict pass and source/currency review.
R15	`briefs/` and run MODEL receipts; exact instruction variants in the prompt supplement: task direction, stage instructions, correction history and prompt identities. The instruction supplement remains the version-1 inspected scope.

ID	Primary sources and responsibility
R16	`reports/overnight/gap1/handoff.md`, `reports/zeta-gap1.md`, `pilot-exit.json`, `final-local-accounting.json`, and the preserved integration review: failed alignment, construction inventory, source limits, timing and fresh usage.
R17	`briefs/sigma-round-12.md` and `briefs/nu4-round-5.md`: historical repair requests, pending at v2; later results are in R18 and R20.
R18	`reports/overnight/gap1r/handoff.md`, `reports/zeta-gap1r.md`, `reports/overnight/gap1r/final-local-accounting.json`, `reports/overnight/gap1r/pilot-exit.json`: Gap1r recovery, frozen inventory, source/padding review, admission refusal, timing and fresh accounting.
R19	`reports/overnight/gap1f/handoff.md`, `reports/sigma-round13.md`, `briefs/zeta-gap-fork2.md`: Gap1f client-window inspection, sigma r13 offline plans/window plumbing, historical packed/unpacked fork brief; completed results in R28.
R20	`reports/nu4-writer.md`, `briefs/nu4-round-7.md`: Nu4 R5–R6 native exports, audit dissent and union batches, citation loss; historical R7 repair brief, completed outcome in R29.
R21	`reports/kappa4-sigma12-review.md`: Kappa4 s12: ID-free response and recovery review, integration fixture blocker.
R22	`reports/kappa4-nu4pi3-review.md`: Kappa4 np: publication screening/title/occurrence blockers, audit and length-policy notes.
R23	`reports/kappa4-r2-review.md`: Kappa4 r2: Pi closure evidence, incomplete adverse-history and citation-stripping blockers.
R24	`reports/kappa4-sigma13-review.md`: Kappa4 s13: stage-window admission mismatch, merged mutation-anchor failure, S12-F1 closure.
R25	`briefs/eps2-L3-novelty.md`: L3 novelty-dossier brief; historical request; completed bounded assessment in R27.
R26	`briefs/meta-credits.md`; per-worker report paths in Appendix B; full inventory and read-only harness roster in `data.json/credits`: complete credits, aliases and model-attribution limits.
R27	`reports/kappa4-xi29-review.md`, `reports/eps2-L3-novelty.md`, `reports/sigma-round14.md`, `reports/pi3-quality5.md`: later completed receipts versus in-progress work, individually qualified in Appendix B (historical rows).
R28	Completed A/B/C fork handoffs: outcomes, source limits, context and fresh accounting. Sources: `reports/overnight/gap1pu/handoff.md`, `reports/overnight/gap1pu/A-handoff.md`, `reports/overnight/gap1pu/B-handoff.md`, `reports/overnight/gap1pu/C-handoff.md`
R29	Nu4 writer report, Round 7: citation/history repair, audit refusals and corrected word-count scope. Sources: `reports/nu4-writer.md`
R30	Kappa4 sigma14 review: S13-F1/F2 closure on the pinned head. Sources: `reports/kappa4-sigma14-review.md`
R31	Kappa4 delivery review, xi round30 and pi3 quality5: R2 closures and DLV-F1 closure boundary. Sources: `reports/kappa4-delivery-review.md`, `reports/xi-round30.md`, `reports/pi3-quality5.md`
R32	Gap2 live checkpoint and grounded-root review; sigma15 review was in progress at the earlier snapshot; R43 supersedes it. Sources: `reports/overnight/gap2/live-status.md`, `reports/overnight/gap2/source-anchor-review.md`, `reports/overnight/gap2/exact-root-prelaunch-check.json`, `reports/kappa4-sigma15-review.md`

ID	Primary sources and responsibility
R33	All-candidates brief and queue; fallback analysis source, status and verification. Sources: `briefs/all-candidates.md`, `reports/candidates/queue.jsonl`, `reports/candidates/gap-T2-analysis/paper.md`, `reports/candidates/gap-T2-analysis/status.json`, `reports/candidates/gap-T2-analysis/verification.json`
R34	`briefs/meta-v5.md`, `correspondence/log.txt`
R35	`reports/overnight/gap2/pre-freeze-check.md`, `reports/overnight/gap2/construction-outcome.md`, `reports/overnight/gap2/live-status.md`, `scaffold/runs/codex-gap-2-control/pre-freeze-operator-check.json`
R36	`reports/overnight/gap3/live-status.md`, `reports/overnight/gap3/work-notes.md`, `reports/overnight/gap3/substantive-construction-review.md`
R37	`briefs/sigma-round-16.md`
R38	`reports/overnight/gap1pu/l3-dossier-mapping.md`, `reports/overnight/gap1pu/l3-dossier-mapping.json`
R39	`briefs/fwd-sweep.md`
R40	`briefs/xi-round-32.md`
R41	`briefs/nu4-proposal-mode.md`
R42	`reports/candidates/gap1pu-B-sign/status.json`, `reports/candidates/gap1pu-B-sign-short/status.json`, `reports/candidates/gap-T2-analysis/status.json`
R43	`reports/kappa4-sigma15-review.md`, `reports/xi-round31.md`
R44	`reports/fwd-sweep.md`, `reports/fwd-sweep-wide/summary.json`: completed truncated census inside an unfinished sweep.
R45	`reports/sigma-round16.md`: historical focused checkpoint and completed combined r16/r17 verification at de4fff8; concurrent windows deferred.
R46	`scaffold/runs/codex-gap-1r/frozen-t.json`, `scaffold/runs/codex-gap-1r/commitment-provenance.json`: independently counted syntax, catalogue reuse and alignment; method in `data.json/vocabulary_bottleneck`.
R47	`briefs/sigma-round-17.md`; correspondence at 06:46:15–06:46:47: shared-vocabulary repair assignment and wider-sweep stop request.
R48	`reports/overnight/gap3/handoff.md`; `reports/overnight/gap3s/handoff.md`
R49	`reports/sigma-round17.md`; `reports/kappa4-sigma17-review.md`
R50	`reports/sigma-round18.md`; `reports/sigma-round19.md`
R51	`reports/sigma-round20.md`; `correspondence/log.txt` (r21 assignment at 21:05:16 UTC)
R52	`reports/xi-round32.md`; `reports/xi-round32b.md`; `reports/xi-round32c.md`; `reports/kappa4-xi32-review.md`
R53	`reports/overnight/gap3p/handoff.md`; `reports/overnight/gap3p/target-route-dossier-review.json`; `reports/overnight/gap3p/target-basis-retrieval-audit.json`
R54	`reports/xi-packing2.md`; `reports/kappa4-packing-review.md`; `reports/kappa5-packing-review.md`; `reports/overnight/gap3p/packing2-checkpoint.md`

ID	Primary sources and responsibility
R55	`reports/candidates/proposal-publication-handoff.md`
R56	`reports/candidates/pi3-original-delivery.md`; `reports/candidates/pi3-original-delivery.json`; `reports/candidates/gap1pu-B-2-proposal/status.json`; `reports/candidates/gap1pu-C-T2-iff-proposal/status.json`
R57	`reports/nu4-audit-inline.md`; `reports/kappa5-nu4-audit-inline-review.md`
R58	`reports/kappa5-inline-combined-review.md`; `reports/kappa5-pi3-inline-review.md`
R59	`reports/fwd-sweep.md`; `reports/fwd-sweep-wide/summary.json`
R60	`reports/gap4a.md`
R61	`reports/overnight/gap3h/gap3h-preflight-result.json`; `reports/overnight/gap3h/preparation.md`; completed `gap3h/handoff.md` and held combined-candidate status
R62	`reports/kappa4-handoff.md`; `reports/kappa5-packing-review.md`
R63	`reports/janitor-evidence.md`; `reports/janitor-secret-triage.md`
R64	`reports/overnight/gap3p/search-ready-checkpoint.md`
R65	`reports/papers/metapaper/v6/research/ai_scientist.response`; `reports/papers/metapaper/v6/research/storm.response`; `reports/papers/metapaper/v6/research/sciagents.response`
R66	`reports/candidates/pi3-audit-inline.md`; T2 `pi3/audit-inline-20261030T214052Z/final-diagnosis.json`: title-only continuation and final refusal.

Related systems (abstract-level comparison only): Lu et al., *The AI Scientist* (2024), arXiv:2408.06292v3; Shao et al., *Assisting in Writing Wikipedia-like Articles From Scratch with Large Language Models* (2024), arXiv:2402.14207v2; Ghafarollahi and Buehler, *SciAgents* (2024), arXiv:2409.05556v1. Retrieved metadata/abstracts and hashes accompany R65.

This report used no new ARA generation or Opus polish calls. Authoring-agent usage is outside the supplied native ledger. Deterministic SVGs are generated from recorded data; local Chromium renders the Markdown/HTML with browser networking blocked. Checks cover source and citation inventories, figure determinism, fonts, captions, page bounds, table rows, anonymity across delivered text and PDF metadata, and preservation of previous deliveries. These checks do not establish an audit pass or philosophical truth. V1–v6 remain historical deliveries; this v7 directory is the current report. Native prompt examples remain historical, not a full current replay package.

Editorial correction in v7. A v6 table-sorting operation treated the construction round label R21 as a source-reference row. It displaced the pending r21 row into the evidence register and R1 into the consolidation table. V7 restores the pending r21 row in Appendix A.4 and the unique R1–R66 register here. Neither underlying evidence nor research status changes. The v6 files remain preserved.

Finding placement. All 84 prior findings are retained, with both the concept summary and detailed incident destination where relevant. The editorial correction above is the additional finding. **85 findings, 85 placed.**

Finding	Report section
1. No substantive novel contribution or qualifying pilot	9.1
2. Historical R4 audit counts versus marker-free words	7.1, A.8
3. Gap1 correct ID refusal and missing continuation	2.4, A.3
4. Recovery and zero unresolved do not certify bridge semantics	2.4, 9.3, A.3
5. Source stance, citation spans, omissions and incomplete L3 anchors	2.2–2.6, A.2
6. SAT freeze excludes R28-F2 here; observed padding remains	4.2
7. Full target request refused despite smaller readiness check	3.2, A.5
8. Pinned window metadata and omitted adapter forwarding	3.2, A.5
9. Offline one-hop plan and actual two-hop fork A are separate evidence	3.3, A.5
10. Audit nondeterminism and k=3 union limits	7.2, A.8
11. Native exports lost all citations; R7 repaired citations but refused both exports	7.2–7.3, A.8
12. Incomplete adverse history can allow unchanged export	7.5, A.9
13. L3 partial anticipation, access limits and bounded completed dossier	5.1–5.2, A.7
14. Isolation and unverified serving identity	8.1, 9.3, A.11
15. Accounting and human record limits	1, 8, Appendix C
16. Target and native report losses	4.1, 8.4, A.6, A.11
17. Privacy failure superseded by later verified status	9.3, Appendix C
18. Intake ambiguity and unverified collection neutrality	9.3, Appendix C
19. S12-F1 fixture failure closed at later integrated head	9.2, A.12
20. NP-F1 imported publication screening closed for publication	9.2, A.12
21. NP-F2 title audit/cache omission closed for publication	9.2, A.12
22. NP-F3 duplicate occurrence permission closed for publication	9.2, A.12
23. S13-F2 stage-window over-admission closed at later reviewed head	3.2, 9.2, A.5, A.12
24. S13-F1 merged mutation control still closed at later reviewed head	9.2, A.12
25. Green scoped tests did not exclude review blockers	9.2, A.12
26. Separate length policies and source versions	7.2–7.4, A.8
27. Named-worker roster, aliases and uncertain model identities	B
28. Later xi29 review closes padding/union findings without a new pilot result	4.2, 9.2, A.12
29. Completed literature dossier is bounded analysis, not a native novel contribution	5.2, A.7

Finding	Report section
30. Current assignments and draft repair reports do not establish completed results	9.3, B.6
31. Completed A/B/C: 100 OPEN, one partial, zero derived/substantive	4.3
32. Same freeze, 159/160 stance failure and 21 skipped windows remain	2.4, 4.3, A.3
33. External T2 atoms are not decomposed or proven false	4.3, 5.2, A.7
34. Missing sufficiency and necessity/exhaustiveness premises	6.1
35. Actual context counts, pre-packing disagreement and experiment confounds	3.3, A.5
36. Conservative admission is not actual serving capacity; recovery disclosed	3.3, A.5
37. Fork-local usage excludes inherited/preflight costs; subscription dollars unknown	8.3, A.11
38. R7 restores citations but refuses both; final k=3 not reached	7.3, A.8
39. Native word counter excludes bibliography; two entries from one paper	7.3, A.8
40. S13/R2 closures and DLV-F1 boundary are exact-head offline evidence	7.5, 9.2, A.9, A.12
41. Gap2 stopped and sigma15 reviewed; plans/anchors are not results	2.5–2.6, 9.2, A.3, A.12
42. Fallback analysis is rendered, not native-audited/exported/polished	7.4
43. Zero substantive novelty; bounded dossier is weaker than priority proof	9.1
44. All-candidates authorization preserves operator selection and worker credit	7.4, B.6, Appendix C
45. Gap2 stopped before freeze: 3/13 accepted, 57 commitments/3 papers; no context refusals	2.5, A.3
46. Changed batch granularity/cache identity; no cut closure or plan exclusion	2.5, A.3
47. Gap2 gained atomic rule but lost safety; correction diagnostic still 94/95	2.5, A.3
48. Gap3 startup/allocation recovery and live counts are not completed coverage	2.6
49. Gap3 anchors atomic; stance and Mighty Failure paraphrase remain qualified	2.6
50. Sigma16 partial/split/attribution/parallel mechanisms pending, running gap3 protected	2.6, 9.3
51. ARA proposed B1/B2 near T2; exact T2 came from eps2 and mappings are partial	5.2, A.7
52. Forward coverage was bounded; wider native sweep pending	5.3
53. Hypothesis premise P is proposed support; conditional derivation would not prove P	6.2
54. Proposal-mode origin labels and audits remain required; four assignments not exports	6.3, 7.4
55. Sign audit refusal and short writer refusal; length policy was orchestrator choice	7.4, Appendix C
56. Sigma15 review and xi31 integrated verification completed but offline; fwd added and omega retired	9.2, B.6, A.12
57. Forward census saturates on Boolean enumeration; configured depth is not coverage; final sweep pending	5.3

Finding	Report section
58. Sigma16 author checkpoint: complexity guard repaired; final proof pending, concurrency deferred and retry coverage distinct	2.6
59. Central T bottleneck: 141/160 bare atoms, 19 syntactically structured formulas, 147 zero-arity symbols	2.3, 5.3
60. Two shared symbols arise only through one bridge; zero direct cross-paper reuse; 149/150 alignment pairs unrelated	2.3
61. Sparse vocabulary unifies bounded padding/OPEN outcomes; sigma17 pending, wider forward stop requested, no impossibility theorem	2.3, 5.3, 9.1
62. Gap3 retains all anchors but source formulas remain disconnected	2.6
63. R17 prompt delivered; gap4a completed construction still shares zero source symbols	2.7, A.4
64. Gap4a alignment disk pause and separate pinned search defect; no C2/C3	2.7, A.4
65. Consolidation preserves source identity but changes formula history in a new lifecycle	2.7, A.4
66. R19 runtime IDs work but r20 cardinality regression prevents freeze; r21 pending	2.7, A.4
67. Gap3/s context failure versus gap3p successful dispatch; unknown exits remain unknown	3.4
68. Packing2 mandatory bases repair request coverage, not an actual useful inference	3.4
69. PACK-F3 pass retraction; repaired/inherited controls are not a fresh full rerun	3.4
70. Gap3p all 46 OPEN and semantic padding/necessity-overstatement flags	4.4
71. Native ten-empty-job pause; historical diagnosis is retrospective	4.5
72. X32 disclosure/padding/source-credit fixes; no actual H truth established	6.2
73. Four proposal thesis origins and literature/audit verdicts remain distinct	6.3, 7.4
74. Gap3h semantic preflight block, no native refusal or completed conditional route	6.4
75. B2/T2 verified originals distinct from refused polished revisions	7.6, A.9
76. Carry-over requires exact source authority; equivalence votes persist	7.6, A.9
77. Meaning probe small live evidence; inline review uses fake providers	7.6, A.9
78. Gap3p local usage not campaign cost	8.3, A.11
79. Second orchestrator stall, corrections ending turns, rotations and cleanup	8.4, A.11
80. Novel mechanism description does not establish first-ever priority	1.1, 9.4, A.1
81. Janitor, kappa5 and xi2 added to complete credits	B.6
82. Gap3h held grounding metadata is not an executable writer packet or licensed queue route	6.4
83. T2 title-only correction verifies citations but final 14-flag union refuses polished revision	7.8, A.10
84. Meaning inventory recovery preserves all 51 adverse pairs; attempted/fallback word counts differ	7.8, A.10
85. V6 table sorting misplaced the pending r21 incident row and R1 source row; v7 restores both without changing evidence	A.4, D

Appendix E. Complete retained-target route atlas

Each row depicts a retained target branching to its two polarity routes. Labels are abbreviated for legibility; `data.json/v6/route_atlas` retains full target texts/formulas and route IDs. The atlas includes all 27 A targets (54 routes), all 20 B targets (40 routes), all three C targets (six routes), and all 23 gap3p targets (46 routes). Every outcome is OPEN. B has one three-node partial argument; all other A/B/C roots and every gap3p route have no accepted inference. These are 73 target records and 146 polarity-route records across distinct experiments, not 73 philosophically distinct theses. Rejected external T2 probes in gap3 have no route to draw (§4.4). Earlier pilot populations are separately shown in the historical route census (§4.1). [R28, R53]



Figure 21. Every retained A target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

A: retained targets → polarity routes

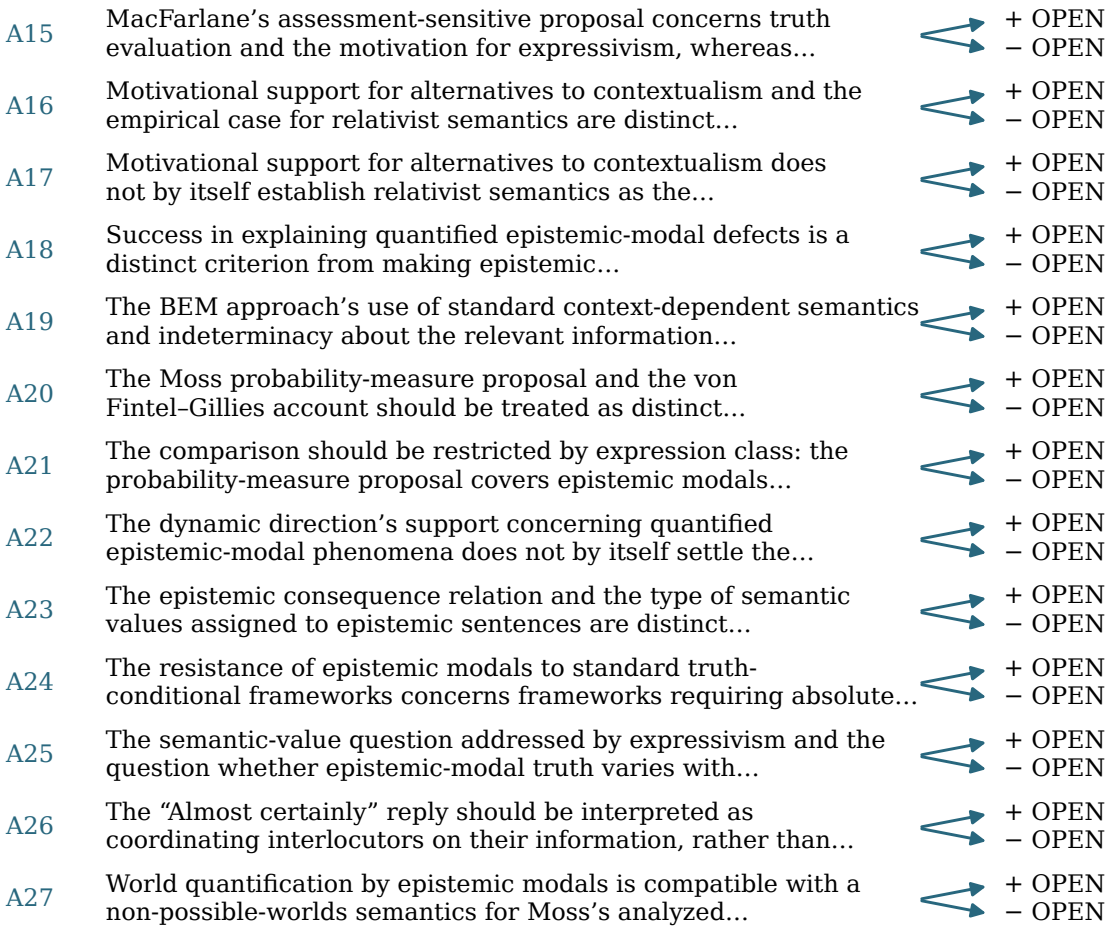


Figure 22. Every retained A target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

B: retained targets → polarity routes



Figure 23. Every retained B target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

B: retained targets → polarity routes

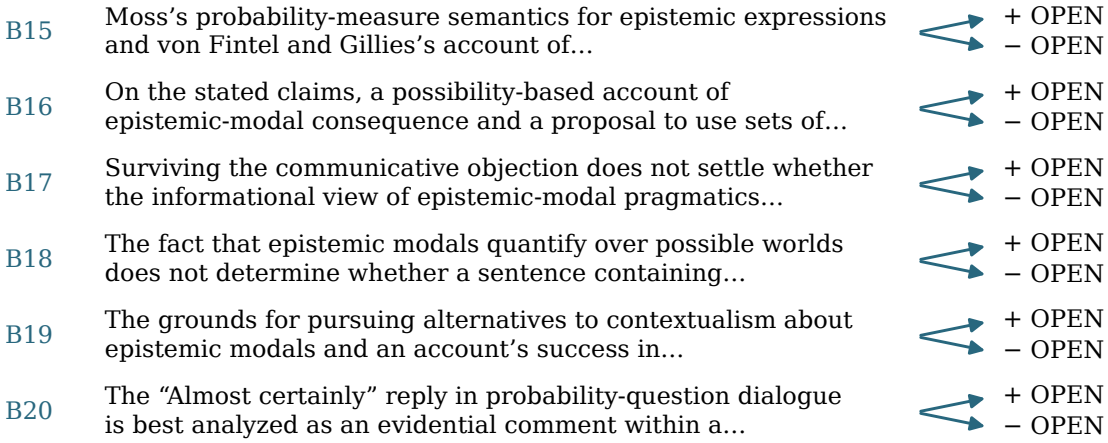


Figure 24. Every retained B target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

C: retained targets → polarity routes

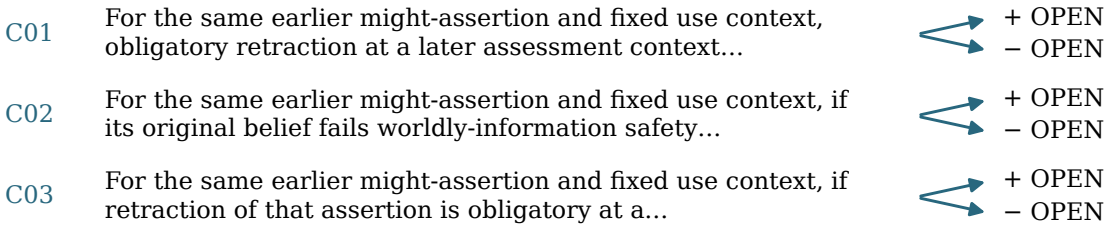


Figure 25. Every retained C target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

gap3p: retained targets → polarity routes

G01	Information-sensitive safety for probabilistic modal knowledge does not by itself license a retraction obligation:…	+ OPEN − OPEN
G02	Agreement and subject-directed attitude-ascription motivations for alternatives to contextualism support an…	+ OPEN − OPEN
G03	Agreement and subject-directed attitude-ascription motivations for alternatives to contextualism can be addressed…	+ OPEN − OPEN
G04	For epistemic modals, the agreement-based case for moving beyond contextualism can be met by making truth…	+ OPEN − OPEN
G05	Assessment-context selection of relevant information is compatible with a bindable covert partition argument: both…	+ OPEN − OPEN
G06	Assessment-context selection of relevant information and binding of a covert partition argument are distinct…	+ OPEN − OPEN
G07	Binding by “however you look at it” supports a bindable covert partition argument, but does not by itself show that…	+ OPEN − OPEN
G08	A possibility-based semantic account of epistemic modals and an adverbialist account of epistemic modal attitudes…	+ OPEN − OPEN
G09	An adverbialist account of a subject’s epistemic-modal attitude does not by itself specify the semantic consequence…	+ OPEN − OPEN
G10	A semantic framework can explain epistemic possibility through assertibility and entailment without treating ◇φ as…	+ OPEN − OPEN
G11	An assertibility clause for epistemic possibility does not by itself commit a theory to a distinct proposition…	+ OPEN − OPEN
G12	The assertibility clause for epistemic possibility and the adverbialist account of the corresponding doxastic…	+ OPEN − OPEN
G13	The epistemic-contradiction pattern is accommodated in the possibility framework by separating compatibility from…	+ OPEN − OPEN
G14	The possibility-framework treatment accommodates epistemic contradiction without making the three constraints…	+ OPEN − OPEN

Figure 26. Every retained gap3p target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.

gap3p: retained targets → polarity routes

G15	The compatibility claim in the possibility framework is not a counterexample to the classical incompatibility…	+ OPEN − OPEN
G16	Relativism surviving the communicative objection and binding effects supporting a contextualist theory concern…	+ OPEN − OPEN
G17	Moss’s binding-effects argument is evidence for a contextualist theory, while Egan’s communicative-test result…	+ OPEN − OPEN
G18	Epistemic possibility talk can be assessed relative to information selected by the context of assessment while…	+ OPEN − OPEN
G19	In epistemic-possibility discourse, the information state selected as relevant by the context of assessment can…	+ OPEN − OPEN
G20	Assessment context can determine which information is relevant to evaluating epistemic possibility talk, while that…	+ OPEN − OPEN
G21	The acceptability test for epistemic “might” should be parameterized by the context of assessment.	+ OPEN − OPEN
G22	Epistemic “might” both tests consistency with an information state and conveys higher-order information about the…	+ OPEN − OPEN
G23	An epistemic “might” test can be evaluated against worldly information distinct from the speaker’s information…	+ OPEN − OPEN

Figure 27. Every retained gap3p target in this panel has both polarity routes. Labels are abbreviated; exact text/formulas and route IDs are retained in the figure data. These rows are records, not a count of distinct novel theses.